# DISTRIBUTED GRADIENT-REGULARIZED NEWTON METHOD: SCHEDULED CONSENSUS AND $\mathcal{O}(\varepsilon^{-1})$ GLOBAL ITERATION COMPLEXITY*

WEI HU†, PENGCHENG XIE‡, YA-XIANG YUAN§, AND LI ZHANG¶

**Abstract.** We propose DisGrem, a fully decentralized second-order method for convex consensus optimization over networks. Each agent solves a local Newton system with vanishing gradient-norm regularization $\lambda_{i,k} = \sqrt{M\|\tilde{g}_{i,k}\|}$ and an eigenvalue-shift stabilizer, communicating through a two-stage gossip-mixing mechanism. We introduce a reference-step framework that reduces the network-wide update to an inexact centralized regularized Newton step, replacing the static Hessian-heterogeneity assumptions of prior work with an increment-based dispersion analysis that imposes no irreducible accuracy floor. Under a bounded-iterates assumption, after a lower-order burn-in phase, DisGrem achieves $\|\nabla f(\bar{x}_k)\| \leq \varepsilon$ in $\mathcal{O}(\varepsilon^{-1})$ iterations—matching the centralized regularized Newton rate—without line search or stepsize tuning; the total communication cost is $\mathcal{O}(\varepsilon^{-1}\log(1/\varepsilon))$ rounds. Under strong convexity and a relative tracking-accuracy condition, we further establish conditional local superlinear convergence of order 3/2. In our nine-problem benchmark suite, the DisGrem family attains relF $\leq 10^{-6}$ on every test instance, while the tested baselines stagnate or diverge on at least one problem.

**Key words.** decentralized optimization, second-order methods, gradient regularization, Newton method, consensus optimization, communication efficiency

**AMS subject classifications.** 90C25, 90C30, 65K05, 68W15

## 1. Introduction.

We consider the decentralized consensus optimization problem

$$\min_{x\in\mathbb{R}^d} f(x) := \frac{1}{N}\sum_{i=1}^{N} f_i(x), \tag{1.1}$$

where each local cost $f_i$ is known only to agent $i$ of a connected network and agents communicate exclusively with their immediate neighbors. Each outer iteration involves one or more gossip rounds, so communication is a primary bottleneck in decentralized implementations; consequently, reducing the iteration count typically lowers the overall communication budget. First-order decentralized methods may need prohibitively many iterations on ill-conditioned problems. Second-order information can cut this count substantially, yet decentralized second-order methods face additional difficulties: curvature data are expensive to transmit, and global convergence typically requires stepsize tuning or line search.

In the centralized setting, the regularized Newton method of Mishchenko [5] takes a different approach, updating the iterate via an explicitly regularized system:

$$x_{k+1} = x_k - \left(\nabla^2 f(x_k) + \lambda_k I\right)^{-1}\nabla f(x_k).$$

*Submitted to the editors March 3, 2026.

† LSEC, ICMSEC, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China; and University of Chinese Academy of Sciences, Beijing 100049, China (huwei@amss.ac.cn). Corresponding author.

‡ Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA (pxie98@gmail.com).

§ LSEC, ICMSEC, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China (yyx@lsec.cc.ac.cn).

¶ LSEC, ICMSEC, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China (zhangli2022@lsec.cc.ac.cn).

With a vanishing regularization parameter $\lambda_k \asymp \sqrt{\|\nabla f(x_k)\|}$, this method achieves the optimal $\mathcal{O}(1/k^2)$ functional rate under convexity and Lipschitz Hessians—equivalently, $\|\nabla f(x_k)\| \le \varepsilon$ in $\mathcal{O}(\varepsilon^{-1})$ iterations—requiring no line search or stepsize tuning. Doikov and Nesterov [18] show that gradient-norm regularization can replace the more expensive cubic model [1, 10] while preserving the same complexity guarantees; Gratton et al. [17] extend the approach to nonconvex objectives via negative-curvature exploitation. A natural question is whether this $\mathcal{O}(\varepsilon^{-1})$ iteration complexity can be preserved in a fully decentralized setting. The difficulty is that the vanishing regularization $\lambda_k \asymp \sqrt{\|\nabla f(x_k)\|}$ relies on globally consistent gradient and curvature information. When each agent sees only a local approximation corrupted by consensus error, the balance between regularization and curvature breaks down, and existing methods resort to line search or static heterogeneity assumptions to recover convergence.

Two specific challenges arise. First, each agent inverts a local regularized system whose average generally differs from the global Newton step; this gap depends on inter-agent agreement and must be controlled without static heterogeneity constants. Second, Hessian trackers can become transiently indefinite due to imperfect consensus, threatening the well-posedness of local solves. Our algorithm DisGrem (Distributed Gradient-Regularized Newton Method) addresses the first challenge through an increment-based dispersion recursion that bounds tracker mismatch via Lipschitz differences, and the second through an eigenvalue-shift stabilizer that ensures well-posedness at every iteration. DisGrem requires only a single scaling parameter $M \ge L_2$ and does not need stepsize tuning or line search. To our knowledge, no prior fully decentralized method achieves $\mathcal{O}(\varepsilon^{-1})$ convergence to arbitrary accuracy with gradient-norm regularization under these conditions.

We summarize our main contributions as follows:

- **A fully decentralized gradient-regularized Newton framework.** We propose DisGrem, in which each agent solves a local Newton system with vanishing regularization $\lambda_{i,k} = \sqrt{M\|\tilde{g}_{i,k}\|}$. A two-stage gossip mechanism and an eigenvalue-shift stabilizer ensure that all local systems remain well posed without line search or explicit positive-semidefinite (PSD) projection.
- **An analytical reduction to an inexact centralized Newton step.** We introduce a reference-step construction that quantifies the gap between the averaged local step and the ideal global Newton step, thereby reducing the decentralized dynamics to an inexact centralized regularized Newton iteration. This framework also replaces the usual static heterogeneity assumption by an increment-based dispersion recursion, avoiding an irreducible heterogeneity floor in the final accuracy bound.
- **Sharp global and local guarantees.** Under convexity, Lipschitz Hessians, and bounded iterates, after a lower-order burn-in phase, we prove an $\mathcal{O}(\varepsilon^{-1})$ global iteration complexity for achieving $\|\nabla f(\bar{x}_k)\| \le \varepsilon$, matching the centralized regularized Newton rate. Under a logarithmic mixing schedule, this leads to $\mathcal{O}(\varepsilon^{-1}\log(1/\varepsilon))$ communication rounds. Under strong convexity and a relative tracking-accuracy condition, we further establish conditional local $Q$-superlinear convergence of order 3/2.
- **Practical variants and empirical robustness.** We develop communication-efficient and adaptive variants, namely CeDisGrem and AdaDisGrem. Across nine benchmark problems, the DisGrem family achieves a strong accuracy–robustness balance among the tested methods, while AdaDisGrem achieves the highest robustness in the multi-start experiments.

Section 2 reviews related work. Sections 3–5 present the notation, assumptions, algorithmic variants, and convergence theory. Section 6 reports numerical results. Appendices A and B collect the longer proofs, while Appendix C reports supplementary experiments.

## 2. Related work.

**First-order decentralized methods.** Decentralized (sub)gradient descent originates with Nedić and Ozdaglar [8]. EXTRA [11] and exact diffusion [14] remove the bias of constant-stepsize methods through correction steps or primal-dual reformulations. Gradient tracking [9, 29, 30] achieves exact convergence with a single doubly stochastic matrix by accumulating gradient increments; Alghunaim et al. [21] unify these two perspectives. These methods converge at $\mathcal{O}(1/k)$ or linearly for strongly convex objectives, but the iteration complexity scales with $(1-\rho)^{-1}$ or $(1-\rho)^{-2}$, a severe penalty on poorly connected graphs. Acceleration [4, 27] and communication compression [23, 26] are orthogonal improvements; we adopt the latter for Hessian data in Section 4.4.

**Second-order and quasi-Newton decentralized methods.** Network Newton [6] approximates the global Newton direction via a truncated Hessian power series; Newton tracking [7] embeds Newton-type updates into gradient tracking. DQM [3] replaces exact Hessians with BFGS surrogates; Bajović et al. [22] studies distributed Newton-type corrections with diagonal approximations, and Li et al. [31] propose communication-efficient approximate Newton and variance-reduced methods for networked optimization. ESOM [28] combines exact second-order information with an alternating direction method of multipliers (ADMM) consensus step. SONATA [12] solves a sequence of strongly convex local surrogates and handles nonconvex objectives, while Network-GIANT [13] constructs a global Newton direction through harmonic-mean Hessian consensus. All of these methods require explicit stepsize tuning, penalty parameter selection, or line search. Furthermore, to the best of our knowledge, none attains the optimal centralized Newton iteration complexity of $\mathcal{O}(1/k^2)$ for general convex problems in a rigorous global sense. While exact methods like ESOM and Newton tracking can reach arbitrary precision for strongly convex objectives, their convergence heavily relies on static Hessian-heterogeneity constants (e.g., $\sigma_H$) to dictate conservative stepsizes. Other approximate decentralized Newton methods simply ignore this heterogeneity or suffer from an irreducible accuracy floor $\|\nabla f(\bar{x})\| \leq \mathcal{O}(\sigma_H)$. Daneshmand et al. [16] pursue a different approach, combining gradient tracking with cubic regularization and local Hessian subsampling to achieve an iteration count comparable to centralized cubic Newton. However, their convergence guarantee holds only up to the statistical precision of the Hessian estimator—an inherent accuracy floor from subsampling—and the method still requires a stepsize parameter. Our increment-based dispersion analysis, combined with exact Hessian tracking and gradient-norm regularization, eliminates both the heterogeneity floor and the statistical precision floor, yielding convergence to arbitrary $\varepsilon$ without any stepsize.

**Communication-efficient and inversion-free approaches.** The $\mathcal{O}(d^2)$ per-round cost of transmitting Hessian data and the $\mathcal{O}(d^3)$ cost of solving Newton systems are the two main bottlenecks of decentralized second-order methods. Zhang et al. [20] combine lazy Hessian updates with compression for distributed cubic Newton but rely on a central parameter server; our CeDisGrem achieves analogous savings in a fully peer-to-peer topology. On the computation side, DINAS [15] avoids Hessian inversion through iterative linear solvers, and INDO [19] proposes an inversion-free method for consensus optimization. Incorporating such inexact solvers into the DisGrem framework is a promising direction, as discussed in Section 7.

**Regularized Newton and cubic regularization.** Our starting point is the centralized regularized Newton method of Mishchenko [5]: setting $\lambda_k \asymp \sqrt{\|\nabla f(x_k)\|}$ yields the tight $\mathcal{O}(1/k^2)$ functional rate under convexity and Lipschitz Hessians, with no stepsize to choose. The idea has roots in cubic regularization [1, 10], which achieves the same rate through an adaptive cubic model. Doikov and Nesterov [18] establish that gradient-norm regularization can replace the cubic model across a broad class of problems. Gratton et al. [17] extend it to nonconvex objectives via negative-curvature exploitation. Doikov and Nesterov [24] study local convergence of higher-order tensor methods, and Doikov et al. [25] analyze lazy Hessian updates in the centralized setting, an idea we also explore in Section 6.3. As far as we know, there is no prior work that combines gradient-norm regularization (as opposed to cubic regularization) with full gradient and Hessian tracking in a decentralized setting, achieving exact $\mathcal{O}(\varepsilon^{-1})$ convergence without stepsize tuning, line search, or static heterogeneity constants.

## 3. Preliminaries and assumptions.

**3.1. Network model and notation.** Unless otherwise stated, $\|\cdot\|$ denotes the Euclidean norm for vectors and stacked vectors. For matrices, $\|\cdot\|_2$ and $\|\cdot\|_F$ denote the spectral and Frobenius norms, respectively.

We model the communication network as an undirected, connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \ldots, N\}$ is the set of agents and $\mathcal{E}$ is the set of communication links. Agents $i$ and $j$ can exchange information if and only if $(i, j) \in \mathcal{E}$. Information mixing over this network is represented by a symmetric, doubly stochastic matrix $W \in \mathbb{R}^{N\times N}$ that respects the graph topology, meaning $W_{ij} > 0$ only if $(i, j) \in \mathcal{E}$ or $i = j$. We assume that $W$ has positive diagonal entries and satisfies $W_{ij} \geq 0$, $W\mathbf{1} = \mathbf{1}$, $\mathbf{1}^\top W = \mathbf{1}^\top$. Let $\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) := \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ denote the averaging projector, $I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) := I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$ the complementary projector, and

$$\rho := \left\| W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) \right\|_2 \in [0, 1).$$

The positive diagonal condition is satisfied by the Metropolis–Hastings weights used in our experiments and rules out the periodic case in which an eigenvalue $-1$ would give $\rho = 1$. For stacked vectors $Z = [z_1; \ldots; z_N] \in \mathbb{R}^{Nd}$ we use the Euclidean norm $\|Z\|$ and write $\bar{z} := \frac{1}{N}\sum_{i=1}^N z_i$ for the average. Stacked primal variables are denoted $X = [x_1; \ldots; x_N] \in \mathbb{R}^{Nd}$, and we define the separable network objective

$$F(X) := \sum_{i=1}^N f_i(x_i).$$

Then

$$\nabla F(X) = [\nabla f_1(x_1); \ldots; \nabla f_N(x_N)] \in \mathbb{R}^{Nd},$$
$$\nabla^2 F(X) = \operatorname{blkdiag}\big(\nabla^2 f_1(x_1), \ldots, \nabla^2 f_N(x_N)\big),$$

where $\operatorname{blkdiag}(\cdot)$ denotes the block-diagonal matrix. We also introduce the stacked Hessian vectorization

$$\mathcal{H}(X) := \big[\operatorname{vec}(\nabla^2 f_1(x_1)); \ldots; \operatorname{vec}(\nabla^2 f_N(x_N))\big] \in \mathbb{R}^{Nd^2},$$

where $\operatorname{vec}(\cdot)$ stacks the columns of a matrix into a single vector. When evaluated at a consensus point $x$, we write $\mathcal{H}(\mathbf{1} \otimes x)$.

Table 1 collects the main symbols used throughout the paper.

Table 1: Notation summary.

| Symbol | Meaning |
|---|---|
| $N$, $d$ | number of agents; problem dimension |
| $x_{i,k}$ | primal iterate of agent $i$ at iteration $k$ |
| $\bar{x}_k$ | average iterate $\frac{1}{N}\sum_{i=1}^{N} x_{i,k}$ |
| $X_k$ | stacked vector $[x_{1,k}; \dots; x_{N,k}] \in \mathbb{R}^{Nd}$ |
| $\tilde{x}_{i,k}$, $\tilde{g}_{i,k}$, $\tilde{H}_{i,k}$ | pre-mixed quantities (after $\tau_k$ gossip rounds) |
| $g_{i,k}$, $H_{i,k}$ | gradient and Hessian trackers |
| $\tilde{G}_k$ | stacked pre-mixed gradient trackers $[\tilde{g}_{1,k}; \dots; \tilde{g}_{N,k}]$ |
| $W$ | doubly stochastic mixing matrix |
| $\rho := \|W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\|_2$ | mixing rate ($1-\rho$ is the spectral gap) |
| $\tau_k$, $t_k$ | pre-mixing and post-mixing depths |
| $D(Z)$ | RMS dispersion $\left(\frac{1}{N}\sum_{i=1}^{N} \|z_i - \bar{z}\|^2\right)^{1/2}$ |
| $M$ | regularization scaling ($M \geq L_2$) |
| $\lambda_{i,k}$ | $\sqrt{M\|\tilde{g}_{i,k}\|}$ (vanishing regularizer) |
| $\delta_{i,k}$ | eigenvalue-shift stabilizer |
| $L_1$, $L_2$ | gradient and Hessian Lipschitz constants |
| $\|\cdot\|$, $\|\cdot\|_2$, $\|\cdot\|_F$ | Euclidean norm for vectors; spectral and Frobenius norms for matrices |

To quantify disagreement among agents we define the root-mean-square (RMS) dispersion (also called consensus error or disagreement in the decentralized optimization literature, see e.g. Nedić and Ozdaglar [8], Nedić et al. [9]) $D(Z) := \left(\frac{1}{N}\sum_{i=1}^{N}\|z_i - \bar{z}\|^2\right)^{1/2}$. In particular, writing $X_k := [x_{1,k}; \dots; x_{N,k}]$ for the stacked iterates and $\tilde{G}_k := [\tilde{g}_{1,k}; \dots; \tilde{g}_{N,k}]$ for the stacked gradient trackers, $D(X_k)$ and $D(\tilde{G}_k)$ measure the primal and gradient-tracker disagreements at iteration $k$.

### 3.2. Assumptions.

ASSUMPTION 3.1. *(i) Each $f_i$ is convex and twice continuously differentiable with $L_1$-Lipschitz gradient and $L_2$-Lipschitz Hessian:*

$$\|\nabla f_i(x) - \nabla f_i(y)\| \leq L_1 \|x - y\|, \qquad \left\|\nabla^2 f_i(x) - \nabla^2 f_i(y)\right\|_2 \leq L_2 \|x - y\|.$$

*(ii) $W$ is symmetric, doubly stochastic, has positive diagonal entries, and $\rho = \left\|W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\right\|_2 < 1$.*

*(iii) The average objective $f$ is coercive: $f(x) \to +\infty$ as $\|x\| \to \infty$. Equivalently, for every $c \in \mathbb{R}$ the sublevel set $\{x : f(x) \leq c\}$ is bounded.*

*Remark* 3.2. Coercivity guarantees existence of minimizers and bounded sublevel sets. It is satisfied, for instance, whenever the objective contains an $\ell_2$ regularizer.

## 4. The DisGrem algorithm.

**4.1. Design motivation.** Lifting the regularized Newton method to a decentralized setting requires mimicking the global Newton step without a central coordinator. This presents three coupled mathematical and algorithmic difficulties, which motivate the design of DisGrem.

**1. The gap between averaging and inversion.** The ideal centralized step solves the global average system

$$s_k^\star = -\Big(\frac{1}{N}\sum_{i=1}^{N} \nabla^2 f_i(\bar{x}_k) + \lambda_k I\Big)^{-1} \frac{1}{N}\sum_{i=1}^{N} \nabla f_i(\bar{x}_k).$$

In a decentralized network, each agent $i$ solves a local regularized system and produces a local step $s_{i,k} = -(\tilde{H}_{i,k} + \tilde{\lambda}_{i,k} I)^{-1} \tilde{g}_{i,k}$. Due to the non-commutativity of averaging and matrix inversion, the average step $\bar{s}_k$ generally differs from the step obtained by inverting the averaged system:

$$\bar{s}_k \;=\; \frac{1}{N}\sum_{i=1}^{N}\big(\tilde{H}_{i,k} + \tilde{\lambda}_{i,k} I\big)^{-1}\tilde{g}_{i,k} \;\neq\; \Big(\frac{1}{N}\sum_{i=1}^{N}\big(\tilde{H}_{i,k} + \tilde{\lambda}_{i,k} I\big)\Big)^{-1}\frac{1}{N}\sum_{i=1}^{N}\tilde{g}_{i,k},$$

in general. This discrepancy is governed by the inter-agent disagreement in $(x_{i,k}, g_{i,k}, H_{i,k})$. DisGrem mitigates it by introducing a multi-round pre-mixing stage that drives the local inputs closer to their network-wide means, tightly controlling the step dispersion.

**2. Transient indefiniteness from imperfect consensus.** Decentralized tracking propagates Hessian increments $\nabla^2 f_j(x_{j,k+1}) - \nabla^2 f_j(x_{j,k})$ across the network. Although the true local Hessians satisfy $\nabla^2 f_i(x) \succeq 0$ for convex objectives, the tracked surrogate decomposes as

$$\tilde{H}_{i,k} \;=\; \underbrace{\nabla^2 f(\bar{x}_k)}_{\succeq 0} \;+\; \underbrace{\big(\tilde{H}_{i,k} - \nabla^2 f(\bar{x}_k)\big)}_{\text{tracking error}},$$

where the tracking error accumulates gossip-mixed Hessian increments and can have negative eigenvalues. During the transient phase before consensus is reached, $\lambda_{\min}(\tilde{H}_{i,k})$ may be negative, rendering the local system ill-posed. We introduce an eigenvalue-shift stabilizer $\delta_{i,k} := \max\{0, -\lambda_{\min}(\tilde{H}_{i,k})\}$. It makes the local coefficient matrix positive definite whenever $\tilde{g}_{i,k} \neq 0$. When $\tilde{g}_{i,k} = 0$, the local step is set to zero.

**3. Stepsize-free local regularization.** The vanishing choice $\lambda_k \asymp \sqrt{\|\nabla f(\bar{x}_k)\|}$ is responsible for the optimal $\mathcal{O}(1/k^2)$ centralized rate. In the absence of global gradient knowledge, DisGrem sets a local vanishing regularizer $\lambda_{i,k} = \sqrt{M\|\tilde{g}_{i,k}\|}$. The scaling parameter $M \geq L_2$ acts as a damping mechanism: it bounds the local step magnitude ($\|s_{i,k}\| \leq \sqrt{\|\tilde{g}_{i,k}\|/M}$) analogous to a trust-region radius, removing the need for per-iteration stepsize tuning or line search.

**4.2. DisGrem: full algorithm.** Each node $i$ stores a primal variable $x_{i,k} \in \mathbb{R}^d$, a gradient tracker $g_{i,k} \in \mathbb{R}^d$, and a Hessian tracker $H_{i,k} \in \mathbb{R}^{d\times d}$. For compactness, the result of $t$ successive neighbor-gossip rounds is written as $\sum_{j=1}^{N}[W^t]_{ij} z_j$. Operationally, with $\mathcal{N}_i^+ := \mathcal{N}_i \cup \{i\}$, this quantity is obtained by $t$ local updates $z_i^{(\ell+1)} = \sum_{j\in\mathcal{N}_i^+} W_{ij} z_j^{(\ell)}$; no direct all-to-all communication is required.

**Algorithm 4.1** DisGrem

1: **Input:** $\{x_{i,0}\}$, $W$, $M > 0$, mixing schedule $\{\tau_k, t_k\}$.
2: **Initialize:** $g_{i,0} \leftarrow \nabla f_i(x_{i,0})$, $H_{i,0} \leftarrow \nabla^2 f_i(x_{i,0})$ for all $i$.
3: **for** $k = 0, 1, 2, \dots$ **do**
4: **(A) Pre-mixing** ($\tau_k \geq 1$ gossip rounds):
$\tilde{x}_{i,k} \leftarrow \sum_{j=1}^N [W^{\tau_k}]_{ij}\, x_{j,k}, \quad \tilde{g}_{i,k} \leftarrow \sum_{j=1}^N [W^{\tau_k}]_{ij}\, g_{j,k}, \quad \tilde{H}_{i,k} \leftarrow \sum_{j=1}^N [W^{\tau_k}]_{ij}\, H_{j,k}$.
5: **(B) Local Newton step** (each agent $i$ in parallel):
$\lambda_{i,k} \leftarrow \sqrt{M\|\tilde{g}_{i,k}\|}, \quad \delta_{i,k} \leftarrow \max\{0, -\lambda_{\min}(\tilde{H}_{i,k})\}$.
If $\tilde{g}_{i,k} = 0$, set $s_{i,k} \leftarrow 0$; otherwise solve $\big(\tilde{H}_{i,k} + (\lambda_{i,k} + \delta_{i,k})I\big)\, s_{i,k} = -\tilde{g}_{i,k}$.
$y_{i,k+1} \leftarrow \tilde{x}_{i,k} + s_{i,k}$.
6: **(C) Post-mixing** ($t_k \geq 1$ gossip rounds):
$x_{i,k+1} \leftarrow \sum_{j=1}^N [W^{t_k}]_{ij}\, y_{j,k+1}$.
7: **(D) Tracker updates:**
$g_{i,k+1} \leftarrow \sum_{j=1}^N [W^{t_k}]_{ij} \big(\tilde{g}_{j,k} + \nabla f_j(x_{j,k+1}) - \nabla f_j(x_{j,k})\big)$.
$H_{i,k+1} \leftarrow \sum_{j=1}^N [W^{t_k}]_{ij} \big(\tilde{H}_{j,k} + \nabla^2 f_j(x_{j,k+1}) - \nabla^2 f_j(x_{j,k})\big)$.
8: **end for**

**4.3. Communication cost per iteration.** Each outer iteration of DisGrem involves $(\tau_k + 2t_k)$ rounds of neighbor communication. In each round every agent $i$ sends messages to and receives messages from its $|\mathcal{N}_i|$ neighbors. The payload differs across the three communication stages. Pre-mixing exchanges $(x, g, H)$, namely two $d$-dimensional vectors and one symmetric Hessian matrix. Post-mixing exchanges only the trial variable $y$. The tracker update exchanges one gradient-tracker input and one Hessian-tracker input. The post-mixing step (C) and the tracker-update step (D) must be performed sequentially. Agent $i$ must first receive the mixed $y$ variables to compute $x_{i,k+1}$; only then can it evaluate the exact gradients and Hessians at $x_{i,k+1}$ to form the tracker inputs. Consequently, step (C) requires $t_k$ rounds and step (D) requires an additional $t_k$ rounds, giving a total of $(\tau_k + 2t_k)$ communication rounds per outer iteration.

With double precision (8 bytes per float), the sent message volume per agent per outer iteration in the uncompressed algorithm is

$$\begin{aligned} C_{\text{iter}}^{\text{send}}(k) &= 8|\mathcal{N}_i|\Big[\tau_k\Big(2d + \frac{d(d+1)}{2}\Big) + t_k d + t_k\Big(d + \frac{d(d+1)}{2}\Big)\Big] \\ &= 8|\mathcal{N}_i|(\tau_k + t_k)\Big(2d + \frac{d(d+1)}{2}\Big) \quad \text{bytes.} \end{aligned} \tag{4.1}$$

All communication figures reported in Section 6 use the same stage-wise payload accounting, summed over the directed off-diagonal entries of the mixing matrix and reported in cumulative MB.

**4.4. Practical variants.** We develop three practical variants of Algorithm 4.1: two address communication cost and parameter selection individually, while a third combines both mechanisms. All variants share the same four-step structure and differ only in the Hessian-tracker and regularization-parameter modules (see Table 2 at the end of this section for a summary).

CeDisGrem (the prefix "Ce" stands for communication-efficient) reduces communication by compressing Hessian data. The dominant communication cost of DisGrem is the $\frac{d(d+1)}{2}$-dimensional Hessian increment $\nabla^2 f_j(x_{j,k+1}) - \nabla^2 f_j(x_{j,k})$. CeDisGrem replaces each symmetric Hessian increment $\Delta H_{j,k} := \nabla^2 f_j(x_{j,k+1}) - \nabla^2 f_j(x_{j,k})$ by a compressed approximation $\mathcal{C}(\Delta H_{j,k})$. The implementation supports both element-wise

Top-$k$ sparsification and low-rank symmetric truncation; the main experiments use Top-$k$ with a 10% element budget, while Appendix C compares the two choices. For the low-rank option, let $\Delta H_{j,k} = V \operatorname{diag}(\mu_1, \ldots, \mu_d) V^\top$ with $|\mu_1| \geq \cdots \geq |\mu_d|$ and $V = [v_1, \ldots, v_d]$ orthogonal. Define the truncated approximation

$$\hat{\Delta} H_{j,k} := \sum_{\ell=1}^{r} \mu_\ell v_\ell v_\ell^\top = V_r \Lambda_r V_r^\top,$$

where $V_r = [v_1, \ldots, v_r] \in \mathbb{R}^{d\times r}$ and $\Lambda_r = \operatorname{diag}(\mu_1, \ldots, \mu_r) \in \mathbb{R}^{r\times r}$. This $\hat{\Delta} H_{j,k}$ is the best rank-$r$ symmetric approximation of $\Delta H_{j,k}$ in Frobenius norm, since the best rank-$r$ approximation of a symmetric matrix retains the $r$ eigenpairs with largest absolute eigenvalues. Thus, transmitting $(V_r, \Lambda_r)$ costs $r(d+1)$ floats instead of $\frac{d(d+1)}{2}$ floats per neighbor per round.

The truncation error satisfies

$$\begin{aligned} \left\| \Delta H_{j,k} - \hat{\Delta} H_{j,k} \right\|_F &= \left( \textstyle\sum_{\ell=r+1}^{d} \mu_\ell^2 \right)^{1/2} \\ &\leq |\mu_{r+1}(\Delta H_{j,k})| \sqrt{d-r}, \end{aligned}$$

where $\mu_{r+1}(\Delta H_{j,k})$ denotes the $(r+1)$-th largest absolute eigenvalue. For Top-$k$, $\mathcal{C}(\Delta H_{j,k})$ retains the largest-magnitude entries and symmetrizes the result. The compression error can be viewed as an additional Hessian-tracker perturbation; in the experiments it is effectively controlled by the regularization. A full theory for the compressed variant is left for future work. Empirically, both Top-$k$ and low-rank compression reduce communication with modest iteration overhead when the compression level is not too aggressive (Section 6.3).

*Remark* 4.1. The exact eigendecomposition of $\Delta H_{j,k} \in \mathbb{R}^{d\times d}$ costs $\mathcal{O}(d^3)$ per agent per iteration—the same order as the Newton-system solve. When $d$ is large, randomized singular value decomposition (SVD) (cost $\mathcal{O}(d^2 r)$) or Nyström-type approximations can replace the exact decomposition. The additional approximation error enters the tracker dispersion additively; a full convergence theory for this perturbation is left for future work.

AdaDisGrem introduces an adaptive scaling factor. When $M$ is fixed, choosing a good value typically relies on problem-specific curvature information (see Section 4.5). AdaDisGrem maintains a running estimate $\hat{M}_{i,k}$ of each agent's local Hessian Lipschitz constant via a secant condition. Specifically, define the local secant ratio

$$\begin{aligned} \hat{L}_{i,k} &\leftarrow \frac{\left\| \nabla^2 f_i(x_{i,k}) - \nabla^2 f_i(x_{i,k-1}) \right\|_2}{\| x_{i,k} - x_{i,k-1} \|}, \\ \hat{M}_{i,k} &\leftarrow \max\!\big(\gamma\, \hat{M}_{i,k-1},\ \zeta\, \min(\hat{L}_{i,k},\, \eta_c\, \hat{M}_{i,0})\big), \end{aligned}$$

where $\zeta \geq 1$ is a safety inflation factor, $\gamma \in (0,1)$ is the decay factor, and $\eta_c > 0$ is the cap parameter. If $x_{i,k} = x_{i,k-1}$, we set $\hat{L}_{i,k} = 0$. The vanishing regularizer becomes $\lambda_{i,k} = \sqrt{\hat{M}_{i,k} \|\tilde{g}_{i,k}\|}$. The decay factor allows $\hat{M}_{i,k}$ to decrease when the local curvature stabilizes, enabling larger Newton steps near the solution; the cap prevents transient large Hessian increments from inflating the estimate. This mechanism removes the need to specify $L_2$ a priori at the cost of a mild adaptation lag. Note that $\hat{L}_{i,k}$ is a per-iterate secant approximation, distinct from the static proxy $H^0_{\max} := \max_i \|\nabla^2 f_i(x_0)\|_2$ used to scale the default $M$ (Section 4.5). Because the secant ratio probes curvature along a single direction, it may underestimate $L_2$; the max-smoothing and cap $\eta_c$ prevent the estimate from collapsing. Section 6.4 provides an empirical study indicating robustness across a $100\times$ range of initial $\hat{M}_{i,0}$.

CeAdaDisGrem applies both Hessian compression (Ce) and adaptive $M$ (Ada) simultaneously. Because the compressed Hessian increments introduce approximation noise into the secant-based Lipschitz estimation, the interaction between the two mechanisms is nontrivial: Section 6.3 shows that this combination performs well at moderate precision but may slow convergence at very high precision.

Table 2 summarizes the full DisGrem family. All members share the core structure of Algorithm 4.1; differences are confined to the Hessian-tracker and regularization-parameter modules.

Table 2: The DisGrem algorithm family. All variants use the same four-step structure (Algorithm 4.1); differences are confined to the Hessian communication and the choice of $M$.

| Variant | Hessian | $M$ | Extra parameter(s) |
|---|---|---|---|
| DisGrem | exact tracking | fixed | $M$ |
| CeDisGrem | compressed + lazy | fixed | $M$, comp., budget, $K_{\text{lazy}}$ |
| AdaDisGrem | exact tracking | adaptive | $\hat{M}_{i,0}$, $\gamma$, $\zeta$, $\eta_c$ |
| CeAdaDisGrem | compressed + lazy | adaptive | Ada params, comp., budget, $K_{\text{lazy}}$ |

**4.5. Practical guidelines for parameter selection.** The theoretical convergence requires $M \geq L_2$ (Theorem 5.20). When the Hessian Lipschitz constant $L_2$ is unknown, $M$ acts as a robustness parameter: larger values increase damping and ensure well-conditioning, though potentially slowing asymptotic convergence. A practical default is $M = M_{\text{fac}} \cdot H^0_{\max}$, where $H^0_{\max} := \max_i \|\nabla^2 f_i(x_0)\|_2$ is a readily computable baseline curvature proxy and $M_{\text{fac}} \in [0.1, 15]$ is a tuning factor that scales with the problem's ill-conditioning. While this heuristic does not strictly enforce $M \geq L_2$, AdaDisGrem fully automates the choice via online secant estimation.

The communication depths are chosen by the logarithmic schedule

$$\tau_k = t_k = \left\lceil \frac{p \log(k+2) + c_{\text{mix}}}{-\log \rho} \right\rceil,$$

with a fixed constant $c_{\text{mix}} \geq 0$. In the numerical experiments we impose a maximum communication depth once the tested accuracy range has been reached.

For Hessian compression in CeDisGrem, Top-$k$ with a 10% element budget is used as the default in the main experiments. For low-rank compression, $r = \lceil d/5 \rceil$ serves as a robust default, while $r = 1$ often suffices for diagonally dominant objectives. To further reduce communication, the Hessian tracker update (step (D) of Algorithm 4.1) can be executed every $K_{\text{lazy}}$-th iteration. Reusing the previous Hessian for $K_{\text{lazy}} \in \{5, 10\}$ consecutive steps reduces payload bytes with minimal impact on convergence for slowly varying problems.

**5. Convergence analysis.** Throughout this section we assume Assumptions 3.1 and 5.1, the parameter requirement $M \geq L_2$, and that Algorithm 4.1 is run with the initialization $g_{i,0} = \nabla f_i(x_{i,0})$, $H_{i,0} = \nabla^2 f_i(x_{i,0})$. Complete proofs are collected in Appendix B, and the dispersion-decay and burn-in construction is detailed in Appendix A.

The convergence argument proceeds through the following chain of reductions. Averaging identities (§5.2) show that the average iterate evolves as $\bar{x}_{k+1} = \bar{x}_k + \bar{s}_k$

and that tracked averages equal true averages. Reference step and bridge bounds (§5.3–§5.4) introduce a "centralized reference" step $s_k^{\mathrm{ref}}$ and bound its gap from $\bar{s}_k$ in terms of tracker dispersions. Stabilizer and step-dispersion control (§5.5–§5.6) relate the eigenvalue shift $\delta_{i,k}$ and the step dispersion to gradient/Hessian tracker dispersions. These three ingredients are combined in §5.7 to verify that the average iterate satisfies an inexact Newton residual bound, from which a 3/2-recursion on the optimality gap (§5.8) yields the $\mathcal{O}(\varepsilon^{-1})$ rate.

In summary, the logical dependency chain is: averaging identities → reference step → stabilizer/dispersion bounds → inexact Newton condition (Proposition 5.14) → 3/2-recursion on the steady subsequence (Lemma 5.18) → global complexity (Theorem 5.20). Appendix A supplies the burn-in index $K_0(\varepsilon)$.

The proof combines two independent stages (detailed in Remark 5.22 after the main theorem). The reader interested only in the final result may skip directly to Theorem 5.20.

### 5.1. Bounded trajectory and local constants.

Assumption 5.1 (Bounded iterates). *There exists a solution $x_\star \in \arg\min f$ and a finite constant $D > 0$ such that the iterates generated by Algorithm 4.1 satisfy*

$$\|\bar{x}_k - x_\star\| \le D, \ \|x_{i,k} - x_\star\| \le D, \ \|\tilde{x}_{i,k} - x_\star\| \le D, \tag{5.1}$$

*for all $k \ge 0$ and all $i \in \{1, \dots, N\}$.*

Under Assumption 5.1, we define all subsequent constants on the compact set $\{x : \|x - x_\star\| \le D\}$. Since each $\nabla f_i$ is globally $L_1$-Lipschitz, the Hessian bound

$$M_{H,\max} := L_1 \tag{5.2}$$

is valid for all iterates and is independent of the compact set.

*Remark* 5.2. Since each $f_i$ is twice continuously differentiable (Assumption 3.1(i)) and the iterates are confined to a compact set by Assumption 5.1, the quantity $\sup_{x\in\mathcal{B}} \max_{1\le i\le N} \|\nabla^2 f_i(x)\|_2$ is automatically finite for any bounded $\mathcal{B} \subset \mathbb{R}^d$. No separate assumption is needed.

*Remark* 5.3. Prior decentralized second-order analyses [12, 13, 28] posit a bounded static heterogeneity constant $\sigma_H := \sup_{\|x-x_\star\|\le D} \frac{1}{\sqrt{N}} \|(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) \otimes I_{d^2})\mathcal{H}(\mathbf{1} \otimes x)\|$. We avoid this entirely: Hessian-tracker dispersion is controlled by an increment-based recursion (Appendix A, Lemma A.8) that tracks mismatch through Lipschitz-continuous differences $\nabla^2 F(X_{k+1}) - \nabla^2 F(X_k)$, so no accuracy floor linked to $\sigma_H$ appears in the final bounds.

**5.2. Averaging identities.** The starting point of the analysis is that doubly stochastic mixing preserves averages, so the mean iterate and mean trackers evolve as if they were computed by a single "virtual agent."

Define averages $\bar{x}_k := \frac{1}{N}\sum_{i=1}^N x_{i,k}$ and similarly for other variables. Define pre-mixed averages

$$\bar{\tilde{x}}_k := \frac{1}{N}\sum_{i=1}^N \tilde{x}_{i,k}, \quad \bar{\tilde{g}}_k := \frac{1}{N}\sum_{i=1}^N \tilde{g}_{i,k},$$

$$\bar{\tilde{H}}_k := \frac{1}{N}\sum_{i=1}^N \tilde{H}_{i,k}.$$

Define the average step $\bar{s}_k := \frac{1}{N}\sum_{i=1}^N s_{i,k}$.

LEMMA 5.4. *For all* $k \ge 0$, $\bar{\tilde{x}}_k = \bar{x}_k$ *and* $\bar{x}_{k+1} = \bar{x}_k + \bar{s}_k$.

*Proof.* Immediate from $W\mathbf{1} = \mathbf{1}$ (doubly stochastic), which gives $\frac{1}{N}\mathbf{1}^\top(W^t \otimes I_d) = \frac{1}{N}\mathbf{1}^\top \otimes I_d$ for every integer $t \ge 1$. ∎

LEMMA 5.5. *Assume Algorithm 4.1 is initialized by* $g_{i,0} = \nabla f_i(x_{i,0})$. *Then for all* $k \ge 0$,

$$\bar{g}_k := \frac{1}{N}\sum_{i=1}^N g_{i,k} = \frac{1}{N}\sum_{i=1}^N \nabla f_i(x_{i,k}), \qquad \bar{\tilde{g}}_k = \bar{g}_k.$$

*Proof.* By induction using $W\mathbf{1} = \mathbf{1}$ and the telescoping form of the tracker update (step (D)). ∎

LEMMA 5.6. *Assume Algorithm 4.1 is initialized by* $H_{i,0} = \nabla^2 f_i(x_{i,0})$. *Then for all* $k \ge 0$,

$$\bar{H}_k := \frac{1}{N}\sum_{i=1}^N H_{i,k} = \frac{1}{N}\sum_{i=1}^N \nabla^2 f_i(x_{i,k}), \qquad \bar{\tilde{H}}_k = \bar{H}_k.$$

*Proof.* The same argument as in the proof of Lemma 5.5, applied to the Hessian tracker, yields the claim. ∎

**5.3. Well-posed local solves and a reference step.** With the averaging identities in hand, we next introduce the reference step $s_k^{\mathrm{ref}}$: the Newton step that a centralized agent would compute using the averaged Hessian and gradient. The gap $\|\bar{s}_k - s_k^{\mathrm{ref}}\|$ then quantifies how much the decentralized updates deviate from this ideal step.

Algorithm 4.1 sets

$$\begin{aligned}
\lambda_{i,k} &:= \sqrt{M\|\tilde{g}_{i,k}\|},\\
\delta_{i,k} &:= \max\{0, -\lambda_{\min}(\tilde{H}_{i,k})\},\\
\tilde{\lambda}_{i,k} &:= \lambda_{i,k} + \delta_{i,k}.
\end{aligned}$$

Define

$$A_{i,k} := \tilde{H}_{i,k} + \tilde{\lambda}_{i,k} I,$$

When $\tilde{g}_{i,k} \ne 0$, the coefficient matrix is positive definite and Algorithm 4.1 sets $s_{i,k} := -A_{i,k}^{-1}\tilde{g}_{i,k}$. When $\tilde{g}_{i,k} = 0$, Algorithm 4.1 uses the convention $s_{i,k} = 0$. All inverse-based estimates below are stated on indices where the corresponding regularization lower bound is positive; in particular, Lemma 5.12 assumes $\underline{\lambda}_k > 0$. Define averaged quantities

$$\begin{aligned}
\bar{\tilde{\lambda}}_k := \tfrac{1}{N}\textstyle\sum_{i=1}^N \tilde{\lambda}_{i,k}, \quad A_k^{\mathrm{ref}} := \bar{\tilde{H}}_k + \bar{\tilde{\lambda}}_k I,\\
s_k^{\mathrm{ref}} := -(A_k^{\mathrm{ref}})^{-1}\bar{\tilde{g}}_k.
\end{aligned}$$

LEMMA 5.7. *For all* $i, k$, *the local step satisfies*

$$M\,\|s_{i,k}\| \le \lambda_{i,k} \qquad \textit{and hence} \qquad L_2\,\|s_{i,k}\| \le \lambda_{i,k}.$$

LEMMA 5.8. *For all* $k$,

$$L_2\|\bar{s}_k\| \le \frac{1}{N}\sum_{i=1}^N \lambda_{i,k} \le \bar{\tilde{\lambda}}_k.$$

**5.4. Bridge bounds: tracked averages versus true quantities.** The reference step uses the tracked averages $\tilde{\bar{g}}_k, \tilde{\bar{H}}_k$ rather than the true gradient $g_k := \nabla f(\bar{x}_k)$ and true Hessian $\nabla^2 f(\bar{x}_k)$. By Lemma 5.5, $\tilde{\bar{g}}_k = \frac{1}{N}\sum_{i=1}^N \nabla f_i(x_{i,k})$, which coincides with $\nabla f(\bar{x}_k)$ only at exact consensus ($x_{i,k} = \bar{x}_k$ for all $i$). Away from consensus the discrepancy is controlled by the bridge lemmas below, which quantify it in terms of the post-mixing spatial disagreement $D(X_k)$. All constants depending on iterate boundedness (e.g., $L_1, L_2$) are evaluated on the compact set fixed by Assumption 5.1.

The post-mixing disagreement (RMS) is $D(X_k) = \left(\frac{1}{N}\sum_{i=1}^N \|x_{i,k} - \bar{x}_k\|^2\right)^{1/2}$, i.e., the dispersion $D(\cdot)$ defined in Section 3 applied to the stacked iterate $X_k$.

LEMMA 5.9. *For every $k$,*

$$\|\nabla f(\bar{x}_k) - \tilde{\bar{g}}_k\| \le L_1\, D(X_k).$$

LEMMA 5.10. *For every $k$,*

$$\|\nabla^2 f(\bar{x}_k) - \tilde{\bar{H}}_k\|_2 \le L_2\, D(X_k).$$

**5.5. The eigenvalue-shift stabilizer $\delta_{i,k}$.** The stabilizer $\delta_{i,k}$ makes the local coefficient matrix positive definite whenever $\tilde{g}_{i,k} \neq 0$. When $\tilde{g}_{i,k} = 0$, Algorithm 4.1 uses the convention $s_{i,k} = 0$. Therefore the local step is well posed in all cases. Because it acts as an additive bias in $\tilde{\lambda}_{i,k}$, we need to show that its average $\bar{\delta}_k$ is controlled by quantities that vanish as consensus improves.

Define the stabilizer average

$$\bar{\delta}_k := \frac{1}{N}\sum_{i=1}^N \delta_{i,k}.$$

LEMMA 5.11. *For every $k$ and every $i$,*

$$0 \le \delta_{i,k} \le \left\|\tilde{H}_{i,k} - \nabla^2 f(\bar{x}_k)\right\|_2.$$

*Consequently,*

$$\bar{\delta}_k \le \Delta_k^H + L_2 D(X_k),$$
$$\Delta_k^H := \tfrac{1}{N}\sum_{i=1}^N \left\|\tilde{H}_{i,k} - \tilde{\bar{H}}_k\right\|_2.$$

**5.6. Step dispersion via a resolvent identity.** The inexact Newton condition (Proposition 5.14 below) requires three quantities to be small: the step dispersion $\|\bar{s}_k - s_k^{\mathrm{ref}}\|$, the gradient bridge error $L_1 D(X_k)$, and the Hessian bridge error $L_2 D(X_k)$. The latter two are already controlled by Lemmas 5.9–5.10; this subsection handles the first. The key tool is a resolvent identity that factorizes the difference of two linear-system solutions through their coefficient matrices.

Define dispersions

$$\Delta_k^g := \tfrac{1}{N}\textstyle\sum_{i=1}^N \|\tilde{g}_{i,k} - \tilde{\bar{g}}_k\|,$$
$$\Delta_k^\lambda := \tfrac{1}{N}\textstyle\sum_{i=1}^N |\tilde{\lambda}_{i,k} - \tilde{\bar{\lambda}}_k|,$$
$$\underline{\lambda}_k := \min_{1\le i\le N} \tilde{\lambda}_{i,k}.$$

LEMMA 5.12. *For every $k$ with $\underline{\lambda}_k > 0$,*

$$\|\bar{s}_k - s_k^{\mathrm{ref}}\| \le \frac{1}{\underline{\lambda}_k}\Delta_k^g + \frac{\|\tilde{\bar{g}}_k\|}{\underline{\lambda}_k\,\tilde{\bar{\lambda}}_k}\big(\Delta_k^H + \Delta_k^\lambda\big).$$

LEMMA 5.13. *Assume $\tilde{\bar{g}}_k \neq 0$ and the relative gradient dispersion condition*

$$\max_{1\le i\le N}\|\tilde{g}_{i,k} - \tilde{\bar{g}}_k\| \le \alpha_d\|\tilde{\bar{g}}_k\| \qquad \text{for some } \alpha_d \in (0,1). \tag{5.3}$$

*Then*

$$\Delta_k^\lambda \le \frac{\sqrt{M}}{\sqrt{1-\alpha_d}} \cdot \frac{\Delta_k^g}{\sqrt{\|\tilde{\bar{g}}_k\|}} + 2\bar{\delta}_k. \tag{5.4}$$

**5.7. Inexact regularized Newton condition for the average step.** Collecting the bridge, stabilizer, and step-dispersion bounds, we verify that the average iterate $\bar{x}_k$ satisfies a standard inexact regularized Newton condition once the burn-in phase is complete, that is, for $k \ge K_0(\varepsilon)$, where $K_0$ is the burn-in index from Proposition A.15. This is the key step, as it permits invoking the one-step descent lemma from the centralized analysis.

Specifically, we define $g_k := \nabla f(\bar{x}_k)$, $\lambda_k := \tilde{\bar{\lambda}}_k$, and $r_k := (\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k + g_k$.

PROPOSITION 5.14. *Assume the following hold at iteration $k$ for some $\eta \in (0,1)$:*
1. *Dispersion control:*

$$\|\bar{s}_k - s_k^{\mathrm{ref}}\| \le \frac{\eta}{8} \cdot \frac{\lambda_k}{M_{H,\max} + \lambda_k}\,\|\bar{s}_k\|.$$

2. *Gradient bridge accuracy:*

$$L_1 D(X_k) \le \frac{\eta}{8}\lambda_k\|\bar{s}_k\|.$$

3. *Hessian bridge accuracy:*

$$L_2 D(X_k) \le \frac{\eta}{8}\lambda_k.$$

*Then*

$$\|r_k\| \le \eta\,\lambda_k\|\bar{s}_k\|, \qquad L_2\|\bar{s}_k\| \le \lambda_k.$$

**5.8. Descent and global rates.** The preceding subsections have established the four building blocks of the proof pipeline: the average iterate satisfies an inexact regularized Newton condition (Proposition 5.14) once the burn-in phase ends. The remaining descent step follows the centralized analysis of Mishchenko [5]: a one-step descent lemma yields a 3/2-power recursion on the optimality gap, from which the global complexity bound follows.

We introduce the optimality gap $\Phi_k := f(\bar{x}_k) - f_\star$.

LEMMA 5.15. *Assume*

$$\|(\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k + g_k\| \le \eta\lambda_k\|\bar{s}_k\|, \qquad L_2\|\bar{s}_k\| \le \lambda_k,$$

*for some $\eta \in (0,1)$. Then*

$$f(\bar{x}_{k+1}) \le f(\bar{x}_k) - \Big(1 - \eta - \frac{1}{6}\Big)\lambda_k\|\bar{s}_k\|^2.$$

LEMMA 5.16. *Under the conditions of Lemma 5.15,*

$$\|g_{k+1}\| \le \Big(1+\eta+\frac{1}{2}\Big)\lambda_k\|\bar{s}_k\|.$$

We partition the iteration indices into *steady* iterations (where the gradient norm does not drop sharply) and *super-descent* iterations (where it drops by at least a factor of 4):

$$\mathcal{I} := \{k \ge 0 :\ \|g_{k+1}\| \ge \tfrac{1}{4}\|g_k\|\},$$
$$\mathcal{S} := \{k \ge 0 :\ \|g_{k+1}\| < \tfrac{1}{4}\|g_k\|\}.$$

The following lemma establishes a sharp decrease in the optimality gap during steady-descent iterations. The auxiliary requirement $\lambda_k \le C_\lambda\sqrt{\|g_k\|}$ is rigorously verified in Appendix B (Lemma B.1).

LEMMA 5.17. *If Proposition 5.14 holds for all $k$ with some $\eta \le 1/12$ and there exists $C_\lambda > 0$ such that*

$$\lambda_k \le C_\lambda\sqrt{\|g_k\|} \qquad \text{for all } k \text{ with } g_k \ne 0.$$

*Then there exists $\nu > 0$ such that for all $k \in \mathcal{I}$,*

$$\Phi_k - \Phi_{k+1} \ge \nu\,\Phi_k^{3/2}.$$

LEMMA 5.18. *If $\Phi_{k+1} \le \Phi_k - \nu\Phi_k^{3/2}$ for some $\nu > 0$ and all $k$, then*

$$\Phi_k \le \frac{4}{\nu^2(k+2)^2}.$$

Throughout we use the logarithmic mixing schedule

$$\tau_k \ = \ t_k \ = \ \left\lceil \frac{p\,\log(k+2)+c_{\mathrm{mix}}}{-\log\rho}\right\rceil, \qquad k \ge 0, \tag{5.5}$$

where $c_{\mathrm{mix}} \ge 0$ is a fixed constant. Then $\rho^{\tau_k} \le e^{-c_{\mathrm{mix}}}(k+2)^{-p}$; details and motivation are given in Appendix A.

*Remark* 5.19. The schedule (5.5) requires the spectral gap $1-\rho$, or equivalently, $\rho = \|W - \big(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\big)\|_2$. Computing $\rho$ exactly is a centralized operation; in a fully decentralized implementation one may use any certified upper bound $\hat{\rho} \ge \rho$ available for the chosen weight matrix. Overestimating $\rho$ increases each $\tau_k$ by a constant factor but does not affect the $\mathcal{O}(\varepsilon^{-1})$ iteration complexity. If only a rough network description is available, one may use a conservative upper bound; the asymptotic statement is unchanged.

THEOREM 5.20. *If Assumptions 3.1 and 5.1 hold and $M \ge L_2$, then the following is true. Fix $p > 2$ and run DISGREM with the logarithmic mixing schedule (5.5). Then for every target accuracy $\varepsilon > 0$, there exist a finite burn-in index $K_0(\varepsilon)$ (depending on the spectral gap, Lipschitz constants, initial dispersions, and $\varepsilon$; see Proposition A.15) and a constant $C_K > 0$ (independent of $\varepsilon$) such that the total number of outer iterations to achieve $\|\nabla f(\bar{x}_k)\| \le \varepsilon$ satisfies*

$$K(\varepsilon) \ \le \ K_0(\varepsilon) + C_K\,\varepsilon^{-1}.$$

*In particular, the post-burn-in phase requires at most $\mathcal{O}(\varepsilon^{-1})$ iterations, matching the centralized regularized Newton complexity of Mishchenko [5], and under the schedule (5.5) the total number of neighbor communication rounds is $\mathcal{O}\big((K_0(\varepsilon)+\varepsilon^{-1})\log(K_0(\varepsilon)+\varepsilon^{-1})\big)$.*

*Remark* 5.21. The dispersions decay as $\mathcal{O}(k^{-(p-1)})$ (Proposition A.14), and the inexact Newton conditions (Proposition 5.14) require them to be at most $\mathcal{O}(\sqrt{\varepsilon})$. Inverting the polynomial decay gives $K_0(\varepsilon)=\mathcal{O}\big(\varepsilon^{-1/(2(p-1))}\big)$. For any $p>2$, the exponent $\frac{1}{2(p-1)}<1$, so $K_0(\varepsilon)=o(\varepsilon^{-1})$ and the total complexity $K_0(\varepsilon)+\mathcal{O}(\varepsilon^{-1})$ is dominated by the $\mathcal{O}(\varepsilon^{-1})$ post-burn-in term. For instance, $p=3$ yields $K_0(\varepsilon)=\mathcal{O}(\varepsilon^{-1/4})$.

The dependence on the spectral gap enters through the mixing schedule: $\tau_k=\lceil(p\log(k+2)+c_{\text{mix}})/(-\log\rho)\rceil$ implies that each outer iteration uses $\mathcal{O}\big((1-\rho)^{-1}\log k\big)$ communication rounds; the constant $c_{\text{mix}}$ affects only the constant. Here $-1/\log\rho\approx(1-\rho)^{-1}$ for $\rho$ near 1. The total communication-round budget is therefore

$$\mathcal{O}\big((1-\rho)^{-1}\,\varepsilon^{-1}\log(1/\varepsilon)\big),$$

exhibiting a linear penalty in $(1-\rho)^{-1}$—the same order as first-order gradient-tracking methods [9]. On very sparse graphs ($\rho\to1$), the per-iteration communication overhead grows, but the iteration count remains network-independent.

*Remark* 5.22. The proof of Theorem 5.20 combines two independent analyses, distinct from the per-iteration reduction chain in Sections 5.2–5.8 (which is used in Stage 2). In Stage 1 (dispersion decay; Proposition A.14, Appendix A), the logarithmic schedule makes tracker dispersions decay polynomially on the bounded trajectory of Assumption 5.1. In Stage 2 (descent), once dispersions are small enough (after $K_0(\varepsilon)$ iterations), the per-iteration chain verifies an inexact regularized Newton recursion (Proposition 5.14), and the 3/2-recursion (Lemma 5.18) gives $\mathcal{O}(\varepsilon^{-1})$ further iterations. The two stages are self-contained and do not depend on each other's conclusions.

**5.9. Strong convexity and local superlinear convergence.** The preceding results require only convexity. When the objective is locally strongly convex near the optimum, the stabilizer term $\bar{\delta}_k$ becomes negligible relative to $\|g_k\|$, recovering the exact Newton regime and, with it, local $Q$-superlinear convergence.

ASSUMPTION 5.23. *The average objective $f$ is $\mu$-strongly convex on the bounded set $\{x:\|x-x_\star\|\le D\}$ (where $D$ is given by Assumption 5.1): $\nabla^2 f(x)\succeq\mu I$ for all $\|x-x_\star\|\le D$.*

LEMMA 5.24. *If Assumptions 3.1, 5.1, and 5.23 hold, then the following is true. Fix any exponent $\gamma>0$. Let*

$$G_g:=\sup_{k\ge0}\max_{1\le i\le N}\|\tilde{g}_{i,k}\|\in(0,\infty),$$

*which is finite by Lemma A.9. Suppose that for all $k$ large enough (and $k\ge1$) the mixing depths satisfy*

$$\max\{\rho^{\tau_k},\rho^{t_{k-1}}\}\le\tfrac{1}{C_\delta}\|g_k\|^{1+\gamma}, \tag{5.6}$$

*where $C_\delta:=\hat{C}_H+L_2(2D+\sqrt{G_g/M})$ and $\hat{C}_H:=B_H+2\sqrt{d}\,L_2D$, where $B_H:=\sup_{k\ge0}D(\tilde{\mathcal{H}}_k)<\infty$ is the uniform Hessian-tracker dispersion bound from Lemma A.10. Then, for all sufficiently large $k$,*

$$\bar{\delta}_k=o(\|g_k\|)\qquad\textit{and more precisely}\qquad\bar{\delta}_k\le\|g_k\|^{1+\gamma}.$$

THEOREM 5.25. *Under the hypotheses of Theorem 5.20, suppose in addition that Assumption 5.23 holds, Proposition 5.14 holds for all sufficiently large $k$ with some $\eta \in (0, 1/12]$, and there exists $\gamma > 0$ such that, for all sufficiently large $k$,*

$$\max\{\rho^{\tau_k}, \rho^{t_{k-1}}\} \le \tfrac{1}{C_\delta}\|g_k\|^{1+\gamma}, \qquad \Delta_k^g \le C_g \|g_k\|^{1+\gamma} \tag{5.7}$$

*for some constant $C_g > 0$, then there exist constants $C_{\rm sc} > 0$ and $k_{\rm sc}$ such that, for all $k \ge k_{\rm sc}$,*

$$\|\nabla f(\bar{x}_{k+1})\| \le C_{\rm sc}\, \|\nabla f(\bar{x}_k)\|^{3/2}.$$

*Remark* 5.26. The logarithmic schedule in Theorem 5.20 is sufficient for the global $\mathcal{O}(\varepsilon^{-1})$ rate. The local superlinear result uses the relative accuracy condition (5.7), which requires the consensus and tracking errors to decay relative to the current gradient norm along the local tail.

## 6. Numerical experiments.

This section evaluates the practical performance of the DisGrem family using the logarithmic communication rule motivated by the analysis. All main convergence experiments report statistics over 20 independent Monte Carlo (MC) trials with randomized starting points and random Erdős–Rényi graphs. Four DisGrem variants are compared against six first- and second-order baselines on nine objectives spanning well-conditioned and ill-conditioned convex problems, real-data logistic regression, and four nonconvex objectives.

### 6.1. Experimental setup.

We use $N$=10 agents communicating over an Erdős–Rényi (ER) random graph with edge probability $p_{\rm er}$=0.5 (regenerated per trial). The Metropolis–Hastings doubly stochastic mixing matrix $W$ yields spectral gap $1-\rho \approx 0.08$ ($\rho \approx 0.92$; typical range $\rho \in [0.88, 0.96]$ across 20 trials), representing a moderately sparse topology. The exact theoretical algorithm uses the same logarithmic depth for all tracked quantities. In the experiments, the vector mixing depths in the DisGrem family are chosen as

$$\tau_k = t_k = \min\left\{10,\ \left\lceil \frac{3\log(k+2)+2}{-\log\rho} \right\rceil\right\},$$

which is the logarithmic rule (5.5) with $p = 3$ and $c_{\rm mix} = 2$, implemented with a maximum depth of 10 over the tested tolerance range. For the full-matrix variants, Hessian pre-mixing is limited to three matrix rounds. For the communication-efficient variants, Hessian-matrix mixing is limited to two matrix rounds while vector mixing uses the full $\tau_k$ and $t_k$ depths. These matrix-round limits are payload-saving implementation choices and are evaluated empirically; the exact-tracking theory corresponds to using the same logarithmic depth for vectors and Hessian trackers. Baselines retain their standard consensus settings, and all comparisons report cumulative communication cost so that different per-iteration communication patterns are accounted for explicitly.

We compare ten methods in three groups. The proposed DisGrem family comprises four variants: DisGrem (Algorithm 4.1), CeDisGrem (communication-efficient Hessian tracking using Top-$k$ sparsification with a 10% default budget, with low-rank alternatives studied in Appendix C), AdaDisGrem (adaptive $M$ via secant-based Lipschitz estimation), and CeAdaDisGrem (adaptive + compressed). In the Ce variants, the lazy Hessian-update period and compression budget are adjusted with the current communication depth so that communication is reduced early while more Hessian information is retained as the requested consensus depth increases. The regularization scaling is $M = M_{\rm fac} \cdot H^0_{\max}$ where $H^0_{\max} = \max_i \|\nabla^2 f_i(x_0)\|_2$; per-function $M_{\rm fac}$ values are listed in Table 4. The theoretical condition $M \geq L_2$ is not explicitly enforced by the heuristic default $M = M_{\mathrm{fac}} \cdot H^0_{\max}$; the adaptive variant AdaDisGrem addresses this through online secant-based estimation (Section 6.4).

Two first-order baselines are included: EXTRA [11] and DIGing [9]. Four second-order baselines complete the comparison: DQM [3], ESOM [28], SONATA [12], and Network-GIANT [13]. First-order baselines use a Lipschitz-scaled stepsize $\alpha = \alpha_{\mathrm{base}}/H^0_{\max}$; for each function, $\alpha_{\mathrm{base}}$ is selected from $\{0.01, 0.1, 0.5, 1.0\}$ as the value giving the fastest convergence without divergence over 5 preliminary runs (Table 4). Second-order baselines (DQM, ESOM, SONATA, Network-GIANT) use their published default parameters; SONATA and Network-GIANT solve local subproblems to machine precision via a direct solver. The baselines use their standard consensus settings. Because the DisGrem family has a three-stage gossip structure and carries Hessian payloads, its per-iteration byte cost can exceed that of single-stage baselines; cross-method fairness is therefore assessed through the cumulative communication cost (MB) reported in Figure 2. Our tuning philosophy is *light, problem-class-level defaults*: the per-function $M_{\mathrm{fac}}$ for the proposed family and the per-function $\alpha_{\mathrm{base}}$ for first-order baselines are both selected from small discrete grids via a handful of preliminary runs, while second-order baselines retain their published defaults. A sensitivity analysis (Appendix C, Figure 16) suggests that exhaustive tuning of the second-order baselines' hyperparameters does not resolve their stagnation on hard instances like LogSumExp. Code to reproduce all experiments will be made publicly available upon acceptance.

We test on nine objective functions (Table 3); all synthetic functions use $d$=30, while the two logistic regression problems inherit $d$=22 from the svmguide3 dataset. See Table 4 for per-function algorithmic parameters. Five objectives are convex: (i) Ridge ($\ell_2$-regularized least squares, $\lambda$=$10^{-3}$); (ii) QuadBad (heterogeneous ill-conditioned quadratic, $\kappa$=$10^3$); (iii) LogSumExp (smooth approximation of the max function [32], $\sigma$=0.5); (iv) Huber (pseudo-Huber loss [33], $\delta$=1); (v) LogReg-real ($\ell_2$-regularized logistic regression on the svmguide3 LibSVM dataset [2], $m$=1243, $d$=22; a standard benchmark in decentralized optimization [11, 12]). The remaining four objectives are nonconvex, not covered by the theory but included to test algorithmic robustness beyond the convex regime: (vi) LinLog (piecewise quadratic-logarithmic loss with flat curvature regions); (vii) Rosenbrock [34]; (viii) Styblinski–Tang [35] (multimodal); (ix) LogReg-NCVR (logistic regression with bounded nonconvex penalty $\alpha \sum_k x_k^2/(1+x_k^2)$ [36] on svmguide3). On nonconvex objectives the eigenvalue-shift stabilizer $\delta_{i,k}$ makes the local coefficient matrix positive definite whenever $\tilde{g}_{i,k} \neq 0$. When $\tilde{g}_{i,k} = 0$, Algorithm 4.1 uses the convention $s_{i,k} = 0$. Thus the algorithm remains well posed and operates as a damped Newton-like method; however, the convergence guarantees of Theorems 5.20–5.25 do not apply. The shift $\delta_{i,k}$ plays two structurally similar but theoretically distinct roles: in the convex analysis it compensates for transient indefiniteness of tracked Hessian matrices (which are PSD in the exact case); on nonconvex objectives it additionally absorbs true negative curvature, acting as a Levenberg–Marquardt damping term. A unified nonconvex convergence analysis is left for future work. For convex objectives, $f_{\mathrm{ref}} := f^\star$ (the global minimum, whose existence is guaranteed by coercivity). For nonconvex objectives, $f_{\mathrm{ref}}$ denotes the best value found by multi-start L-BFGS-B (50 restarts, tolerance $10^{-15}$); it is not a certified global optimum.

We define the consensus residual

$$\mathrm{cons}_k \;:=\; D(X_k) \;=\; \sqrt{\tfrac{1}{N}\textstyle\sum_{i=1}^N \|x_{i,k} - \bar{x}_k\|^2},$$

Table 3: Test function definitions. Each $f_i\colon \mathbb{R}^d \to \mathbb{R}$ is the local objective of agent $i$. For synthetic problems (Ridge–LinLog), data $(A_i, b_i)$ are independently generated per agent with i.i.d. Gaussian entries; for logistic regression, the SVMGUIDE3 dataset [2] ($m$=1243, $d$=22) is randomly partitioned across agents. Rosenbrock and Styblinski–Tang are homogeneous (all agents share the same $f_i$).

| Name | Local objective $f_i(x)$ | Data / parameters |
|---|---|---|
| *Convex* | | |
| Ridge | $\frac{1}{2}\lVert A_i x - y_i\rVert^2 + \frac{\lambda}{2}\lVert x\rVert^2$ | $A_i \in \mathbb{R}^{150\times d}$, $y_i = A_i x_{\mathrm{true}} + 0.05\varepsilon_i$, $\lambda = 10^{-3}$ |
| QuadBad | $\frac{1}{2}x^\top Q_i x + b_i^\top x$ | $Q_i$ diagonal, eigenvalues log-spaced in $[1, \chi_i]$, $\chi_i \approx \kappa$; $b_i \sim \mathcal{N}(0, I)$ |
| LogSumExp | $\sigma \log\Big(\sum_{j=1}^p e^{(A_i^\top x - b_i)_j/\sigma}\Big)$ | $A_i \in \mathbb{R}^{d\times p}$, $p = \max(d+2, 12)$, $\sigma = 0.5$ |
| Huber | $\sum_{j=1}^p \delta^2\Big(\sqrt{1+(r_j/\delta)^2} - 1\Big)$, $r = A_i x - b_i$ | $A_i \in \mathbb{R}^{5\times d}$, $\delta = 1$ |
| LogReg-real | $\frac{\iota}{2}\lVert x\rVert^2 + \frac{1}{m_i}\sum_{j=1}^{m_i} \log(1+e^{-b_j a_j^\top x})$ | SVMGUIDE3 data split; $\iota = 10^{-2}$ |
| *Nonconvex* | | |
| LinLog | $\sum_{j=1}^d \ell\Big((A_i x - b_i)_j\Big)$; see † below | $A_i \in \mathbb{R}^{d\times d}$ |
| Rosenbrock | $\sum_{j=1}^{d/2}\Big[100(x_{2j} - x_{2j-1}^2)^2 + (x_{2j-1} - 1)^2\Big]$ | shared ($f_i = f \ \ \forall\, i$) |
| Styblinski | $\sum_{j=1}^d (x_j^4 - 16x_j^2 + 5x_j)$ | shared ($f_i = f \ \ \forall\, i$) |
| LogReg-NCVR | $\frac{1}{m_i}\sum_{j=1}^{m_i} \log(1+e^{-b_j a_j^\top x}) + \alpha\sum_{k=1}^d \frac{x_k^2}{1+x_k^2}$ | SVMGUIDE3 data split; $\alpha = 0.05$ |

†LinLog: $\ell(r) = r^2/2$ for $|r| \le 1$; $\ell(r) = \ln|r| + \frac{1}{2}$ for $|r| > 1$.

Table 4: Per-function experimental configuration. $M_{\mathrm{fac}}$: regularization scaling for DISGREM ($M = M_{\mathrm{fac}} \cdot H^0_{\max}$); $\alpha_{\mathrm{base}}$: stepsize coefficient for first-order baselines (actual step $\alpha_{\mathrm{base}}/H^0_{\max}$); $K_{\max}$: iteration budget (shared by all algorithms); Decay: whether a $1/\sqrt{k}$ stepsize decay is applied to first-order baselines (relevant only for nonconvex problems).

| Function | $M_{\mathrm{fac}}$ | $\alpha_{\mathrm{base}}$ | $K_{\max}$ | Decay | Notes |
|---|---|---|---|---|---|
| Ridge | 0.1 | 0.20 | 200 | No | $\lambda$=$10^{-3}$ |
| QuadBad | 0.1 | 0.10 | 1500 | No | $\kappa$=$10^3$ |
| LogSumExp | 5.0 | 0.30 | 400 | No | |
| Huber | 1.5 | 0.30 | 800 | No | Pseudo-Huber, $\delta$=1 |
| LogReg-real | 3.0 | 1.00 | 600 | No | svmguide3, $d$=22 |
| LinLog | 1.0 | 0.20 | 1500 | No | Nonconvex |
| Rosenbrock | 3.0 | 0.10 | 300 | Yes | Nonconvex |
| Styblinski | 15.0 | 0.05 | 100 | Yes | Multimodal |
| LogReg-NCVR | 3.0 | 1.00 | 1000 | Yes | svmguide3, $d$=22 |

and the combo stopping criterion

$$\text{combo}_k \;:=\; \|\nabla f(\bar{x}_k)\| + \text{cons}_k,$$

where $\nabla f(\bar{x}_k) = \frac{1}{N}\sum_{i=1}^{N} \nabla f_i(\bar{x}_k)$ is the true gradient at the average iterate (computed offline, not from the tracker), ensuring a fair comparison across algorithms with different internal tracking mechanisms. Each algorithm runs to $K_{\max}$ iterations or until $\text{combo}_k < 10^{-12}$. We report four metrics: (a) combo $= \min_k \text{combo}_k$; (b) relF $= \min_k |f(\bar{x}_k) - f_{\text{ref}}|/|f(\bar{x}_0) - f_{\text{ref}}|$; (c) wall-clock time (seconds; for reference only—all implementations use Python/NumPy with comparable vectorization); (d) cumulative communication cost (MB), defined by summing the stage-wise directed neighbor-message payloads over all outer iterations (consistent with the accounting in Section 4.3). All convergence curves show the 20-run median with interquartile shading, after applying the standard running-best envelope to monotone accuracy metrics. Analytical gradients and Hessians are used for all functions. A run is deemed successful at threshold $\varepsilon$ if it reaches relF $\le \varepsilon$ within $K_{\max}$ iterations without encountering NaN or overflow; for nonconvex problems, relF is computed relative to $f_{\text{ref}}$. Unless otherwise noted, the Hessian is updated every iteration ($K_{\text{lazy}}$=1); Section 6.3 studies the effect of lazy updates.

The theoretical results correspond to the logarithmic rule without a maximum-depth restriction; the reported experiments use the displayed maximum depth because the tolerances in the figures and tables are reached within that range.

**6.2. Convergence benchmarks.** The main empirical finding is that the DisGrem family is consistently accurate across the full benchmark suite: it attains high accuracy on all nine test problems, whereas the baselines exhibit stagnation, divergence, or strong problem dependence on at least one instance (Figures 1–2).

Table 5 reports key numerical results on four representative functions covering both favorable and challenging cases; per-problem details follow.

Table 5: Summary on four representative functions ($N$=10, $d$=30, 20 MC runs). Column keys: Time (s), $K$ (iterations), min(relF), Comm (MB).

| | Huber ($K_{\max}$=800) | | | | LogSumExp ($K_{\max}$=400) | | | | LinLog ($K_{\max}$=1500) | | | | Styblinski–Tang ($K_{\max}$=100) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Algorithm | Time | $K$ | min(relF) | MB | Time | $K$ | min(relF) | MB | Time | $K$ | min(relF) | MB | Time | $K$ | min(relF) | MB |
| Proposed (DisGrem family) | | | | | | | | | | | | | | | | |
| DisGrem | 0.16 | 85 | $3e^{-13}$ | 39 | 0.57 | 259 | $1e^{-13}$ | 121 | 0.08 | 41 | $2e^{-13}$ | 19 | 0.14 | 94 | $7e^{-16}$ | 44 |
| CeDisGrem | 1.13 | 800 | $2e^{-12}$ | 260 | 0.43 | 259 | $1e^{-13}$ | 98 | 0.23 | 140 | $1e^{-12}$ | 57 | 0.10 | 93 | $1e^{-13}$ | 37 |
| AdaDisGrem | 1.33 | 800 | $4e^{-12}$ | 426 | 0.07 | 33 | $9e^{-13}$ | 15 | 0.08 | 42 | $6e^{-13}$ | 19 | 0.02 | 17 | $5e^{-16}$ | 8 |
| CeAdaDisGrem | 1.21 | 800 | $5e^{-10}$ | 313 | 0.07 | 35 | $3e^{-13}$ | 13 | 0.26 | 156 | $6e^{-13}$ | 64 | 0.02 | 18 | $5e^{-16}$ | 7 |
| First-order baselines | | | | | | | | | | | | | | | | |
| EXTRA | 5.9 | 800 | $5e^{-3}$ | 8 | 3.5 | 400 | $4e^{-1}$ | 4 | 16.9 | 1500 | $7e^{-9}$ | 15 | 0.70 | 100 | $7e^{-1}$ | 1 |
| DIGing | 6.0 | 800 | $3e^{-4}$ | 25 | 3.3 | 400 | $2e^{-1}$ | 12 | 10.3 | 989 | $1e^{-12}$ | 31 | 0.72 | 100 | $9e^{-1}$ | 3 |
| Second-order baselines | | | | | | | | | | | | | | | | |
| DQM | 0.8 | 800 | $2e^{-1}$ | 12 | 0.5 | 400 | $3e^{-3}$ | 6 | 1.7 | 1500 | $2e^{0}$ | 23 | 0.09 | 100 | $2e^{-1}$ | 2 |
| ESOM | 1.4 | 800 | $1e^{-1}$ | 74 | 0.7 | 400 | $2e^{-3}$ | 37 | 3.5 | 1500 | $1e^{1}$ | 139 | 0.19 | 100 | $1e^{1}$ | 9 |
| SONATA | 1.0 | 800 | $1e^{-1}$ | 37 | 0.5 | 400 | $3e^{-2}$ | 19 | 2.3 | 1500 | $1e^{0}$ | 69 | 0.03 | 12 | $5e^{-16}$ | 1 |
| Net-GIANT | 1.0 | 800 | $4e^{-2}$ | 37 | 0.6 | 400 | $1e^{-2}$ | 19 | 2.5 | 1500 | $1e^{0}$ | 69 | 0.01 | 12 | $5e^{-16}$ | 1 |

Figure 3 shows relF versus iteration for all nine benchmark functions; additional metrics (combo, wall-clock time, communication cost) are collected in Appendix C, Figures 8–10.

We now detail the per-problem behavior. On the harder convex problems the accuracy gap is significant. On LogSumExp, the DisGrem family reaches relF $\lesssim 10^{-12}$, whereas every baseline stagnates above $10^{-3}$ (first-order) or $10^{-2}$ (SONATA, Net-GIANT). On Huber the proposed methods attain $10^{-12}$–$10^{-10}$, well below the $> 10^{-1}$ plateau of DQM/ESOM/SONATA. On ill-conditioned problems (QuadBad, $\kappa$=$10^3$),

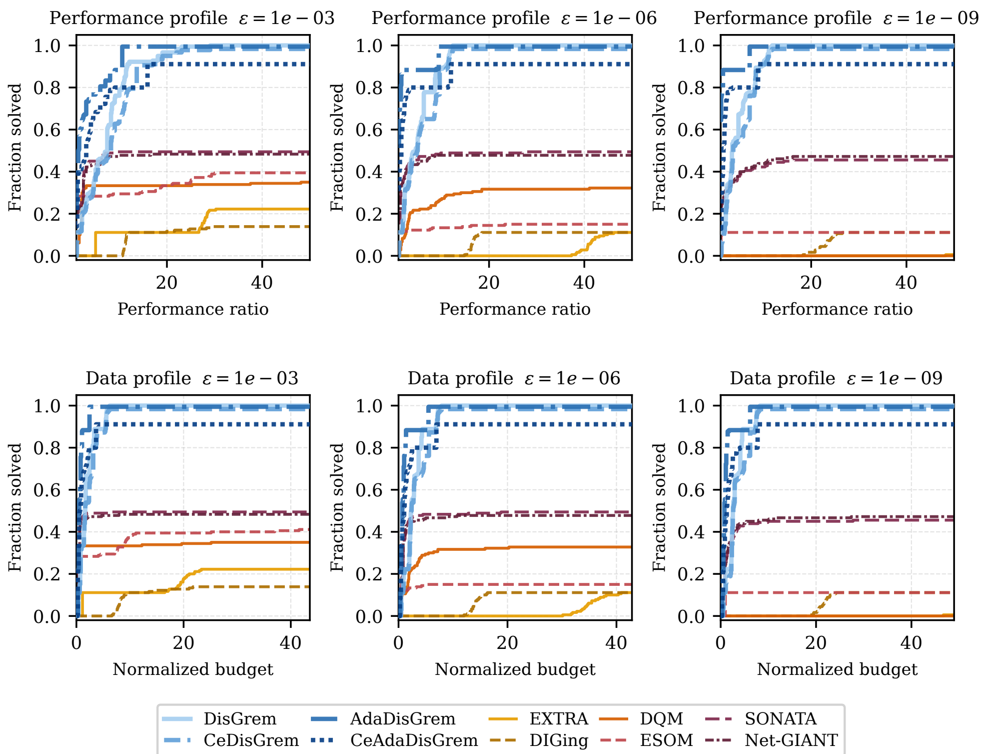


Fig. 1: Iteration-budget profiles at three precision levels $\varepsilon \in \{10^{-3}, 10^{-6}, 10^{-9}\}$. Top: performance profiles; bottom: data profiles. A solver's curve at ordinate $\pi$ indicates it solves a $\pi$-fraction of the problem–MC instances within the given budget.

the relF curve exhibits a clear two-phase pattern: an initial plateau of $\sim$50–100 iterations during which the tracker dispersions contract but the objective barely decreases, followed by a rapid descent phase consistent with the $\mathcal{O}(1/k^2)$ theoretical rate. This matches the predicted burn-in/descent structure of Theorem 5.20.

We note two exceptions where individual baselines outperform. DIGing on LinLog reaches relF $\approx 10^{-12}$, comparable to the DisGrem family: LinLog exhibits very flat curvature regions where the Hessian-based regularization overestimates curvature, whereas DIGing's simpler dynamics are less affected. However, DIGing is much slower in wall-clock time and fails on the majority of other functions. SONATA and Net-GIANT reach $\sim 10^{-16}$ on Styblinski–Tang in 12 steps, but stagnate or diverge on Huber and LinLog (relF $> 1$).

Regarding the adaptive versus fixed-$M$ trade-off, AdaDisGrem is fastest on LogSumExp (33 versus 259 iterations for fixed-$M$ DisGrem) and Styblinski–Tang (17 versus 94 iterations), indicating that the secant estimate tracks the local curvature well. In summary, adaptation removes manual tuning (Section 6.4) and often reduces the iteration count, although fixed $M$ may still use less communication on some problems. Section 6.3 compares total communication volume to a target precision, the more informative metric given the $\mathcal{O}(d^2)$ per-iteration Hessian payload of DisGrem.

Theorem 5.20 assumes convexity, yet the DisGrem family converges on all four nonconvex benchmarks in our tests: all four proposed variants reach relF $\lesssim 10^{-12}$ on LinLog; all four variants achieve relF $< 10^{-12}$ on Rosenbrock within 300 iterations;

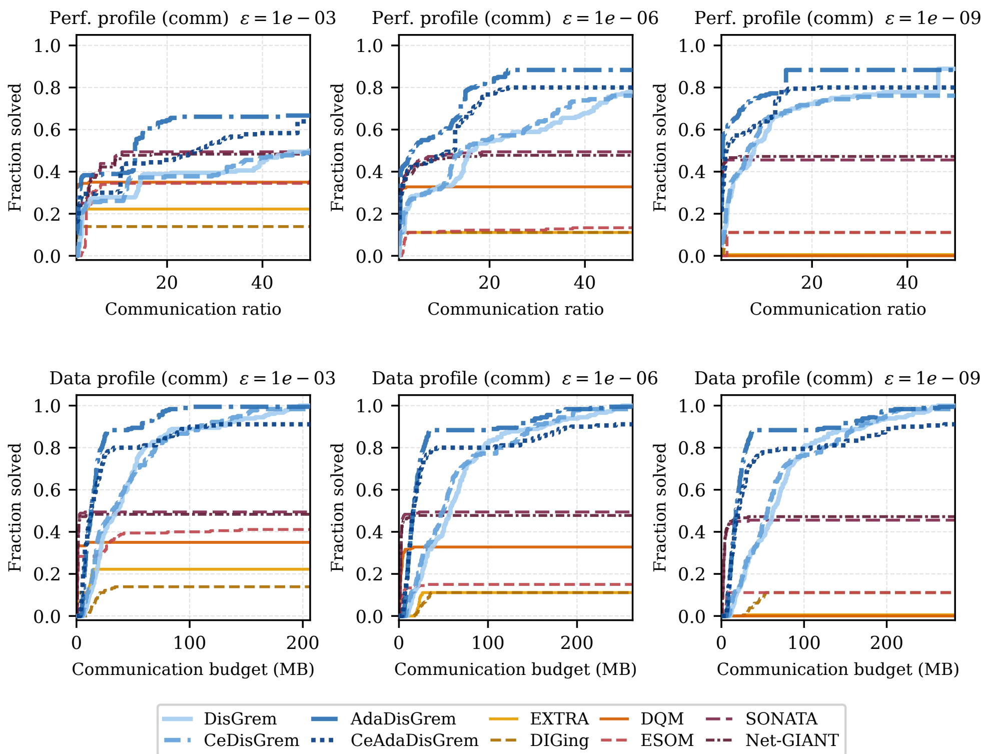


Fig. 2: Communication-budget profiles at three precision levels $\varepsilon \in \{10^{-3}, 10^{-6}, 10^{-9}\}$. Top: performance profiles with cumulative communication cost (MB) as the ratio axis. Bottom: data profiles with communication cost on the $\kappa$-axis.

AdaDisGrem attains $\sim 10^{-16}$ on Styblinski–Tang in 17 steps; and the family converges to $\sim 10^{-12}$ on LogReg-NCVR within 128 steps for the fixed-$M$ pair and within 114 steps for CeAdaDisGrem. The implicit damping provided by the regularization term appears sufficient for these nonconvex test cases; extending the theory to this regime is an open problem.

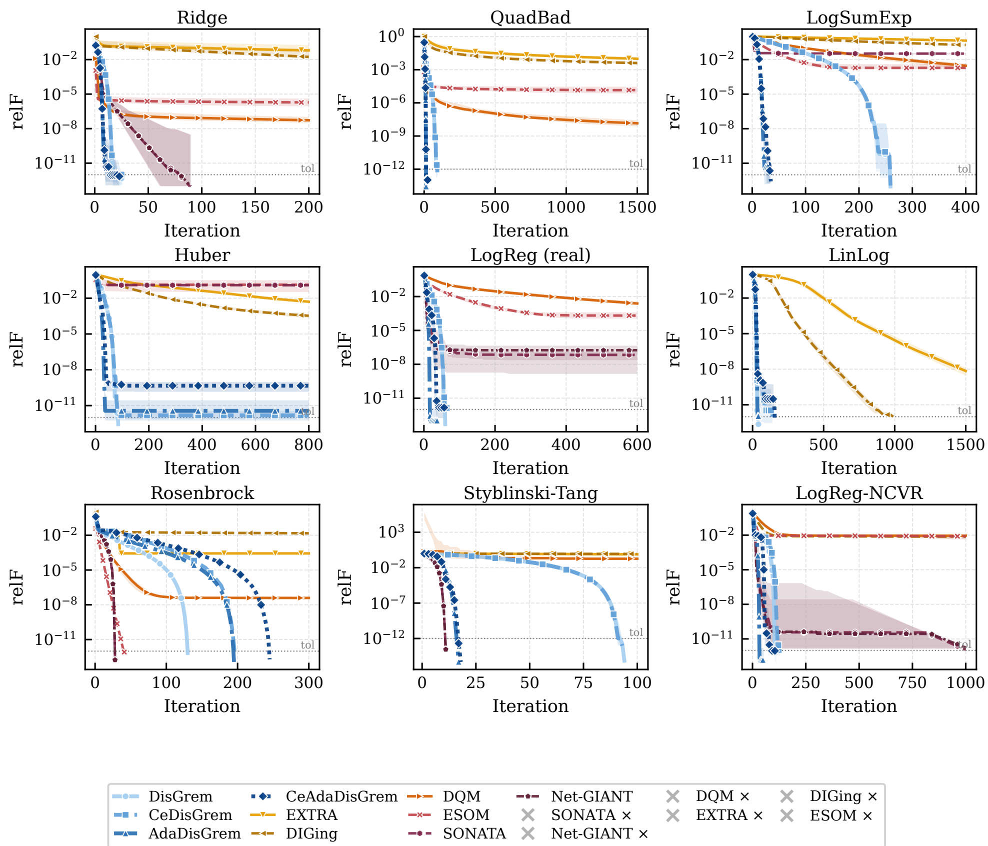


Fig. 3: relF vs. iteration on all nine test functions ($N$=10, $d$=30, 20 MC runs; logarithmic communication implementation). Solid curves: DisGrem family; dashed: baselines. Shaded bands: interquartile range. Legend entries marked by × indicate algorithms whose median curve is not drawn because the corresponding runs terminated by NaN, overflow, or divergence.

**6.3. Communication cost.** The profiles in Section 6.2 use iteration count as the budget axis; here we take a communication-volume perspective. We focus on four representative functions (Ridge, LogSumExp, Huber, and LogReg-real) with 5 MC runs; the synthetic problems use $d = 30$, while LogReg-real uses the SVMGUIDE3 dimension $d = 22$.

**6.3.1. Benefit of communication-efficient (Ce) variants.** Figure 4 and Table 6 compare the total communication cost needed to reach precision levels $\varepsilon \in \{10^{-3}, 10^{-6}, 10^{-8}, 10^{-10}\}$. For the fixed-$M$ pair (DisGrem vs. CeDisGrem), the Ce variant saves about 18–25% communication across the four representative functions and the tested precision range.

For the adaptive pair (AdaDisGrem vs. CeAdaDisGrem), the picture is less clear-cut. At low precision, CeAdaDisGrem provides savings on Ridge, LogSumExp, and LogReg-real, but at high precision it can require more communication on LogReg-real; on Huber it does not reach the selected precision levels in this study. This suggests that the Ce mechanism is most beneficial when combined with fixed $M$, where the compression noise does not corrupt the curvature-tracking feedback loop.

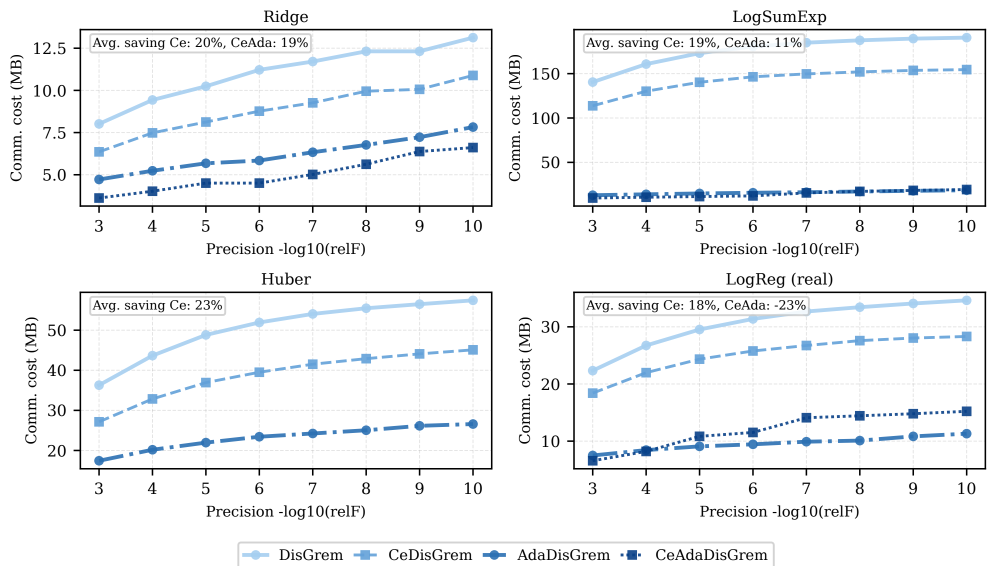


Fig. 4: Communication cost (MB) to reach target precision $\varepsilon \in \{10^{-3}, \ldots, 10^{-10}\}$ for full vs. Ce variants. Missing markers: target not reached.

Table 6: Communication savings (%) of Ce variants over their full counterparts at selected precision levels. Positive values indicate savings; negative values indicate that Ce requires more communication (due to slower convergence erasing per-iteration byte savings).

| Function | Pair | $\varepsilon$=$10^{-3}$ | $\varepsilon$=$10^{-6}$ | $\varepsilon$=$10^{-8}$ | $\varepsilon$=$10^{-10}$ |
|---|---|---|---|---|---|
| Ridge | DisGrem $\to$ Ce | +21% | +22% | +19% | +17% |
| | AdaDisGrem $\to$ Ce | +23% | +23% | +17% | +16% |
| LogSumExp | DisGrem $\to$ Ce | +19% | +19% | +19% | +19% |
| | AdaDisGrem $\to$ Ce | +24% | +22% | −1% | −4% |
| Huber | DisGrem $\to$ Ce | +25% | +24% | +23% | +22% |
| | AdaDisGrem $\to$ Ce | — | — | — | — |
| LogReg-real | DisGrem $\to$ Ce | +18% | +18% | +18% | +18% |
| | AdaDisGrem $\to$ Ce | +12% | −22% | −43% | −35% |

In summary, the Ce mechanism reliably reduces per-iteration payload (by compressing the $\mathcal{O}(d^2)$ Hessian exchange), but total communication savings depend on the convergence-speed trade-off: with fixed $M$, savings of 18–25% are typical in the fixed-$M$ setting; with adaptive $M$, compression noise can still slow convergence enough to increase total bytes at high precision on some problems (Table 6).

**6.3.2. $K_{\text{lazy}}$ and compression ablation.** Appendix C (Figures 11–12) reports a full sweep of $K_{\text{lazy}} \in \{1, 5, 10, 20, 40, 80\}$ and compression methods (Top-$k$, Low-Rank). Moderate values $K_{\text{lazy}} \in \{5, 10\}$ reduce communication by 40–60% on well-conditioned problems with negligible precision loss. Low-rank compression with $r$=$d/5$ provides a good balance across all tested functions; aggressive compression ($r$=1 or Top-$k$ at 5%) degrades convergence on Huber and LogSumExp.

**6.4. Adaptive mechanism.** We study the adaptive $M$ mechanism of AdaDisGrem on four representative functions (Ridge, LogSumExp, LogReg-real, LogReg-NCVR) with 5 Monte Carlo runs each.

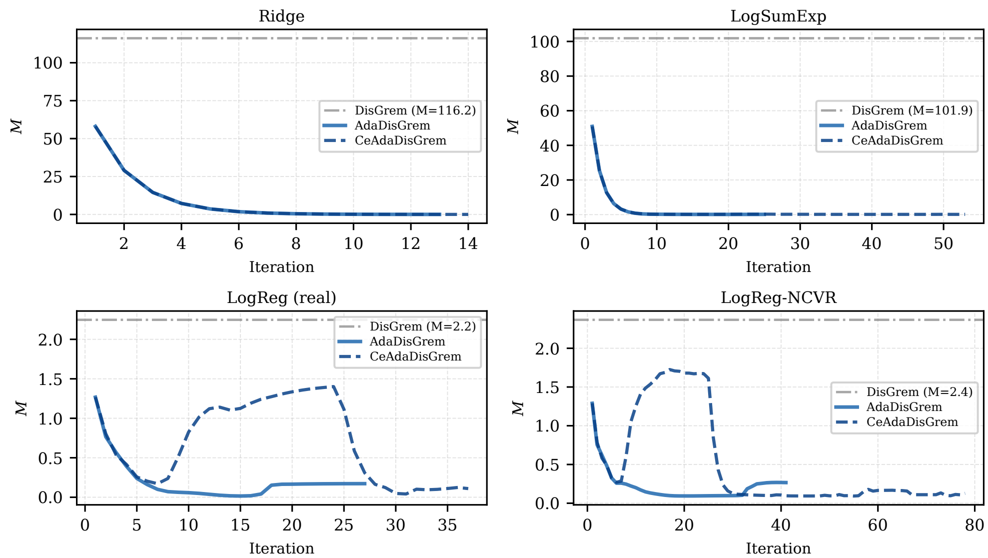


Fig. 5: Regularization parameter $M$ trajectory for AdaDisGrem (solid) vs. the fixed $M$ of DisGrem (dashed). Curves averaged over 5 MC runs.

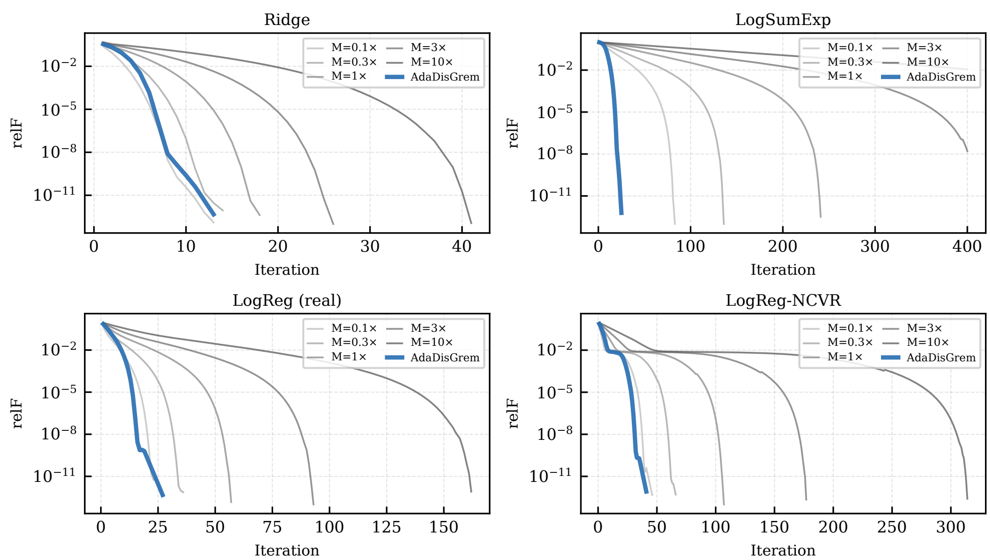


Fig. 6: AdaDisGrem (bold blue) vs. DisGrem at five fixed-$M$ values ($0.1\times$–$10\times$ the default $M^*$; gray curves light-to-dark). The adaptive variant is competitive with the best fixed choice on most functions and avoids divergence at extreme values.

Figure 5 shows the $M$ trajectory. The adaptive estimate tracks the local curvature: it starts near or above the fixed default, then decays as the iterate approaches stationarity. Figure 6 compares AdaDisGrem against five manually chosen fixed-$M$

settings spanning a 100× range: AdaDisGrem is competitive with the best fixed choice on most functions while avoiding the divergence that extreme values can cause. Furthermore, all initializations in $[0.1M^*, 10M^*]$ converge to a common $M$ trajectory within 50–100 iterations (Appendix C, Figure 13), suggesting that the choice of initial $M$ has limited effect in these tests.

**6.5. Robustness.** We investigate robustness along two complementary axes: (i) sensitivity to the starting point, and (ii) sensitivity to the key algorithmic parameter ($M_{\text{fac}}$ for the DisGrem family, stepsize $\alpha$ for first-order methods, penalty or regularization for second-order baselines). All runs use $N$=10, $d$=30, and the same communication settings as the convergence benchmarks.

**6.5.1. Starting-point robustness.** Each algorithm is run from 100 independent random initial points sampled uniformly on a ball of radius $r$ centered at the default initialization. We test two regimes: *near* ($r$=1) and *far* ($r$=3). A run is counted as successful if it reaches relF $< 10^{-6}$ within the iteration budget $K_{\text{max}}$. (This threshold is deliberately less stringent than the combo $< 10^{-12}$ criterion used in the convergence benchmarks, meaning that iteration counts between the two sections are not directly comparable.) Table 7 reports success rates (%) across all nine functions and ten algorithms.

Table 7: Starting-point robustness: success rate (%) at two radii ($r$=1 near, $r$=3 far; 100 MC runs). Success: relF $< 10^{-6}$. Bold: best value in each row; "—": 0%.

| | DisGrem family | | | | 1st-order | | 2nd-order baselines | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Function | DG | CeDG | AdaDG | CeAdaDG | EXTRA | DIGing | DQM | ESOM | SON | N-GI |
| Near initialization ($r$=1) | | | | | | | | | | |
| Ridge | **100** | **100** | **100** | **100** | — | — | **100** | 59 | **100** | **100** |
| QuadBad | **100** | **100** | **100** | **100** | — | — | 98 | — | 1 | 1 |
| LogSumExp | **100** | **100** | 98 | **100** | — | — | — | — | — | — |
| Huber | **100** | 68 | 96 | 17 | — | — | — | — | — | — |
| LinLog | 99 | **100** | 99 | **100** | **100** | **100** | — | — | — | — |
| LogReg-real | **100** | **100** | **100** | **100** | — | — | — | — | 53 | 49 |
| Rosenbrock | **100** | **100** | **100** | **100** | — | — | 97 | **100** | **100** | **100** |
| Styblinski | **100** | **100** | **100** | **100** | — | — | — | — | **100** | **100** |
| LogReg-NCVR | **100** | **100** | **100** | 99 | — | — | — | — | 98 | 96 |
| *Avg* | ***100*** | *96* | *99* | *91* | *11* | *11* | *33* | *18* | *50* | *50* |
| Far initialization ($r$=3) | | | | | | | | | | |
| Ridge | **100** | **100** | **100** | **100** | — | — | **100** | 59 | **100** | **100** |
| QuadBad | **100** | **100** | **100** | **100** | — | — | 98 | 1 | 1 | 1 |
| LogSumExp | **100** | **100** | 98 | **100** | — | — | — | — | — | — |
| Huber | **100** | 80 | 96 | 24 | — | — | — | — | — | — |
| LinLog | 99 | **100** | 99 | **100** | 89 | **100** | — | — | — | — |
| LogReg-real | **100** | **100** | **100** | **100** | — | — | — | — | 51 | 48 |
| Rosenbrock | **100** | **100** | **100** | **100** | — | — | 82 | 96 | **100** | **100** |
| Styblinski | 57 | 61 | **100** | **100** | — | — | — | — | **100** | **100** |
| LogReg-NCVR | **100** | **100** | **100** | **100** | — | 1 | — | — | 99 | 97 |
| *Avg* | *95* | *93* | ***99*** | *92* | *10* | *11* | *31* | *17* | *50* | *50* |

AdaDisGrem is the most robust variant (about 99 % average success, near and far), followed closely by DisGrem. Both maintain nearly identical rates across the two initialization radii, indicating stable regions of convergence. The adaptive mechanism is especially valuable on Styblinski–Tang, where AdaDisGrem and CeAdaDisGrem maintain 100% success even from far starts. Hessian compression is mostly benign but can reduce robustness on Huber, where CeAdaDisGrem succeeds on only 18–24% of trials. The drop is consistent with compression noise accumulating in the Hessian tracker and, for CeAdaDisGrem, perturbing the $\hat{M}$ estimation. The baselines are function-specific: first-order methods succeed only on LinLog; DQM only on Ridge/QuadBad/Rosenbrock; SONATA/Net-GIANT only on Ridge/Rosenbrock/Styblinski–Tang. Huber remains the hardest function for the compressed adaptive variant, due to its near-linear tails and flat curvature regions.

**6.5.2. Parameter sensitivity.** We sweep the key parameter ($M_{\text{fac}}$ for the DisGrem family; stepsize $\alpha$ for first-order methods; penalty/regularization for second-order baselines) over 10 values on four representative functions. Full curves are in Appendix C, Figures 14–16.

The DisGrem family converges across a wide range of $M_{\text{fac}}$ on well-conditioned problems; only below a problem-dependent threshold does it diverge (e.g., $M_{\text{fac}}<1$ on LogSumExp). First-order baselines are highly sensitive to $\alpha$: too large triggers divergence, too small causes stagnation, and even the best $\alpha$ fails to break the relF $>$ $10^{-1}$ barrier on LogSumExp. Second-order baselines exhibit moderate sensitivity; DQM and ESOM diverge for extreme penalty values, while SONATA and Net-GIANT are more stable but stagnate on LogSumExp.

**6.6. Scalability with problem dimension.** To assess how the DisGrem family scales with problem dimension, we repeat the benchmark on three representative functions (Ridge, LogSumExp, Rosenbrock) at $d = 30$, 100, and 200, keeping $N = 10$ agents and the same logarithmic communication implementation as in Section 6.1, with 5 Monte Carlo runs (reduced from the 20 runs in Section 6.2 due to the higher per-run cost at $d$=200; the deterministic objective functions ensure low inter-run variance, so the median curves are highly stable). We compare DisGrem, AdaDisGrem, EXTRA, and SONATA.

Figure 7 reports relF versus iteration. On Ridge and Rosenbrock, the DisGrem family reaches relF $\leq 10^{-12}$ within a similar iteration count across all three dimensions, suggesting an empirically near dimension-insensitive iteration count, consistent with the network-independent iteration bound of Theorem 5.20. On LogSumExp, the final accuracy degrades slightly at $d = 200$, yet DisGrem still outperforms both baselines by 3–5 orders of magnitude in relF. EXTRA stagnates above relF $\sim 1$ on all problems regardless of dimension, while SONATA converges but requires substantially more iterations.

The per-iteration wall-clock time scales as $\mathcal{O}(d^3)$, because each agent solves a dense $d \times d$ linear system (e.g., on Ridge, the cumulative time for all 5 MC runs increases from 2.3 s at $d$=30 to 118 s at $d$=200). Since per-iteration communication volume is $\mathcal{O}(d^2)$ (already analyzed in §4.3), the dominant bottleneck at high dimensions shifts from communication to local computation.

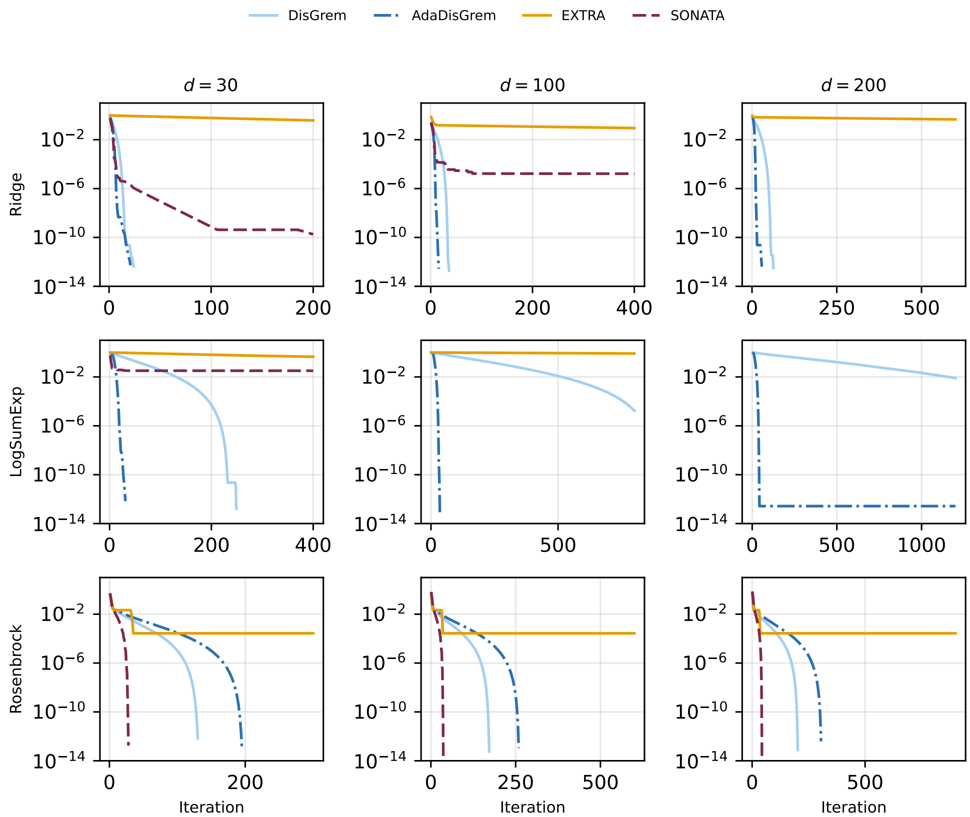


Fig. 7: Dimension scalability: relF vs. iteration for $d \in \{30, 100, 200\}$ on three functions (5 MC runs, median).

**7. Conclusion and future directions.** Under the bounded-trajectory and smoothness assumptions used in the analysis, DisGrem retains the centralized regularized Newton rate, up to a lower-order burn-in phase, in a fully decentralized setting without line search, stepsize tuning, or static Hessian-heterogeneity constants. Algorithmic stability is achieved by combining a local eigenvalue-shift stabilizer with a two-stage mixing protocol. Analytically, the key ingredients are a virtual reference-step construction—which reduces the decentralized dynamics to an inexact centralized update—and an increment-based dispersion analysis. By bounding tracker mismatch through Lipschitz differences rather than fixed heterogeneity constants, the consensus error becomes transient and imposes no accuracy floor. Under the logarithmic mixing schedule used in the analysis, the total communication cost is $\mathcal{O}(\varepsilon^{-1}\log(1/\varepsilon))$ rounds. The experiments use the corresponding implementation over the precision range reported in the figures and tables.

The experiments are consistent with the theoretical picture and provide three further findings. First, AdaDisGrem attains the highest success rate (about 99% average over nine functions and two initialization radii) without per-problem tuning of $M$. Second, Hessian compression (CeDisGrem) saves 18–25% communication at moderate precision, though compression noise can offset these savings at extremely high precision. Third, dimension-scalability experiments ($d \in \{30, 100, 200\}$) demonstrate a nearly dimension-insensitive iteration count, consistent with the network-independent theoretical bound.

Several questions remain open. On the theoretical front, while DisGrem performs well empirically on nonconvex objectives, extending these guarantees to the nonconvex setting is a natural open problem; the recent centralized nonconvex extension of Gratton et al. [17] provides a natural starting point. Establishing a convergence guarantee for AdaDisGrem that does not require a priori knowledge of $L_2$ is of theoretical interest. More broadly, establishing formal lower bounds to determine whether the $\mathcal{O}(\varepsilon^{-1})$ iteration complexity and $\mathcal{O}(\varepsilon^{-1}\log(1/\varepsilon))$ communication rounds are globally optimal for decentralized second-order methods remains open, as does the extension to directed or time-varying graphs.

On the algorithmic and scalability front, the $\mathcal{O}(d^3)$ cost of solving the local Newton system limits the practical regime to moderate dimensions ($d \lesssim 200$–$500$). Inversion-free approaches, such as those using iterative linear solvers [15] or inversion-free tracking [19], could be integrated into the DisGrem framework to reduce the per-iteration computational cost to $\mathcal{O}(d^2)$ or less. Finally, replacing exact local gradients and Hessians with stochastic or variance-reduced estimators would extend the method to large-scale stochastic settings.

**Acknowledgments.** The work of W. Hu, Y.-X. Yuan, and L. Zhang was supported in part by NSFC and the Chinese Academy of Sciences. The work of P. Xie was supported in part by the U.S. Department of Energy, Office of Science.

**References.**

[1] C. Cartis, N. I. M. Gould, and Ph. L. Toint. Adaptive cubic regularisation methods for unconstrained optimization. Part I: motivation, convergence and numerical results. Mathematical Programming, 127:245–295, 2011.
[2] C.-C. Chang and C.-J. Lin. LIBSVM: a library for support vector machines. ACM Transactions on Intelligent Systems and Technology, 2(3):27:1–27:27, 2011. Software available at https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/.
[3] M. Eisen, A. Mokhtari, and A. Ribeiro. Decentralized quasi-Newton methods. IEEE Transactions on Signal Processing, 65(10):2613–2628, 2017.
[4] D. Jakovetić, J. Xavier, and J. M. F. Moura. Fast distributed gradient methods. IEEE Transactions on Automatic Control, 59(5):1131–1146, 2014.
[5] K. Mishchenko. Regularized Newton method with global $O(1/k^2)$ convergence. SIAM Journal on Optimization, 33(3):1440–1462, 2023.
[6] A. Mokhtari, Q. Ling, and A. Ribeiro. Network Newton distributed optimization methods. IEEE Transactions on Signal Processing, 65(1):146–161, 2017.
[7] J. Zhang, Q. Ling, and A. M.-C. So. A Newton tracking algorithm with exact linear convergence for decentralized consensus optimization. IEEE Transactions on Signal and Information Processing over Networks, 7:346–358, 2021.
[8] A. Nedić and A. Ozdaglar. Distributed subgradient methods for multi-agent optimization. IEEE Transactions on Automatic Control, 54(1):48–61, 2009.
[9] A. Nedić, A. Olshevsky, and W. Shi. Achieving geometric convergence for distributed optimization over time-varying graphs. SIAM Journal on Optimization, 27(4):2597–2633, 2017.
[10] Y. Nesterov and B. T. Polyak. Cubic regularization of Newton method and its global performance. Mathematical Programming, 108:177–205, 2006.
[11] W. Shi, Q. Ling, G. Wu, and W. Yin. EXTRA: an exact first-order algorithm for decentralized consensus optimization. SIAM Journal on Optimization, 25(2):944–966, 2015.
[12] Y. Sun, G. Scutari, and D. P. Palomar. Distributed nonconvex optimization and

learning based on successive convex approximation. IEEE Transactions on Signal Processing, 70:5900–5915, 2022.

[13] A. Maritan, G. Sharma, L. Schenato, and S. Dey. Network-GIANT: fully distributed Newton-type optimization via harmonic Hessian consensus. arXiv preprint arXiv:2305.07898, 2023.

[14] K. Yuan, B. Ying, and A. H. Sayed. Exact diffusion for distributed optimization and learning—Part I: algorithm development. IEEE Transactions on Signal Processing, 67(3):708–723, 2019.

[15] D. Jakovetić, N. Krejić, and G. Malaspina. DINAS: Distributed inexact Newton method with adaptive step sizes. Computational Optimization and Applications, 91:683–715, 2025.

[16] A. Daneshmand, G. Scutari, P. Dvurechensky, and A. Gasnikov. Newton method over networks is fast up to the statistical precision. In Proceedings of the 38th International Conference on Machine Learning (ICML), volume 139 of PMLR, pp. 2398–2409, 2021.

[17] S. Gratton, S. Jerad, and Ph. L. Toint. Yet another fast variant of Newton's method for nonconvex optimization. arXiv preprint arXiv:2302.10065, 2023.

[18] N. Doikov and Y. Nesterov. Gradient regularization of Newton method with Bregman distances. Mathematical Programming, 204:1–25, 2024.

[19] G. Yuan, X. Li, and Q. Ling. INDO: INversion-free Distributed second-Order method for consensus optimization. Optimization Online preprint, 2022.

[20] Z. Zhang, K. Che, S. Yang, et al. Communication-efficient distributed cubic Newton with compressed lazy Hessian. Neural Networks, 174:106212, 2024.

[21] S. A. Alghunaim, E. K. Ryu, K. Yuan, and A. H. Sayed. Decentralized proximal gradient algorithms with linear convergence rates. IEEE Transactions on Automatic Control, 66(6):2787–2794, 2021.

[22] D. Bajović, D. Jakovetić, N. Krejić, and N. Krklec Jerinkić. Newton-like method with diagonal correction for distributed optimization. SIAM Journal on Optimization, 27(2):1171–1203, 2017.

[23] A. Beznosikov, P. Richt
'arik, M. Diskin, et al. Distributed methods with compressed communication for solving variational inequalities, with theoretical guarantees. In Advances in Neural Information Processing Systems (NeurIPS), 35:14013–14029, 2022.

[24] N. Doikov and Y. Nesterov. Local convergence of tensor methods. Mathematical Programming, 193:315–336, 2022.

[25] N. Doikov, E. M. Chayti, and M. Jaggi. Second-order optimization with lazy Hessians. In Proceedings of the International Conference on Machine Learning (ICML), 2023.

[26] A. Koloskova, S. U. Stich, and M. Jaggi. Decentralized stochastic optimization and gossip algorithms with compressed communication. In Proceedings of the International Conference on Machine Learning (ICML), pp. 3478–3487, 2019.

[27] H. Li and Z. Lin. Accelerated gradient tracking over time-varying graphs for decentralized optimization. Journal of Machine Learning Research, 25(274):1–52, 2024.

[28] A. Mokhtari, W. Shi, Q. Ling, and A. Ribeiro. ESOM: A second-order method for exact decentralized optimization over networks. IEEE Transactions on Signal and Information Processing over Networks, 2(4):507–522, 2016.

[29] S. Pu and A. Nedić. Distributed stochastic gradient tracking methods. Mathematical Programming, 187:409–457, 2021.

[30] R. Xin and U. A. Khan. Distributed heavy-ball: a generalization and acceleration

of first-order methods with gradient tracking. IEEE Transactions on Automatic Control, 65(6):2627–2633, 2020.

[31] B. Li, S. Cen, Y. Chen, et al. Communication-efficient distributed optimization in networks with gradient tracking and variance reduction. Journal of Machine Learning Research, 21(180):1–51, 2020.

[32] Y. Nesterov. Smooth minimization of non-smooth functions. Mathematical Programming, 103(1):127–152, 2005.

[33] P. Charbonnier, L. Blanc-Féraud, G. Aubert, and M. Barlaud. Deterministic edge-preserving regularization in computed imaging. IEEE Transactions on Image Processing, 6(2):298–311, 1997.

[34] H. H. Rosenbrock. An automatic method for finding the greatest or least value of a function. The Computer Journal, 3(3):175–184, 1960.

[35] M. A. Styblinski and T. S. Tang. Experiments in nonconvex optimization: Stochastic approximation with function smoothing and simulated annealing. Neural Networks, 3(4):467–483, 1990.

[36] D. Geman and C. Yang. Nonlinear image recovery with half-quadratic regularization. IEEE Transactions on Image Processing, 4(7):932–946, 1995.

## Appendix A. Dispersion recursions, burn-in analysis, and logarithmic mixing.

### A.1. Basic inequalities.

LEMMA A.1. *Let $h : \mathbb{R}^d \to \mathbb{R}$ have $L_2$-Lipschitz Hessian. Then for all $x, s$,*

$$h(x+s) \le h(x) + \langle \nabla h(x), s\rangle + \frac{1}{2} s^\top \nabla^2 h(x) s + \frac{L_2}{6}\|s\|^3 .$$

LEMMA A.2. *For any $z \in \mathbb{R}^N$ and integer $t \ge 1$,*

$$\left\|(W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))z\right\|_2 \le \rho^t \|z\|_2 .$$

*Consequently, for any stacked $Z \in \mathbb{R}^{Nd}$,*

$$\left\|((W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))\otimes I_d)Z\right\| \le \rho^t \|Z\| .$$

LEMMA A.3. *Let $h$ be convex with $L$-Lipschitz gradient. Then for any minimizer $x_\star$,*

$$\|\nabla h(x)\|^2 \le 2L\big(h(x) - h(x_\star)\big).$$

This appendix develops the tracker-dispersion analysis in full detail, eliminating the non-vanishing heterogeneity constants present in prior work. The central idea is to bound tracker increments using Lipschitz continuity of differences $\nabla F(X_{k+1}) - \nabla F(X_k)$ and $\nabla^2 F(X_{k+1}) - \nabla^2 F(X_k)$.

**A.2. Dispersion operators and a contraction identity.** For $Z = [z_1; \dots; z_N] \in \mathbb{R}^{Nd}$ define the (RMS) dispersion

$$D(Z) := \sqrt{\tfrac{1}{N}\textstyle\sum_{i=1}^N \|z_i - \bar z\|^2} = \tfrac{1}{\sqrt{N}}\|(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\otimes I_d)Z\|,$$

$$\bar z := \tfrac{1}{N}\textstyle\sum_{i=1}^N z_i.$$

Since $\|A\|_F = \|\operatorname{vec}(A)\|_2$, the standard vector dispersion $D(\cdot)$ naturally applies to the stacked vectorized Hessians $\mathcal{H} \in \mathbb{R}^{Nd^2}$.

LEMMA A.4. *For any stacked vectors $Z \in \mathbb{R}^{Nd}$ and any integer $t \geq 1$,*

$$D\big((W^t \otimes I_d)Z\big) \leq \rho^t\, D(Z).$$

*Likewise, for any stacked matrices $\mathcal{H} \in \mathbb{R}^{Nd^2}$,*

$$D\big((W^t \otimes I_{d^2})\mathcal{H}\big) \leq \rho^t\, D(\mathcal{H}).$$

*Proof.* We prove the vector case; the matrix case follows identically with $I_d$ replaced by $I_{d^2}$. Let $Z^\perp := ((I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)Z$. Since $W$ is doubly stochastic, $W^t\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$ and $\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)W^t = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$, hence $I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)W^t = (W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))$. Therefore

$$\begin{aligned}
&(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) \otimes I_d)(W^t \otimes I_d)Z \\
&\quad = ((W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)Z = ((W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)Z^\perp,
\end{aligned}$$

because $(W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) = 0$ implies $(W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) = (W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$. Taking norms and using Lemma A.2 gives

$$\begin{aligned}
\|(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) \otimes I_d)(W^t \otimes I_d)Z\| &= \|((W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)Z^\perp\| \\
&\leq \|W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\|_2\, \|Z^\perp\| \leq \rho^t \|Z^\perp\|.
\end{aligned}$$

Divide by $\sqrt{N}$ to obtain the claim. ∎

### A.3. A uniform step bound.

LEMMA A.5. *For every $k \geq 0$ and $i \in \{1, \ldots, N\}$,*

$$\|s_{i,k}\| \leq \sqrt{\frac{\|\tilde{g}_{i,k}\|}{M}}.$$

*Consequently, using the boundedness from Assumption 5.1,*

$$\|x_{i,k+1} - x_{i,k}\| \leq 2D \qquad \textit{and} \qquad \|X_{k+1} - X_k\| \leq 2\sqrt{N}\, D,$$

*where $X_k = [x_{1,k}; \ldots; x_{N,k}]$.*

*Proof.* Fix $i, k$. The local linear system is

$$(\tilde{H}_{i,k} + (\lambda_{i,k} + \delta_{i,k})I)s_{i,k} = -\tilde{g}_{i,k}.$$

By definition of $\delta_{i,k}$, the matrix $\tilde{H}_{i,k} + \delta_{i,k}I$ is PSD, hence

$$\tilde{H}_{i,k} + (\lambda_{i,k} + \delta_{i,k})I \succeq \lambda_{i,k} I.$$

Taking norms, we obtain

$$\lambda_{i,k}\|s_{i,k}\| \leq \|(\tilde{H}_{i,k} + (\lambda_{i,k} + \delta_{i,k})I)s_{i,k}\| = \|\tilde{g}_{i,k}\|.$$

If $\tilde{g}_{i,k} = 0$, then $s_{i,k} = 0$ and the bound holds. Otherwise divide by $\lambda_{i,k} > 0$:

$$\|s_{i,k}\| \leq \frac{\|\tilde{g}_{i,k}\|}{\lambda_{i,k}} = \frac{\|\tilde{g}_{i,k}\|}{\sqrt{M\|\tilde{g}_{i,k}\|}} = \sqrt{\frac{\|\tilde{g}_{i,k}\|}{M}}.$$

For the second part, Assumption 5.1 gives $\|x_{i,k}\| \leq \|x_\star\| + D$ and in particular $\|x_{i,k+1} - x_{i,k}\| \leq \|x_{i,k+1} - x_\star\| + \|x_{i,k} - x_\star\| \leq 2D$. Summing squares yields $\|X_{k+1} - X_k\| \leq 2\sqrt{N}\, D$. ∎

**A.4. Primal dispersion recursion (explicit).** Recall the stacked post-mixing update is

$$X_{k+1} = (W^{t_k} \otimes I_d)(\tilde{X}_k + S_k), \qquad \tilde{X}_{k+1} = (W^{\tau_{k+1}} \otimes I_d)X_{k+1}.$$

LEMMA A.6. *For all $k \geq 0$,*

$$D(\tilde{X}_{k+1}) \leq \rho^{\tau_{k+1}} \rho^{t_k} \Big( D(\tilde{X}_k) + \frac{\|S_k\|}{\sqrt{N}} \Big).$$

*Also,*

$$D(X_{k+1}) \leq \rho^{t_k} \Big( D(\tilde{X}_k) + \frac{\|S_k\|}{\sqrt{N}} \Big).$$

*Proof.* By Lemma A.4 applied to $Z = \tilde{X}_k + S_k$,

$$D(X_{k+1}) = D((W^{t_k} \otimes I_d)(\tilde{X}_k + S_k)) \leq \rho^{t_k} D(\tilde{X}_k + S_k).$$

Next we bound $D(\tilde{X}_k + S_k)$ by the triangle inequality in the Hilbert space:

$$\begin{aligned} D(\tilde{X}_k + S_k) &= \tfrac{1}{\sqrt{N}} \|(I - \Big(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\Big) \otimes I_d)(\tilde{X}_k + S_k)\| \\ &\leq \tfrac{1}{\sqrt{N}} \|(I - \Big(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\Big) \otimes I_d)\tilde{X}_k\| \\ &\quad + \tfrac{1}{\sqrt{N}} \|(I - \Big(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\Big) \otimes I_d)S_k\|. \end{aligned}$$

The first term equals $D(\tilde{X}_k)$. For the second, $\|((I - (\frac{1}{N}\mathbf{1}\mathbf{1}^\top)) \otimes I_d)S_k\| \leq \|S_k\|$, hence it is bounded by $\|S_k\|/\sqrt{N}$. This proves the post-mixing inequality. Finally, apply Lemma A.4 to the pre-mixing step: $D(\tilde{X}_{k+1}) \leq \rho^{\tau_{k+1}} D(X_{k+1})$ and combine. ◻

**A.5. Gradient-tracker dispersion recursion (increment-based, fully expanded).** Write stacked variables

$$\begin{aligned} G_k &:= [g_{1,k}; \ldots; g_{N,k}], \ \tilde{G}_k := [\tilde{g}_{1,k}; \ldots; \tilde{g}_{N,k}], \\ \Delta \nabla F_k &:= \nabla F(X_{k+1}) - \nabla F(X_k). \end{aligned}$$

Algorithm 4.1 implies the stacked recursion

$$\begin{aligned} G_{k+1} &= (W^{t_k} \otimes I_d)(\tilde{G}_k + \Delta \nabla F_k), \\ \tilde{G}_{k+1} &= (W^{\tau_{k+1}} \otimes I_d) G_{k+1}. \end{aligned} \tag{A.1}$$

LEMMA A.7. *For all $k \geq 0$,*

$$D(\tilde{G}_{k+1}) \leq \rho^{\tau_{k+1}} \rho^{t_k} \Big( D(\tilde{G}_k) + \frac{L_1}{\sqrt{N}} \|X_{k+1} - X_k\| \Big). \tag{A.2}$$

*In particular, using Lemma A.5,*

$$D(\tilde{G}_{k+1}) \leq \rho^{\tau_{k+1}} \rho^{t_k} \Big( D(\tilde{G}_k) + 2L_1 D \Big). \tag{A.3}$$

*Proof.* Starting from (A.1), substitute $G_{k+1}$ into $\tilde{G}_{k+1}$:

$$\begin{aligned}\tilde{G}_{k+1} &= (W^{\tau_{k+1}}\otimes I_d)(W^{t_k}\otimes I_d)(\tilde{G}_k+\Delta\nabla F_k)\\ &= (W^{\tau_{k+1}+t_k}\otimes I_d)(\tilde{G}_k+\Delta\nabla F_k).\end{aligned}$$

Apply $(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) \otimes I_d)$:

$$\begin{aligned}&(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\otimes I_d)\tilde{G}_{k+1}\\ &\quad= (I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) W^{\tau_{k+1}+t_k}\otimes I_d)(\tilde{G}_k + \Delta\nabla F_k)\\ &\quad= ((W^{\tau_{k+1}+t_k} - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))\otimes I_d)(\tilde{G}_k + \Delta\nabla F_k),\end{aligned}$$

since $I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) W^m = W^m - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$ for any $m \geq 0$. Taking norms and using $\|W^m - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\|_2 \leq \rho^m$ from Lemma A.2,

$$\|((I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)\tilde{G}_{k+1}\| \leq \rho^{\tau_{k+1}+t_k}\, \|(\tilde{G}_k + \Delta\nabla F_k)\|.$$

Divide by $\sqrt{N}$ and bound the RHS by dispersion plus increment:

$$\begin{aligned}D(\tilde{G}_{k+1}) &= \tfrac{1}{\sqrt{N}}\|(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\otimes I_d)\tilde{G}_{k+1}\|\\ &\leq \rho^{\tau_{k+1}+t_k}\big(\underbrace{\tfrac{1}{\sqrt{N}}\|(I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\otimes I_d)\tilde{G}_k\|}_{=D(\tilde{G}_k)}\\ &\qquad + \tfrac{1}{\sqrt{N}}\|\Delta\nabla F_k\|\big),\end{aligned}$$

where we used $\|((I - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)u\| \leq \|u\|$. It remains to bound $\|\Delta\nabla F_k\|$. By block structure,

$$\begin{aligned}\|\Delta\nabla F_k\|^2 &= \textstyle\sum_{i=1}^N \|\nabla f_i(x_{i,k+1})-\nabla f_i(x_{i,k})\|^2\\ &\leq \textstyle\sum_{i=1}^N (L_1\|x_{i,k+1}-x_{i,k}\|)^2 = L_1^2\|X_{k+1}-X_k\|^2.\end{aligned}$$

Taking square roots gives $\|\Delta\nabla F_k\| \leq L_1\|X_{k+1} - X_k\|$ and yields (A.2). Finally, Lemma A.5 gives $\|X_{k+1} - X_k\| \leq 2\sqrt{N}D$, which proves (A.3). ∎

**A.6. Hessian-tracker dispersion recursion (increment-based, fully expanded).** Define stacked Hessian trackers $\mathcal{H}_k := [H_{1,k}; \ldots; H_{N,k}] \in \mathbb{R}^{Nd^2}$ and $\tilde{\mathcal{H}}_k := [\tilde{H}_{1,k}; \ldots; \tilde{H}_{N,k}]$. Define the stacked Hessian increment

$$\Delta\nabla^2 F_k := \nabla^2 F(X_{k+1}) - \nabla^2 F(X_k),$$

viewed as a block vector in $\mathbb{R}^{Nd^2}$ with Frobenius norm per block. The Hessian-tracker recursion is

$$\begin{aligned}\mathcal{H}_{k+1} &= (W^{t_k}\otimes I_{d^2})(\tilde{\mathcal{H}}_k+\Delta\nabla^2 F_k),\\ \tilde{\mathcal{H}}_{k+1} &= (W^{\tau_{k+1}}\otimes I_{d^2})\mathcal{H}_{k+1}.\end{aligned} \tag{A.4}$$

LEMMA A.8. *For all $k \geq 0$,*

$$D(\tilde{\mathcal{H}}_{k+1}) \leq \rho^{\tau_{k+1}}\rho^{t_k}\Big(D(\tilde{\mathcal{H}}_k) + \frac{\sqrt{d}\,L_2}{\sqrt{N}}\|X_{k+1}-X_k\|\Big). \tag{A.5}$$

*In particular, using Lemma A.5,*

$$D(\tilde{\mathcal{H}}_{k+1}) \leq \rho^{\tau_{k+1}}\rho^{t_k}\Big(D(\tilde{\mathcal{H}}_k) + 2\sqrt{d}\,L_2 D\Big). \tag{A.6}$$

*Proof.* By the same argument as in the proof of Lemma A.7, we have

$$\tilde{\mathcal{H}}_{k+1} = (W^{\tau_{k+1}} \otimes I_{d^2})\mathcal{H}_{k+1} \implies D(\tilde{\mathcal{H}}_{k+1}) \leq \rho^{\tau_{k+1}} D(\mathcal{H}_{k+1}),$$

where we used Lemma A.4. From the linear tracker update,

$$\mathcal{H}_{k+1} = (W^{t_k} \otimes I_{d^2})(\tilde{\mathcal{H}}_k + \Delta\nabla^2 F_k),$$

hence Lemma A.4 yields

$$\begin{aligned} D(\mathcal{H}_{k+1}) &\leq \rho^{t_k} D(\tilde{\mathcal{H}}_k + \Delta\nabla^2 F_k) \\ &\leq \rho^{t_k}\Big(D(\tilde{\mathcal{H}}_k) + \tfrac{1}{\sqrt{N}}\|\Delta\nabla^2 F_k\|_F\Big). \end{aligned}$$

Combining the last two displays yields (A.5) once $\|\Delta\nabla^2 F_k\|_F$ is bounded. By block structure and the inequality $\|A\|_F \leq \sqrt{d}\,\|A\|_2$,

$$\begin{aligned} \|\Delta\nabla^2 F_k\|_F^2 &= \textstyle\sum_{i=1}^N \|\nabla^2 f_i(x_{i,k+1}) - \nabla^2 f_i(x_{i,k})\|_F^2 \\ &\leq \textstyle\sum_{i=1}^N d\,\|\nabla^2 f_i(x_{i,k+1}) - \nabla^2 f_i(x_{i,k})\|_2^2. \end{aligned}$$

Using $L_2$-Lipschitzness of $\nabla^2 f_i$,

$$\|\nabla^2 f_i(x_{i,k+1}) - \nabla^2 f_i(x_{i,k})\|_2 \leq L_2\|x_{i,k+1} - x_{i,k}\|,$$

hence

$$\begin{aligned} \|\Delta\nabla^2 F_k\|_F^2 &\leq \textstyle\sum_{i=1}^N dL_2^2\|x_{i,k+1} - x_{i,k}\|^2 \\ &= dL_2^2\|X_{k+1} - X_k\|^2. \end{aligned}$$

Taking square roots gives $\|\Delta\nabla^2 F_k\|_F \leq \sqrt{d}\,L_2\,\|X_{k+1} - X_k\|$, which yields (A.5). Finally use Lemma A.5 to obtain (A.6). ∎

### A.7. Uniform tracker bounds and post-mixing primal decay.

LEMMA A.9. *If Assumptions 3.1 and 5.1 hold and $\tau_k, t_k \geq 1$ for all $k$, then there exists a finite constant $B_G > 0$ such that*

$$\sup_{k\geq 0} D(\tilde{G}_k) \leq B_G,$$

*and consequently there exists a finite constant $G_g > 0$ such that*

$$\sup_{k\geq 0}\max_{1\leq i\leq N} \|\tilde{g}_{i,k}\| \leq G_g.$$

*Proof.* From (A.3) we have

$$D(\tilde{G}_{k+1}) \le \rho^{\tau_{k+1}}\rho^{t_k}\big(D(\tilde{G}_k)+2L_1D\big) \le \rho^2\big(D(\tilde{G}_k)+2L_1D\big),$$

since $\tau_{k+1}, t_k \ge 1$ and $\rho \in [0,1)$. Set $q := \rho^2 \in [0,1)$ and $b := 2L_1D$. Unrolling yields

$$D(\tilde{G}_k) \le q^k D(\tilde{G}_0) + \frac{qb}{1-q} \quad \forall k \ge 0,$$

so we may take $B_G := D(\tilde{G}_0) + \frac{qb}{1-q}$. Next, Lemma 5.5 and the iterate boundedness from Assumption 5.1 imply that $\|\bar{\tilde{g}}_k\| = \|\bar{g}_k\| = \big\|\frac{1}{N}\sum_{i=1}^N \nabla f_i(x_{i,k})\big\|$ is uniformly bounded on the level set; let $G_{\mathrm{avg}} := \sup_{k\ge 0}\|\bar{\tilde{g}}_k\| < \infty$. For any $i$,

$$\|\tilde{g}_{i,k}\| \le \|\bar{\tilde{g}}_k\| + \|\tilde{g}_{i,k} - \bar{\tilde{g}}_k\| \le G_{\mathrm{avg}} + \sqrt{N}\,D(\tilde{G}_k) \le G_{\mathrm{avg}} + \sqrt{N}\,B_G.$$

Thus we may take $G_g := G_{\mathrm{avg}} + \sqrt{N}\,B_G$. ◻

For later reference, we record the uniform gradient bound from Lemma A.9:

$$G_g := \sup_{k\ge 0}\max_{1\le i\le N}\|\tilde{g}_{i,k}\| < \infty, \tag{A.7}$$

where finiteness follows from Lemma A.9.

LEMMA A.10. *If Assumptions 3.1 and 5.1 hold and $\tau_k, t_k \ge 1$ for all $k$, then there exists a finite constant $B_H > 0$ such that*

$$\sup_{k\ge 0} D(\tilde{\mathcal{H}}_k) \le B_H.$$

*Consequently, there exists a finite constant $G_H > 0$ such that*

$$\sup_{k\ge 0}\max_{1\le i\le N}\big\|\tilde{H}_{i,k}\big\|_2 \le G_H, \qquad \sup_{k\ge 0}\bar{\delta}_k \le G_H.$$

*Proof.* From Lemma A.8 and $\tau_{k+1}, t_k \ge 1$, we have

$$D(\tilde{\mathcal{H}}_{k+1}) \le \rho^{\tau_{k+1}}\rho^{t_k}\Big(D(\tilde{\mathcal{H}}_k) + 2\sqrt{d}\,L_2D\Big) \le \rho^2\Big(D(\tilde{\mathcal{H}}_k) + 2\sqrt{d}\,L_2D\Big).$$

Let $q := \rho^2 \in [0,1)$ and $b_H := 2\sqrt{d}\,L_2D$. Unrolling yields

$$D(\tilde{\mathcal{H}}_k) \le q^k D(\tilde{\mathcal{H}}_0) + \frac{qb_H}{1-q} \quad \forall k \ge 0,$$

so we may take $B_H := D(\tilde{\mathcal{H}}_0) + qb_H/(1-q)$.

Next, by Lemma 5.6, $\bar{\tilde{H}}_k = \bar{H}_k = \frac{1}{N}\sum_{i=1}^N \nabla^2 f_i(x_{i,k})$, hence $\|\bar{\tilde{H}}_k\|_2 \le M_{H,\max}$. For any $i$,

$$\|\tilde{H}_{i,k}\|_2 \le \|\bar{\tilde{H}}_k\|_2 + \|\tilde{H}_{i,k} - \bar{\tilde{H}}_k\|_2 \le M_{H,\max} + \sqrt{N}\,B_H.$$

Define $G_H := M_{H,\max} + \sqrt{N}\,B_H$.

Finally, $\delta_{i,k} = \max\{0, -\lambda_{\min}(\tilde{H}_{i,k})\} \le \|\tilde{H}_{i,k}\|_2 \le G_H$, and averaging yields $\bar{\delta}_k \le G_H$. ◻

LEMMA A.11. *For all $k \ge 0$,*

$$D(X_{k+1}) \le \rho^{t_k}\Big(D(\tilde{X}_k) + \frac{\|S_k\|}{\sqrt{N}}\Big).$$

*Proof.* Recall $X_{k+1} = (W^{t_k} \otimes I_d)(\tilde{X}_k + S_k)$ and that $D(Z) = \|Z^\perp\|/\sqrt{N}$ with $Z^\perp := ((I - (\frac{1}{N}\mathbf{1}\mathbf{1}^\top)) \otimes I_d)Z$. Since $(W^{t_k} - (\frac{1}{N}\mathbf{1}\mathbf{1}^\top))$ annihilates consensus components, we have

$$(X_{k+1})^\perp = ((W^{t_k} - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \otimes I_d)(\tilde{X}_k + S_k),$$

and by Lemma A.2 (applied in stacked form) and the triangle inequality,

$$\|(X_{k+1})^\perp\| \le \rho^{t_k}\|(\tilde{X}_k + S_k)^\perp\| \le \rho^{t_k}\left(\|\tilde{X}_k^\perp\| + \|S_k^\perp\|\right) \le \rho^{t_k}\left(\|\tilde{X}_k^\perp\| + \|S_k\|\right).$$

Divide by $\sqrt{N}$ to obtain the claim. ◻

Lemma A.12. *Under Assumptions 3.1 and 5.1 and schedule (A.8), there exists a constant $C_X^+ > 0$ such that for all $k \ge 0$,*

$$D(X_k) \le \frac{C_X^+}{(k+1)^p}.$$

*Proof.* From Lemma A.11 and $\rho^{t_k} \le e^{-c_{\rm mix}}(k+2)^{-p} \le (k+2)^{-p}$ under (A.8), it remains to bound $D(\tilde{X}_k)$ and $\|S_k\|/\sqrt{N}$ uniformly. By Assumption 5.1, each $\tilde{x}_{i,k}$ is a convex combination of $\{x_{j,k}\}$, hence $\|\tilde{x}_{i,k} - x_\star\| \le D$ and

$$D(\tilde{X}_k) \le \max_{1\le i\le N}\|\tilde{x}_{i,k} - \bar{x}_k\| \le \max_i \|\tilde{x}_{i,k} - x_\star\| + \|\bar{x}_k - x_\star\| \le 2D.$$

Moreover, Lemma A.5 yields $\|S_k\|/\sqrt{N} \le \sqrt{G_g/M}$, where $G_g$ is defined in (A.7). Therefore,

$$D(X_{k+1}) \le (k+2)^{-p}\Big(2D + \sqrt{\frac{G_g}{M}}\Big),$$

and the claim follows with $C_X^+ := 2D + \sqrt{G_g/M}$ after shifting indices. ◻

**A.8. Polynomial decay under a logarithmic schedule.** Fix $p > 2$ and set the logarithmic schedule

$$\tau_k = t_k = \left\lceil \frac{p\log(k+2) + c_{\rm mix}}{-\log\rho} \right\rceil, \qquad k \ge 0, \tag{A.8}$$

so that $\rho^{\tau_k} = \rho^{t_k} \le e^{-c_{\rm mix}}(k+2)^{-p}$.

Lemma A.13. *Let $\{u_k\}_{k\ge0}$ satisfy $u_{k+1} \le a_k(u_k + b)$ with $b \ge 0$, where $a_k \le q < 1$ for all $k$ and $a_k \le (k+2)^{-p}$ for all $k$ with $p > 2$. Then there exists $C > 0$ such that $u_k \le C/(k+2)^{p-1}$ for all $k \ge 0$.*

*Proof.* Since $a_k \le q < 1$, we have $u_{k+1} \le q(u_k + b)$ and hence by induction

$$u_k \le q^k u_0 + \sum_{j=0}^{k-1} q^{k-j} qb \le u_0 + \frac{qb}{1-q} =: u_{\max}, \qquad \forall k \ge 0.$$

Since $a_k \le (k+2)^{-p}$, for every $k \ge 0$ we have

$$u_{k+1} \le a_k(u_k + b) \le \frac{u_{\max} + b}{(k+2)^p} \le \frac{u_{\max} + b}{(k+2)^{p-1}}.$$

Since $(k+3)/(k+2) \le 2$ for all $k \ge 0$, we obtain $u_{k+1} \le 2^{p-1}(u_{\max} + b)/(k+3)^{p-1}$. For $k = 0$, the base case $u_0 \le u_{\max} \le C/2^{p-1}$ holds by definition. The claim follows with $C := 2^{p-1}(u_{\max} + b)$. ◻

PROPOSITION A.14. *Under Assumptions 3.1 and 5.1 and schedule (A.8), there exist constants $C_X, C_G, C_H > 0$ such that for all $k \geq 0$,*

$$D(\tilde{X}_k) \leq \frac{C_X}{(k+2)^{p-1}}, \qquad D(\tilde{G}_k) \leq \frac{C_G}{(k+2)^{p-1}}, \qquad D(\tilde{\mathcal{H}}_k) \leq \frac{C_H}{(k+2)^{p-1}}.$$

*Proof.* Apply Lemma A.13 to the recursion in Lemma A.6 and the bounds (A.3) and (A.6), using the bounded forcing constants $b_X := \sup_k \|S_k\|/\sqrt{N} < \infty$ (Lemma A.5), $b_G := 2L_1 D$, and $b_H := 2\sqrt{d}\, L_2 D$. ∎

**A.9. Burn-in conditions for Proposition 5.14.** We now establish that for every target accuracy $\varepsilon > 0$, there exists a finite index $K_0(\varepsilon)$ such that whenever $\|g_k\| \geq \varepsilon$ and $k \geq K_0(\varepsilon)$, the three hypotheses of Proposition 5.14 are satisfied with $\eta = 1/12$.

PROPOSITION A.15. *Fix $p > 2$ and the schedule (A.8). For every $\varepsilon > 0$ there exists an integer $K_0(\varepsilon) \geq 0$ such that for every $k \geq K_0(\varepsilon)$ with $\|g_k\| = \|\nabla f(\bar{x}_k)\| \geq \varepsilon$, all three items of Proposition 5.14 hold with $\eta := 1/12$.*

***Moreover (auxiliary conditions).*** *With the same choice of $K_0(\varepsilon)$, we may further guarantee that for all $k \geq K_0(\varepsilon)$,*

$$\Delta_k^g \leq \varepsilon \qquad \text{and} \qquad \bar{\delta}_k \leq \sqrt{\varepsilon}, \tag{A.9}$$

*where $\Delta_k^g := \frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k} - \bar{\tilde{g}}_k\|$ and $\bar{\delta}_k := \frac{1}{N}\sum_{i=1}^N \delta_{i,k}$.*

*Proof.* Fix $\varepsilon > 0$ and set $\eta := 1/12$. We will produce an explicit index $K_0(\varepsilon)$ such that for every $k \geq K_0(\varepsilon)$ with $\|g_k\| \geq \varepsilon$, all three conditions of Proposition 5.14 hold.

**Step 1: post-mixing disagreement and bridge terms.** Recall $D(X_k) = \|X_k^\perp\|/\sqrt{N}$. By Lemma A.12, there exists $C_X^+ > 0$ such that for all $k \geq 0$,

$$D(X_k) \leq \frac{C_X^+}{(k+1)^p}. \tag{A.10}$$

Therefore, for all $k \geq 0$,

$$L_1 D(X_k) \leq \frac{L_1 C_X^+}{(k+1)^p}, \qquad L_2 D(X_k) \leq \frac{L_2 C_X^+}{(k+1)^p}. \tag{A.11}$$

**Step 2: tracker dispersions.** From Proposition A.14, there exist constants $C_G, C_H > 0$ such that for all $k \geq 0$,

$$D(\tilde{G}_k) \leq \frac{C_G}{(k+2)^{p-1}}, \qquad D(\tilde{\mathcal{H}}_k) \leq \frac{C_H}{(k+2)^{p-1}}. \tag{A.12}$$

By Cauchy–Schwarz and $\|\cdot\|_2 \leq \|\cdot\|_F$,

$$\Delta_k^g = \frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k} - \bar{\tilde{g}}_k\| \leq \sqrt{\frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k} - \bar{\tilde{g}}_k\|^2} = D(\tilde{G}_k),$$

and

$$\Delta_k^H = \tfrac{1}{N}\textstyle\sum_{i=1}^N \left\|\tilde{H}_{i,k} - \bar{\tilde{H}}_k\right\|_2$$

$$\leq \tfrac{1}{N}\sum_{i=1}^{N}\left\|\tilde H_{i,k}-\bar{\tilde H}_k\right\|_F$$
$$\leq D(\tilde{\mathcal H}_k).$$

Hence, using Proposition A.14,

$$\Delta_k^g \leq \frac{C_G}{(k+2)^{p-1}}, \qquad \Delta_k^H \leq \frac{C_H}{(k+2)^{p-1}}. \tag{A.13}$$

**Step 3: a usable lower bound on the reference step.** Assume $\|g_k\| \geq \varepsilon$ and suppose $k$ is large enough such that $L_1 D(X_k) \leq \varepsilon/2$. Then Lemma 5.9 implies $\|\bar{\tilde g}_k\| \geq \varepsilon/2$. Moreover, since $A_k^{\mathrm{ref}} \preceq (M_{H,\max} + \bar{\tilde\lambda}_k)I$,

$$\|s_k^{\mathrm{ref}}\| \geq \frac{\|\bar{\tilde g}_k\|}{M_{H,\max} + \bar{\tilde\lambda}_k} \geq \frac{\varepsilon/2}{M_{H,\max} + \sqrt{MG_g} + G_H} =: \underline{s}(\varepsilon),$$

where we used $\bar{\tilde\lambda}_k = \frac{1}{N}\sum_{i=1}^{N}(\lambda_{i,k} + \delta_{i,k}) \leq \sqrt{MG_g} + G_H$, with $G_g$ from (A.7) and $G_H$ from Lemma A.10.

**Step 4: ensure dispersion control (Item 1).** We first verify the relative dispersion condition (5.3) with $\alpha_d = 1/2$. Since $D(\tilde G_k) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|\tilde g_{i,k} - \bar{\tilde g}_k\|^2}$, we have

$$\max_{1\leq i\leq N}\|\tilde g_{i,k} - \bar{\tilde g}_k\| \leq \sqrt{N}\, D(\tilde G_k).$$

From Step 3, for all $k$ large enough with $\|g_k\| \geq \varepsilon$ we have $\|\bar{\tilde g}_k\| \geq \varepsilon/2$. Thus, choosing $k$ large enough so that $\sqrt{N}\, D(\tilde G_k) \leq \frac{1}{2}\|\bar{\tilde g}_k\|$, condition (5.3) holds with $\alpha_d = 1/2$. Using Lemma 5.12 and Lemma 5.13 with $\alpha_d := 1/2$, for all sufficiently large $k$ with $\|g_k\| \geq \varepsilon$ we have

$$\|\bar s_k - s_k^{\mathrm{ref}}\| \leq \frac{6}{\sqrt{M\varepsilon}}\Delta_k^g + \frac{2}{M}\Delta_k^H + \frac{4}{M}\bar\delta_k.$$

By Lemma 5.11, $\bar\delta_k \leq \Delta_k^H + L_2 D(X_k)$. Combining with (A.13)–(A.11), there exists $K_1(\varepsilon)$ such that for all $k \geq K_1(\varepsilon)$,

$$\|\bar s_k - s_k^{\mathrm{ref}}\| \leq \frac{\eta}{16}\cdot\frac{\lambda_k}{M_{H,\max} + \lambda_k}\|s_k^{\mathrm{ref}}\|.$$

This implies $\|s_k^{\mathrm{ref}}\| \leq 2\|\bar s_k\|$ by the triangle inequality, and hence Item 1 of Proposition 5.14 holds.

**Step 5: ensure Items 2–3.** After the relative dispersion condition has been enforced with $\alpha_d = 1/2$, Step 3 gives $\|\tilde g_{i,k}\| \geq \frac{1}{2}\|\bar{\tilde g}_k\| \geq \varepsilon/4$ for every $i$. Hence

$$\lambda_k = \frac{1}{N}\sum_{i=1}^{N}\left(\sqrt{M\|\tilde g_{i,k}\|} + \delta_{i,k}\right) \geq \frac{1}{2}\sqrt{M\varepsilon}.$$

Since $\|s_k^{\mathrm{ref}}\| \leq 2\|\bar s_k\|$, it is enough to ensure

$$L_1 D(X_k) \leq \frac{\eta}{16}\lambda_k\|s_k^{\mathrm{ref}}\|, \qquad L_2 D(X_k) \leq \frac{\eta}{8}\lambda_k.$$

By (A.11) and $\|s_k^{\mathrm{ref}}\| \geq \underline{s}(\varepsilon)$, there exists $K_2(\varepsilon)$ such that both inequalities hold for all $k \geq K_2(\varepsilon)$.

**Step 6: ensure $\Delta_k^g \le \varepsilon$.** By (A.13), $\Delta_k^g \le \frac{C_G}{(k+2)^{p-1}}$. Define

$$K_\Delta(\varepsilon) := \min\Big\{k \ge 0 :\ \frac{C_G}{(k+2)^{p-1}} \le \varepsilon\Big\}.$$

Then for all $k \ge K_\Delta(\varepsilon)$, we have $\Delta_k^g \le \varepsilon$.

**Step 7: ensure $\bar{\delta}_k \le \sqrt{\varepsilon}$.** By Lemma 5.11, we have for all $k$,

$$\bar{\delta}_k \le \Delta_k^H + L_2 D(X_k).$$

Using (A.13) and (A.10), we obtain for all $k \ge 0$,

$$\bar{\delta}_k \le \frac{C_H}{(k+2)^{p-1}} + \frac{L_2 C_X^+}{(k+1)^p} \le \frac{\widetilde{C}_\delta}{(k+1)^{p-1}},$$

where $\widetilde{C}_\delta := C_H + L_2 C_X^+$. Define

$$K_\delta(\varepsilon) := \min\Big\{k \ge 1 :\ \widetilde{C}_\delta (k+1)^{-(p-1)} \le \sqrt{\varepsilon}\Big\}.$$

Then for all $k \ge K_\delta(\varepsilon)$, we have $\bar{\delta}_k \le \sqrt{\varepsilon}$.

Finally set

$$K_0(\varepsilon) := \max\{K_1(\varepsilon), K_2(\varepsilon), K_\Delta(\varepsilon), K_\delta(\varepsilon)\}.$$

Then for every $k \ge K_0(\varepsilon)$ with $\|g_k\| \ge \varepsilon$, all three items of Proposition 5.14 hold with $\eta = 1/12$, and (A.9) also holds. ∎

THEOREM A.16. *Assume the hypotheses of Theorem 5.20. Then*

$$\sum_{k=0}^{K(\varepsilon)-1} (\tau_k + 2t_k) = \mathcal{O}\Big((K_0(\varepsilon) + \varepsilon^{-1}) \log(K_0(\varepsilon) + \varepsilon^{-1})\Big).$$

*Remark* A.17. The burn-in index $K_0(\varepsilon)$ in Theorem 5.20 is determined by requiring the tracker dispersions to decay below $\mathcal{O}(\sqrt{\varepsilon})$ (see Proposition A.15, Steps 3–4). Under the logarithmic schedule (A.8) with $\tau_k, t_k = \lceil (p \log(k+2) + c_{\mathrm{mix}})/(-\log\rho) \rceil$, Proposition A.14 yields $D(\tilde{G}_k), D(\tilde{\mathcal{H}}_k) = \mathcal{O}(k^{-(p-1)})$. It therefore suffices to choose $k$ such that $k^{-(p-1)} \lesssim \sqrt{\varepsilon}$, giving

$$K_0(\varepsilon) = \mathcal{O}\big(\varepsilon^{-1/(2(p-1))}\big).$$

Since $p > 2$ in the logarithmic schedule, we have $p - 1 > 1$ and hence $1/(2(p-1)) < 1/2 < 1$. In particular, $K_0(\varepsilon) = o(\varepsilon^{-1})$ and therefore does not dominate the $\mathcal{O}(\varepsilon^{-1})$ post-burn-in term. The total iteration complexity in Theorem 5.20 (and communication complexity in Theorem A.16) has the post-burn-in phase as its leading term:

$$K(\varepsilon) = \mathcal{O}(\varepsilon^{-1}), \qquad \sum_{k=0}^{K(\varepsilon)-1} (\tau_k + 2t_k) = \mathcal{O}(\varepsilon^{-1} \log(1/\varepsilon)).$$

For $p = 3$, the exponent is $1/4$, so the burn-in term has a lower asymptotic order than the $\mathcal{O}(\varepsilon^{-1})$ post-burn-in term. The hidden constants still determine the finite-accuracy burn-in length.

## Appendix B. Proofs.

*Proof of Lemma A.1.* Fix $x, s$ and define $\phi(t) := h(x+ts)$ for $t \in [0,1]$. Then $\phi'(t) = \langle \nabla h(x+ts), s\rangle$ and $\phi''(t) = s^\top \nabla^2 h(x+ts)s$. By $L_2$-Lipschitzness of $\nabla^2 h$,

$$
\begin{aligned}
|\phi''(t) - \phi''(0)| &= |s^\top(\nabla^2 h(x+ts) - \nabla^2 h(x))s| \\
&\le \|\nabla^2 h(x+ts) - \nabla^2 h(x)\|_2 \|s\|^2 \\
&\le L_2 t \|s\|^3.
\end{aligned}
$$

Hence $\phi''(t) \le \phi''(0) + L_2 t\|s\|^3$. Integrating twice,

$$
\begin{aligned}
\phi(1) &= \phi(0) + \phi'(0) + \int_0^1 (1-t)\phi''(t)\,dt \\
&\le \phi(0) + \phi'(0) + \int_0^1 (1-t)\big(\phi''(0) + L_2 t\|s\|^3\big)dt.
\end{aligned}
$$

Compute $\int_0^1 (1-t)\,dt = \frac{1}{2}$ and $\int_0^1 (1-t)t\,dt = \frac{1}{6}$ to obtain

$$
\phi(1) \le \phi(0) + \phi'(0) + \tfrac{1}{2}\phi''(0) + \tfrac{L_2}{6}\|s\|^3.
$$

Substitute $\phi(0) = h(x)$, $\phi'(0) = \langle \nabla h(x), s\rangle$, and $\phi''(0) = s^\top \nabla^2 h(x)s$. ∎

*Proof of Lemma A.2.* Because $W$ is symmetric and doubly stochastic, it is diagonalizable with eigenvalues $1 = \lambda_1 > \lambda_2 \ge \cdots \ge \lambda_N \ge -1$ and eigenvector $\mathbf{1}$ for $\lambda_1$. The projector $\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) = \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ is the orthogonal projector onto $\mathrm{span}\{\mathbf{1}\}$ and satisfies $W\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)W = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$. We show $W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) = (W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))^t$ by induction: for $t=1$ trivial. If true for $t$, then

$$
\begin{aligned}
(W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))^{t+1} &= (W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))^t (W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \\
&= (W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))(W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)) \\
&= W^{t+1} - W^t\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)W + \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)^2 \\
&= W^{t+1} - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right),
\end{aligned}
$$

using $W^t\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right) = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$, $\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)W = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$, and $\left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)^2 = \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)$. Thus $\|W^t - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\|_2 = \|(W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right))^t\|_2 \le \|W - \left(\frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\|_2^t = \rho^t$. For stacked vectors, $\|(A \otimes I_d)\|_2 = \|A\|_2$. ∎

*Proof of Lemma A.3.* By $L$-smoothness,

$$
h(y) \le h(x) + \langle \nabla h(x), y - x\rangle + \frac{L}{2}\|y - x\|^2.
$$

Choose $y = x - \frac{1}{L}\nabla h(x)$ to obtain

$$
h\Big(x - \frac{1}{L}\nabla h(x)\Big) \le h(x) - \frac{1}{2L}\|\nabla h(x)\|^2.
$$

Since $x_\star$ minimizes $h$, $h(x_\star) \le h(x - \frac{1}{L}\nabla h(x))$. Rearranging yields the claim. ∎

*Proof of Lemma 5.4.* Because $W^{\tau_k}$ is doubly stochastic,

$$\tilde{\bar{x}}_k = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}[W^{\tau_k}]_{ij}x_{j,k} = \frac{1}{N}\sum_{j=1}^{N}\Big(\sum_{i=1}^{N}[W^{\tau_k}]_{ij}\Big)x_{j,k} = \frac{1}{N}\sum_{j=1}^{N}x_{j,k} = \bar{x}_k.$$

Similarly, $W^{t_k}$ is doubly stochastic and $y_{i,k+1} = \tilde{x}_{i,k} + s_{i,k}$, hence

$$\begin{aligned}\bar{x}_{k+1} &= \tfrac{1}{N}\textstyle\sum_{i=1}^{N} x_{i,k+1} = \tfrac{1}{N}\sum_{i,j}[W^{t_k}]_{ij}y_{j,k+1}\\ &= \tfrac{1}{N}\textstyle\sum_{j=1}^{N} y_{j,k+1} = \tilde{\bar{x}}_k + \bar{s}_k = \bar{x}_k + \bar{s}_k.\end{aligned}$$ ∎

*Proof of Lemma 5.5.* Average the tracker update in Algorithm 4.1:

$$\bar{g}_{k+1} = \frac{1}{N}\sum_{i=1}^{N} g_{i,k+1} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}[W^{t_k}]_{ij}\Big(\tilde{g}_{j,k} + \nabla f_j(x_{j,k+1}) - \nabla f_j(x_{j,k})\Big).$$

Swap sums and use column-stochasticity $\sum_{i=1}^{N}[W^{t_k}]_{ij} = 1$:

$$\begin{aligned}\bar{g}_{k+1} &= \tfrac{1}{N}\textstyle\sum_{j=1}^{N}\Big(\tilde{g}_{j,k} + \nabla f_j(x_{j,k+1}) - \nabla f_j(x_{j,k})\Big)\\ &= \tilde{\bar{g}}_k + \tfrac{1}{N}\textstyle\sum_{j=1}^{N}\big(\nabla f_j(x_{j,k+1}) - \nabla f_j(x_{j,k})\big).\end{aligned}$$

Pre-mixing preserves averages, hence $\tilde{\bar{g}}_k = \bar{g}_k$. Starting from

$$\bar{g}_0 = \frac{1}{N}\sum_{i=1}^{N}\nabla f_i(x_{i,0}),$$

induction gives $\bar{g}_k = N^{-1}\sum_{i=1}^{N}\nabla f_i(x_{i,k})$ for all $k$, and thus $\tilde{\bar{g}}_k = \bar{g}_k$. ∎

*Proof of Lemma 5.6.* Repeating the analysis in the proof of Lemma 5.5 with gradients replaced by Hessians yields the claim. Averaging the Hessian update gives

$$\bar{H}_{k+1} = \tilde{\bar{H}}_k + \frac{1}{N}\sum_{i=1}^{N}\big(\nabla^2 f_i(x_{i,k+1}) - \nabla^2 f_i(x_{i,k})\big),$$

and $\tilde{\bar{H}}_k = \bar{H}_k$. Induction yields $\bar{H}_k = \frac{1}{N}\sum_{i=1}^{N}\nabla^2 f_i(x_{i,k})$. ∎

*Proof of Lemma 5.7.* As in Appendix A, the local system satisfies $(\tilde{H}_{i,k} + (\lambda_{i,k} + \delta_{i,k})I) \succeq \lambda_{i,k}I$. Thus $\lambda_{i,k}\|s_{i,k}\| \le \|\tilde{g}_{i,k}\|$. Since $\lambda_{i,k}^2 = M\|\tilde{g}_{i,k}\|$, we have $\lambda_{i,k}\|s_{i,k}\| \le \lambda_{i,k}^2/M$, hence $M\|s_{i,k}\| \le \lambda_{i,k}$. Using $M \ge L_2$ gives $L_2\|s_{i,k}\| \le \lambda_{i,k}$. ∎

*Proof of Lemma 5.8.* From Lemma 5.7, $M\|s_{i,k}\| \le \lambda_{i,k}$ for all $i$, hence

$$\frac{1}{N}\sum_{i=1}^{N}\lambda_{i,k} \ge \frac{M}{N}\sum_{i=1}^{N}\|s_{i,k}\| \ge M\Big\|\frac{1}{N}\sum_{i=1}^{N}s_{i,k}\Big\| = M\|\bar{s}_k\|,$$

using $\|\frac{1}{N}\sum_{i=1}^{N}s_i\| \le \frac{1}{N}\sum_{i=1}^{N}\|s_i\|$. Since $M \ge L_2$, $L_2\|\bar{s}_k\| \le \frac{1}{N}\sum_{i=1}^{N}\lambda_{i,k}$. Finally $\tilde{\bar{\lambda}}_k = \frac{1}{N}\sum_{i=1}^{N}(\lambda_{i,k} + \delta_{i,k}) \ge \frac{1}{N}\sum_{i=1}^{N}\lambda_{i,k}$. ∎

*Proof of Lemma 5.9.* By Lemma 5.5, $\bar{\tilde{g}}_k = \frac{1}{N}\sum_{i=1}^N \nabla f_i(x_{i,k})$. Therefore

$$\begin{aligned}\|\nabla f(\bar{x}_k)-\bar{\tilde{g}}_k\| &= \left\| \tfrac{1}{N}\textstyle\sum_{i=1}^N \big(\nabla f_i(\bar{x}_k)-\nabla f_i(x_{i,k})\big)\right\| \\ &\le \tfrac{1}{N}\textstyle\sum_{i=1}^N \|\nabla f_i(\bar{x}_k)-\nabla f_i(x_{i,k})\|.\end{aligned}$$

Using $L_1$-Lipschitzness, $\|\nabla f_i(\bar{x}_k)-\nabla f_i(x_{i,k})\| \le L_1\|x_{i,k}-\bar{x}_k\|$. Average and apply Cauchy–Schwarz:

$$\frac{1}{N}\sum_{i=1}^N \|x_{i,k}-\bar{x}_k\| \le \sqrt{\frac{1}{N}\sum_{i=1}^N \|x_{i,k}-\bar{x}_k\|^2} = D(X_k).$$

Combining the last two displays yields the claim. ∎

*Proof of Lemma 5.10.* By Lemma 5.6, $\bar{\tilde{H}}_k = \frac{1}{N}\sum_{i=1}^N \nabla^2 f_i(x_{i,k})$. Thus

$$\nabla^2 f(\bar{x}_k) - \bar{\tilde{H}}_k = \frac{1}{N}\sum_{i=1}^N \big(\nabla^2 f_i(\bar{x}_k) - \nabla^2 f_i(x_{i,k})\big),$$

and so

$$\begin{aligned}&\|\nabla^2 f(\bar{x}_k)-\bar{\tilde{H}}_k\|_2 \\ &\le \tfrac{1}{N}\textstyle\sum_{i=1}^N \|\nabla^2 f_i(\bar{x}_k)-\nabla^2 f_i(x_{i,k})\|_2 \\ &\le \tfrac{L_2}{N}\textstyle\sum_{i=1}^N \|x_{i,k}-\bar{x}_k\|.\end{aligned}$$

Applying Cauchy–Schwarz to the last average, we obtain $\|\nabla^2 f(\bar{x}_k) - \bar{\tilde{H}}_k\|_2 \le L_2 D(X_k)$. ∎

*Proof of Lemma 5.11.* Let $A := \tilde{H}_{i,k}$ and $B := \nabla^2 f(\bar{x}_k) \succeq 0$. By Weyl's inequality,

$$\lambda_{\min}(A) \ge \lambda_{\min}(B) - \|A-B\|_2 \ge -\|A-B\|_2.$$

Therefore $-\lambda_{\min}(A) \le \|A-B\|_2$, and since $\delta_{i,k} = \max\{0, -\lambda_{\min}(A)\}$, we obtain $0 \le \delta_{i,k} \le \|A-B\|_2$, proving the first claim. For the second,

$$\|A-B\|_2 \le \|A-\bar{\tilde{H}}_k\|_2 + \|\bar{\tilde{H}}_k - B\|_2.$$

Averaging over $i$ gives $\bar{\delta}_k \le \Delta_k^H + \|\bar{\tilde{H}}_k - \nabla^2 f(\bar{x}_k)\|_2$, and apply Lemma 5.10. ∎

*Proof of Lemma 5.12.* Fix $k$ and abbreviate $A_i := A_{i,k}$, $A := A_k^{\mathrm{ref}}$, $\tilde{g}_i := \tilde{g}_{i,k}$, $\bar{\tilde{g}} := \bar{\tilde{g}}_k$. Then $s_i = -A_i^{-1}\tilde{g}_i$ and $s^{\mathrm{ref}} = -A^{-1}\bar{\tilde{g}}$. Decompose:

$$s_i - s^{\mathrm{ref}} = -A_i^{-1}(\tilde{g}_i - \bar{\tilde{g}}) + (A^{-1} - A_i^{-1})\bar{\tilde{g}}.$$

Since $A_i \succeq \tilde{\lambda}_{i,k} I \succeq \underline{\lambda}_k I$, we have $\|A_i^{-1}\|_2 \le 1/\underline{\lambda}_k$. Thus

$$\|s_i - s^{\mathrm{ref}}\| \le \frac{1}{\underline{\lambda}_k}\|\tilde{g}_i - \bar{\tilde{g}}\| + \|A^{-1} - A_i^{-1}\|_2\,\|\bar{\tilde{g}}\|.$$

Use the resolvent identity $A^{-1} - A_i^{-1} = A_i^{-1}(A_i - A)A^{-1}$ to get

$$\|A^{-1} - A_i^{-1}\|_2 \le \|A_i^{-1}\|_2\,\|A_i - A\|_2\,\|A^{-1}\|_2 \le \frac{1}{\underline{\lambda}_k}\cdot\frac{1}{\bar{\tilde{\lambda}}_k}\,\|A_i - A\|_2,$$

since $A \succeq \tilde{\bar{\lambda}}_k I$ implies $\|A^{-1}\|_2 \le 1/\tilde{\bar{\lambda}}_k$. Furthermore,

$$A_i - A = \big(\tilde{H}_{i,k} - \tilde{\bar{H}}_k\big) + (\tilde{\lambda}_{i,k} - \tilde{\bar{\lambda}}_k)I,$$

so $\|A_i - A\|_2 \le \|\tilde{H}_{i,k} - \tilde{\bar{H}}_k\|_2 + |\tilde{\lambda}_{i,k} - \tilde{\bar{\lambda}}_k|$. Combining the last two displays and averaging over $i$ using $\|\bar{s} - s^{\mathrm{ref}}\| \le \frac{1}{N}\sum_{i=1}^N \|s_i - s^{\mathrm{ref}}\|$, we obtain the claim. ∎

*Proof of Lemma 5.13.* Write $\tilde{\lambda}_{i,k} - \tilde{\bar{\lambda}}_k = (\lambda_{i,k} - \bar{\lambda}_k) + (\delta_{i,k} - \bar{\delta}_k)$, where $\bar{\lambda}_k := \frac{1}{N}\sum_{i=1}^N \lambda_{i,k}$. Then

$$\Delta_k^\lambda = \frac{1}{N}\sum_{i=1}^N |\tilde{\lambda}_{i,k} - \tilde{\bar{\lambda}}_k| \le \underbrace{\frac{1}{N}\sum_{i=1}^N |\lambda_{i,k} - \bar{\lambda}_k|}_{=:T_1} + \underbrace{\frac{1}{N}\sum_{i=1}^N |\delta_{i,k} - \bar{\delta}_k|}_{=:T_2}.$$

**Step 1: bound $T_2$.** Since $\delta_{i,k} \ge 0$, a standard counting argument yields

$$\sum_{i=1}^N |\delta_{i,k} - \bar{\delta}_k| = 2\sum_{i:\delta_{i,k} > \bar{\delta}_k} (\delta_{i,k} - \bar{\delta}_k) \le 2\sum_{i:\delta_{i,k} > \bar{\delta}_k} \delta_{i,k} \le 2\sum_{i=1}^N \delta_{i,k} = 2N\bar{\delta}_k,$$

so $T_2 \le 2\bar{\delta}_k$.

**Step 2: bound $T_1$.** For any scalars $\{u_i\}$ with $\bar{u} = \frac{1}{N}\sum_{i=1}^N u_i$, $|u_i - \bar{u}| = \big|\frac{1}{N}\sum_{j=1}^N (u_i - u_j)\big| \le \frac{1}{N}\sum_{j=1}^N |u_i - u_j|$. Averaging over $i$ gives

$$T_1 \le \frac{1}{N^2}\sum_{i=1}^N\sum_{j=1}^N |\lambda_{i,k} - \lambda_{j,k}|.$$

Let $a_i := \|\tilde{g}_{i,k}\|$. Then

$$|\lambda_{i,k} - \lambda_{j,k}| = \sqrt{M}\,|\sqrt{a_i} - \sqrt{a_j}| = \sqrt{M}\,\frac{|a_i - a_j|}{\sqrt{a_i} + \sqrt{a_j}}.$$

Moreover $|a_i - a_j| \le \|\tilde{g}_{i,k} - \tilde{g}_{j,k}\|$. By triangle inequality, $\|\tilde{g}_{i,k} - \tilde{g}_{j,k}\| \le \|\tilde{g}_{i,k} - \tilde{\bar{g}}_k\| + \|\tilde{g}_{j,k} - \tilde{\bar{g}}_k\|$, hence

$$\frac{1}{N^2}\sum_{i,j} \|\tilde{g}_{i,k} - \tilde{g}_{j,k}\| \le \frac{1}{N^2}\sum_{i,j} \big(\|\tilde{g}_{i,k} - \tilde{\bar{g}}_k\| + \|\tilde{g}_{j,k} - \tilde{\bar{g}}_k\|\big) = 2\Delta_k^g.$$

Under (5.3), $a_i \ge \|\tilde{\bar{g}}_k\| - \|\tilde{g}_{i,k} - \tilde{\bar{g}}_k\| \ge (1-\alpha_d)\|\tilde{\bar{g}}_k\|$ for all $i$, so $\sqrt{a_i} + \sqrt{a_j} \ge 2\sqrt{(1-\alpha_d)\|\tilde{\bar{g}}_k\|}$. Therefore

$$T_1 \le \sqrt{M} \cdot \frac{2\Delta_k^g}{2\sqrt{(1-\alpha_d)\|\tilde{\bar{g}}_k\|}} = \frac{\sqrt{M}}{\sqrt{1-\alpha_d}} \cdot \frac{\Delta_k^g}{\sqrt{\|\tilde{\bar{g}}_k\|}}.$$

Combining this with $T_2 \le 2\bar{\delta}_k$ gives the claim. ∎

*Proof of Proposition 5.14.* Write

$$r_k = (\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k + g_k.$$

Add and subtract the reference step:

$$r_k = (\nabla^2 f(\bar{x}_k) + \lambda_k I)(\bar{s}_k - s_k^{\mathrm{ref}}) + (\nabla^2 f(\bar{x}_k) + \lambda_k I)s_k^{\mathrm{ref}} + g_k.$$

Using $(\tilde{\bar{H}}_k + \lambda_k I)s_k^{\mathrm{ref}} = -\tilde{\bar{g}}_k$, we have

$$(\nabla^2 f(\bar{x}_k) + \lambda_k I)s_k^{\mathrm{ref}} + g_k = (\nabla^2 f(\bar{x}_k) - \tilde{\bar{H}}_k)s_k^{\mathrm{ref}} + (g_k - \tilde{\bar{g}}_k).$$

Therefore

$$r_k = (\nabla^2 f(\bar{x}_k) + \lambda_k I)(\bar{s}_k - s_k^{\mathrm{ref}}) + (\nabla^2 f(\bar{x}_k) - \tilde{\bar{H}}_k)s_k^{\mathrm{ref}} + (g_k - \tilde{\bar{g}}_k).$$

Take norms:

$$\begin{aligned}\|r_k\| &\le \|\nabla^2 f(\bar{x}_k)+\lambda_k I\|_2\|\bar{s}_k - s_k^{\mathrm{ref}}\| \\ &\quad + \|\nabla^2 f(\bar{x}_k) - \tilde{\bar{H}}_k\|_2\|s_k^{\mathrm{ref}}\| + \|g_k - \tilde{\bar{g}}_k\|.\end{aligned}$$

By the global Hessian bound (5.2), $\|\nabla^2 f(\bar{x}_k)\|_2 \le M_{H,\max}$, hence $\|\nabla^2 f(\bar{x}_k)+\lambda_k I\|_2 \le M_{H,\max} + \lambda_k$. Using item (1),

$$\begin{aligned}&\|\nabla^2 f(\bar{x}_k)+\lambda_k I\|_2\,\|\bar{s}_k - s_k^{\mathrm{ref}}\| \\ &\le (M_{H,\max}+\lambda_k)\cdot\tfrac{\eta}{8}\cdot\tfrac{\lambda_k}{M_{H,\max}+\lambda_k}\|\bar{s}_k\| = \tfrac{\eta}{8}\lambda_k\|\bar{s}_k\|.\end{aligned}$$

Next, item (3) and Lemma 5.10 imply $\|\nabla^2 f(\bar{x}_k) - \tilde{\bar{H}}_k\|_2 \le \frac{\eta}{8}\lambda_k$. Moreover,

$$\|s_k^{\mathrm{ref}}\| \le \|\bar{s}_k\| + \|\bar{s}_k - s_k^{\mathrm{ref}}\| \le \left(1 + \frac{\eta}{8}\right)\|\bar{s}_k\| \le \frac{9}{8}\|\bar{s}_k\|,$$

since $\eta \le 1/12$ implies $1 + \eta/8 \le 9/8$. Therefore the second term is bounded by

$$\|\nabla^2 f(\bar{x}_k) - \tilde{\bar{H}}_k\|_2\,\|s_k^{\mathrm{ref}}\| \le \frac{\eta}{8}\lambda_k \cdot \frac{9}{8}\|\bar{s}_k\| = \frac{9\eta}{64}\lambda_k\|\bar{s}_k\|.$$

Finally, item (2) and Lemma 5.9 imply $\|g_k - \tilde{\bar{g}}_k\| \le \frac{\eta}{8}\lambda_k\|\bar{s}_k\|$. Summing yields

$$\|r_k\| \le \left(\frac{\eta}{8} + \frac{9\eta}{64} + \frac{\eta}{8}\right)\lambda_k\|\bar{s}_k\| = \frac{25\eta}{64}\lambda_k\|\bar{s}_k\| \le \eta\lambda_k\|\bar{s}_k\|.$$

The inequality $L_2\|\bar{s}_k\| \le \lambda_k$ is Lemma 5.8. ∎

*Proof of Lemma 5.15.* Apply Lemma A.1 to $f$ at $(\bar{x}_k, \bar{s}_k)$:

$$f(\bar{x}_k + \bar{s}_k) \le f(\bar{x}_k) + \langle g_k, \bar{s}_k\rangle + \frac{1}{2}\bar{s}_k^\top \nabla^2 f(\bar{x}_k)\bar{s}_k + \frac{L_2}{6}\|\bar{s}_k\|^3.$$

Let $d_k := (\nabla^2 f(\bar{x}_k)+\lambda_k I)\bar{s}_k + g_k$. Then $\langle g_k, \bar{s}_k\rangle = -\bar{s}_k^\top \nabla^2 f(\bar{x}_k)\bar{s}_k - \lambda_k\|\bar{s}_k\|^2 + \langle d_k, \bar{s}_k\rangle$. Substitute:

$$f(\bar{x}_{k+1}) \le f(\bar{x}_k) - \frac{1}{2}\bar{s}_k^\top \nabla^2 f(\bar{x}_k)\bar{s}_k - \lambda_k\|\bar{s}_k\|^2 + \langle d_k, \bar{s}_k\rangle + \frac{L_2}{6}\|\bar{s}_k\|^3.$$

Since $\nabla^2 f(\bar{x}_k) \succeq 0$, drop the nonpositive term. Also $\langle d_k, \bar{s}_k\rangle \le \|d_k\|\,\|\bar{s}_k\| \le \eta\lambda_k\|\bar{s}_k\|^2$ and $L_2\|\bar{s}_k\| \le \lambda_k$ implies $\frac{L_2}{6}\|\bar{s}_k\|^3 \le \frac{1}{6}\lambda_k\|\bar{s}_k\|^2$. Combining these three bounds yields the claim. ∎

*Proof of Lemma 5.16.* By the integral form of Taylor's theorem,

$$\nabla f(\bar{x}_k + \bar{s}_k) = g_k + \nabla^2 f(\bar{x}_k)\bar{s}_k + r_k, \qquad \|r_k\| \le \frac{L_2}{2}\|\bar{s}_k\|^2.$$

Using $d_k = (\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k + g_k$, we have $g_k + \nabla^2 f(\bar{x}_k)\bar{s}_k = -\lambda_k \bar{s}_k + d_k$. Thus

$$\|g_{k+1}\| \le \lambda_k\|\bar{s}_k\| + \|d_k\| + \|r_k\| \le \lambda_k\|\bar{s}_k\| + \eta\lambda_k\|\bar{s}_k\| + \frac{L_2}{2}\|\bar{s}_k\|^2.$$

Finally $L_2\|\bar{s}_k\| \le \lambda_k$ implies $\frac{L_2}{2}\|\bar{s}_k\|^2 \le \frac{1}{2}\lambda_k\|\bar{s}_k\|$. ∎

LEMMA B.1. *If Assumptions 3.1 and 5.1 hold and the schedule (A.8) is used, then the following is true. Fix any $\varepsilon > 0$ and let $K_0(\varepsilon)$ be as in Proposition A.15. Then there exists a constant $C_\lambda > 0$ such that for all $k \ge K_0(\varepsilon)$ with $\|g_k\| \ge \varepsilon$,*

$$\lambda_k \le C_\lambda \sqrt{\|g_k\|}.$$

*Proof.* Recall $\lambda_k = \tilde{\bar{\lambda}}_k = \frac{1}{N}\sum_{i=1}^N(\lambda_{i,k} + \delta_{i,k})$ with $\lambda_{i,k} = \sqrt{M\|\tilde{g}_{i,k}\|}$ and $\delta_{i,k} \ge 0$. By Jensen's inequality (concavity of $x \mapsto \sqrt{x}$),

$$\frac{1}{N}\sum_{i=1}^N \sqrt{\|\tilde{g}_{i,k}\|} \le \sqrt{\frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k}\|}.$$

Moreover, by the triangle inequality and the definition of $\Delta_k^g$,

$$\frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k}\| \le \|\tilde{\bar{g}}_k\| + \Delta_k^g.$$

By Lemma 5.9, $\|\tilde{\bar{g}}_k - g_k\| \le L_1 D(X_k)$. Using Item 2 of Proposition 5.14 (which holds for all $k \ge K_0(\varepsilon)$ with $\|g_k\| \ge \varepsilon$), together with $\|(\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k\| = \|g_k - r_k\| \ge \lambda_k\|\bar{s}_k\|$ and $\|r_k\| \le \eta\lambda_k\|\bar{s}_k\|$, we obtain $(1-\eta)\lambda_k\|\bar{s}_k\| \le \|g_k\|$ and hence

$$L_1 D(X_k) \le \frac{\eta}{8}\lambda_k\|\bar{s}_k\| \le \frac{\eta}{8(1-\eta)}\|g_k\|.$$

Therefore,

$$\|\tilde{\bar{g}}_k\| \le \|g_k\| + L_1 D(X_k) \le \Big(1 + \frac{\eta}{8(1-\eta)}\Big)\|g_k\|.$$

In addition, Proposition A.15 ensures $\Delta_k^g \le \varepsilon \le \|g_k\|$ for all $k \ge K_0(\varepsilon)$ with $\|g_k\| \ge \varepsilon$. Thus,

$$\frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k}\| \le \Big(2 + \frac{\eta}{8(1-\eta)}\Big)\|g_k\|.$$

Hence,

$$\lambda_k = \frac{1}{N}\sum_{i=1}^N \big(\sqrt{M\|\tilde{g}_{i,k}\|} + \delta_{i,k}\big) \le \sqrt{M}\sqrt{\frac{1}{N}\sum_{i=1}^N \|\tilde{g}_{i,k}\|} + \bar{\delta}_k \le C_0\sqrt{\|g_k\|} + \bar{\delta}_k,$$

with $C_0 := \sqrt{M\big(2 + \frac{\eta}{8(1-\eta)}\big)}$. Finally, Proposition A.15 also ensures $\bar{\delta}_k \le \sqrt{\varepsilon} \le \sqrt{\|g_k\|}$ on the same index set. Absorbing this into the constant yields $\lambda_k \le (C_0 + 1)\sqrt{\|g_k\|}$, completing the proof. ∎

*Proof of Lemma 5.17.* Assume $\eta \le 1/12$, so $c_d := 1-\eta-\frac{1}{6} \ge \frac{3}{4}$ in Lemma 5.15. Also from Lemma 5.16, $C_g := 1+\eta+\frac{1}{2} \le 2$. For $k \in \mathcal{I}$, $\|g_{k+1}\| \ge \frac{1}{4}\|g_k\|$ and $\|g_{k+1}\| \le C_g\lambda_k\|\bar{s}_k\|$ imply $\|\bar{s}_k\| \ge \|g_k\|/(4C_g\lambda_k)$. Then Lemma 5.15 yields

$$\Phi_k - \Phi_{k+1} \ge c_d\lambda_k\|\bar{s}_k\|^2 \ge c_d\lambda_k\left(\frac{\|g_k\|}{4C_g\lambda_k}\right)^2 = \frac{c_d}{16C_g^2}\frac{\|g_k\|^2}{\lambda_k}.$$

By the additional assumption in Lemma 5.17, we have $\lambda_k \le C_\lambda\sqrt{\|g_k\|}$. Therefore,

$$\frac{\|g_k\|^2}{\lambda_k} \ge \frac{1}{C_\lambda}\|g_k\|^{3/2}.$$

Finally, convexity and the bounded level set imply $\Phi_k \le \langle g_k, \bar{x}_k - x_\star\rangle \le D\|g_k\|$, hence $\|g_k\|^{3/2} \ge D^{-3/2}\Phi_k^{3/2}$. Collect constants into $\nu$. ∎

*Proof of Lemma 5.18.* If $\Phi_k = 0$ for some $k$, then $\Phi_{k'} = 0$ for all $k' \ge k$ and the claim is trivial. Assume $\Phi_k > 0$ for all $k$.

From $\Phi_{k+1} \le \Phi_k - \nu\Phi_k^{3/2} = \Phi_k(1-\nu\sqrt{\Phi_k})$, we consider two cases.

Case 1: $\nu\sqrt{\Phi_k} \ge 1$. Then $\Phi_{k+1} \le 0$. Since $\Phi_{k+1} \ge 0$, we must have $\Phi_{k+1} = 0$, contradicting $\Phi_{k+1} > 0$. Hence this case cannot occur under the assumption $\Phi_k > 0$.

Therefore, necessarily $\nu\sqrt{\Phi_k} \in (0,1)$ for all $k$.

Let $\psi_k := \Phi_k^{-1/2}$. Then

$$\begin{aligned}\psi_{k+1} = \Phi_{k+1}^{-1/2} &\ge \Phi_k^{-1/2}(1-\nu\sqrt{\Phi_k})^{-1/2}\\ &= \psi_k(1-u_k)^{-1/2}, \quad u_k := \nu\sqrt{\Phi_k} \in (0,1).\end{aligned}$$

Using convexity of $(1-u)^{-1/2}$ on $[0,1)$ and $(1-u)^{-1/2} \ge 1+\frac{1}{2}u$, we obtain

$$\psi_{k+1} \ge \psi_k\Big(1+\frac{1}{2}u_k\Big) = \psi_k + \frac{\nu}{2}.$$

The positive-sequence case also implies $\nu\sqrt{\Phi_0} < 1$, hence $\psi_0 = \Phi_0^{-1/2} > \nu$. Therefore,

$$\psi_k \ge \psi_0 + \frac{\nu}{2}k > \nu + \frac{\nu}{2}k = \frac{\nu}{2}(k+2),$$

and hence

$$\Phi_k = \psi_k^{-2} \le \frac{4}{\nu^2(k+2)^2}.$$ ∎

*Proof of Theorem 5.20.* We establish the $\mathcal{O}(\varepsilon^{-1})$ post-burn-in rate by deriving the 3/2-recursion under the assumption that Proposition 5.14 holds from some index onwards. Concretely, fix $\eta \le 1/12$ and let $K_0 = K_0(\varepsilon)$ be the burn-in index from Proposition A.15, so that Proposition 5.14 holds for all $k \ge K_0$ with $\|g_k\| \ge \varepsilon$. Define $\Phi_k := f(\bar{x}_k) - f_\star$ and $g_k := \nabla f(\bar{x}_k)$. Set

$$\hat{x}_j := \bar{x}_{K_0+j}, \qquad \hat{g}_j := g_{K_0+j}, \qquad \hat{\Phi}_j := \Phi_{K_0+j}, \qquad j \ge 0.$$

We write the argument for the shifted sequence.

**Step 0: descent along the shifted sequence.** By Proposition A.15, the hypotheses of Lemma 5.15 hold at every shifted index $j \ge 0$ satisfying $\|\hat{g}_j\| \ge \varepsilon$. Hence, with $c_d := 1-\eta-\frac{1}{6} > 0$,

$$\text{(B.1)} \qquad \hat{\Phi}_{j+1} \le \hat{\Phi}_j - c_d\,\lambda_{K_0+j}\|\bar{s}_{K_0+j}\|^2 \le \hat{\Phi}_j, \qquad \forall j \ge 0 \text{ with } \|\hat{g}_j\| \ge \varepsilon.$$

Since we seek the first shifted index $j$ with $\|\hat{g}_j\| \le \varepsilon$, descent applies at every step of interest. For notational economy, the following steps relabel the shifted sequence as $(\bar{x}_k, g_k, \Phi_k)$.

**Step 1: a uniform gradient growth bound.** Let $r_k := (\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k + g_k$ as defined in the paper. From Proposition 5.14, $\|r_k\| \le \eta\lambda_k\|\bar{s}_k\|$. Moreover,

$$\lambda_k\|\bar{s}_k\| \le \|(\nabla^2 f(\bar{x}_k) + \lambda_k I)\bar{s}_k\| = \|g_k - r_k\| \le \|g_k\| + \|r_k\| \le \|g_k\| + \eta\lambda_k\|\bar{s}_k\|.$$

Rearranging yields

$$\lambda_k\|\bar{s}_k\| \le \frac{1}{1-\eta}\|g_k\|, \qquad \forall k \ge 0. \tag{B.2}$$

By Lemma 5.16 (which applies at every iteration under Proposition 5.14),

$$\|g_{k+1}\| \le C_g\,\lambda_k\|\bar{s}_k\|, \qquad C_g := 1 + \eta + \frac{1}{2}.$$

Combining with (B.2) gives the uniform growth bound

$$\|g_{k+1}\| \le \kappa_{\mathrm{inc}}\|g_k\|, \qquad \kappa_{\mathrm{inc}} := \frac{C_g}{1-\eta}. \tag{B.3}$$

Since $\eta \le 1/12$, we have $C_g \le 2$ and thus $\kappa_{\mathrm{inc}} \le 24/11 < 4$.

**Step 2: decay along the steady-step subsequence.** Recall the index sets

$$\mathcal{I} := \{k \ge 0 :\ \|g_{k+1}\| \ge \tfrac{1}{4}\|g_k\|\}, \qquad \mathcal{S} := \{k \ge 0 :\ \|g_{k+1}\| < \tfrac{1}{4}\|g_k\|\}.$$

Let $n_k := |\mathcal{I} \cap \{0, 1, \ldots, k-1\}|$ be the number of steady indices among the first $k$ iterations. List these indices increasingly as $0 \le i_0 < i_1 < \cdots < i_{n_k-1} \le k-1$ (if $n_k = 0$ skip this step). Define the subsequence $u_j := \Phi_{i_j}$. By Lemma 5.17, for each steady index $i_j \in \mathcal{I}$,

$$\Phi_{i_j} - \Phi_{i_j+1} \ge \nu\,\Phi_{i_j}^{3/2} = \nu\,u_j^{3/2}.$$

Since $i_{j+1} \ge i_j + 1$ and $\Phi_k$ is nonincreasing along the indices before termination, we have

$$u_{j+1} = \Phi_{i_{j+1}} \le \Phi_{i_j+1} \le \Phi_{i_j} - \nu\,\Phi_{i_j}^{3/2} = u_j - \nu\,u_j^{3/2}.$$

Applying Lemma 5.18 to $\{u_j\}$ yields

$$u_j \le \frac{4}{\nu^2(j+2)^2}, \qquad \forall j \ge 0. \tag{B.4}$$

**Step 3: two-case bound yielding $\mathcal{O}(\varepsilon^{-1})$ iteration complexity.** Let $m_k := |\mathcal{S} \cap \{0, 1, \ldots, k-1\}| = k - n_k$. We consider two cases.

Case 1: $n_k \ge k/2$. For $k \ge 2$, this implies $n_k \ge 1$, so the last steady index $i_{n_k-1}$ is well defined. By monotonicity, $\Phi_k \le \Phi_{i_{n_k-1}} = u_{n_k-1}$, and (B.4) gives

$$\Phi_k \le \frac{4}{\nu^2(n_k+1)^2} \le \frac{16}{\nu^2(k+2)^2}.$$

The finitely many cases $k < 2$ are absorbed into the final constant.

Case 2: $n_k < k/2$ (hence $m_k > k/2$). For every sharp index $\ell \in \mathcal{S}$, we have $\|g_{\ell+1}\| \le \frac{1}{4}\|g_\ell\|$ by definition. For every other index, we have the growth bound (B.3). Therefore, after $k$ iterations,

$$\|g_k\| \le \kappa_{\text{inc}}^{n_k} \left(\frac{1}{4}\right)^{m_k} \|g_0\| \le \kappa_{\text{inc}}^{k/2} \left(\frac{1}{4}\right)^{k/2} \|g_0\| = \left(\frac{\kappa_{\text{inc}}}{4}\right)^{k/2} \|g_0\|.$$

Since $\kappa_{\text{inc}}/4 < 1$, the bounded level set and convexity imply

$$\Phi_k = f(\bar{x}_k) - f_\star \le \langle g_k, \bar{x}_k - x_\star\rangle \le \|\bar{x}_k - x_\star\| \, \|g_k\| \le D \, \|g_k\|.$$

Therefore,

$$\Phi_k \le D\, G_\star \left(\frac{\kappa_{\text{inc}}}{4}\right)^{k/2}, \qquad G_\star := \sup_{\ell \ge 0} \|g_\ell\| < \infty.$$

The finiteness of $G_\star$ follows from Assumption 5.1 and the Lipschitz continuity of $\nabla f$ on the bounded trajectory. Because $\sup_{k\ge 0}(k+2)^2 \left(\frac{\kappa_{\text{inc}}}{4}\right)^{k/2} < \infty$, there exists a finite constant $C_{\exp} > 0$ such that

$$\Phi_k \le \frac{C_{\exp}}{(k+2)^2} \qquad \forall k \ge 0.$$

Combining both cases, there exists $C_\Phi > 0$ such that for all $k \ge 0$,

$$\Phi_k \le \frac{C_\Phi}{(k+2)^2},$$

which proves $\Phi_k = \mathcal{O}(1/k^2)$.

**Step 4: gradient rate and $\varepsilon$-complexity.** By Lemma A.3 with $h = f$ and $L = L_1$,

$$\|g_k\|^2 \le 2L_1 \Phi_k \le \frac{2L_1 C_\Phi}{(k+2)^2},$$

hence $\|g_k\| \le \sqrt{2L_1 C_\Phi}/(k+2) = \mathcal{O}(1/k)$. To ensure $\|g_{K_0(\varepsilon)+j}\| \le \varepsilon$, it suffices to take

$$j \;\ge\; \frac{\sqrt{2L_1 C_\Phi}}{\varepsilon} - 2 = \mathcal{O}(\varepsilon^{-1}).$$

**Step 5: total complexity with burn-in.** Let $K(\varepsilon)$ be the first original iteration where $\|g_k\| \le \varepsilon$. Translating the shifted-index bound back to the original index, it takes at most $\mathcal{O}(\varepsilon^{-1})$ additional steps after $K_0(\varepsilon)$. Thus, the total iteration complexity is bounded by:

$$K(\varepsilon) \le K_0(\varepsilon) + C_K\, \varepsilon^{-1},$$

for some constant $C_K > 0$. This completes the proof of the global rate. ∎

*Proof of Lemma 5.24.* From Lemma 5.11, $\bar{\delta}_k \le \Delta_k^H + L_2 D(X_k)$.

**Bound $\Delta_k^H$.** By Jensen's inequality and $\|\cdot\|_2 \le \|\cdot\|_F$,

$$\Delta_k^H \le D(\tilde{\mathcal{H}}_k).$$

We apply the Hessian-tracker dispersion recursion (Lemma A.8), noting that $\mathcal{H}_k$ is obtained from $\tilde{\mathcal{H}}_{k-1}$ via the post-mixing update at iteration $k-1$:

$$D(\tilde{\mathcal{H}}_k) \le \rho^{\tau_k} D(\mathcal{H}_k)$$

$$\le \rho^{\tau_k}\rho^{t_{k-1}}\big(D(\tilde{\mathcal{H}}_{k-1})+2\sqrt{d}L_2D\big)$$
$$\le \rho^{\tau_k}(B_H+2\sqrt{d}L_2D),$$

where $B_H := \sup_{j\ge 0} D(\tilde{\mathcal{H}}_j) < \infty$ is the uniform bound established in Lemma A.10. In particular, since $B_H$ is finite, we obtain

$$\Delta_k^H \le \rho^{\tau_k}\big(B_H + 2\sqrt{d}\,L_2D\big) =: \hat{C}_H\,\rho^{\tau_k}.$$

**Bound $D(X_k)$.** By Lemma A.11,

$$D(X_k) \le \rho^{t_{k-1}}\Big(D(\tilde{X}_{k-1}) + \frac{\|S_{k-1}\|}{\sqrt{N}}\Big).$$

Moreover, $D(\tilde{X}_{k-1}) \le 2D$ and Lemma A.5 gives $\|S_{k-1}\|/\sqrt{N} \le \sqrt{G_g/M}$. Thus

$$D(X_k) \le \rho^{t_{k-1}}\Big(2D + \sqrt{\frac{G_g}{M}}\Big).$$

Therefore,

$$\bar{\delta}_k \le \hat{C}_H\,\rho^{\tau_k} + L_2\rho^{t_{k-1}}\Big(2D + \sqrt{\frac{G_g}{M}}\Big) \le C_\delta \max\{\rho^{\tau_k}, \rho^{t_{k-1}}\}.$$

Using (5.6), we obtain $\bar{\delta}_k \le \|g_k\|^{1+\gamma}$ for all sufficiently large $k$. ∎

*Proof of Theorem 5.25.* Let $x_k := \bar{x}_k$, $g_k := \nabla f(x_k)$, and $H_k := \nabla^2 f(x_k)$. Under Assumption 5.23, for all $k$ we have $H_k \succeq \mu I$ on the level set.

By definition, $r_k := (H_k + \lambda_k I)\bar{s}_k + g_k$ and Proposition 5.14 gives for all large $k$

$$(H_k + \lambda_k I)\bar{s}_k = -g_k + r_k, \qquad \|r_k\| \le \eta\lambda_k\|\bar{s}_k\|, \qquad L_2\|\bar{s}_k\| \le \lambda_k,$$

with $\eta \in (0, 1/12]$. Under (5.7) and the bound $D(X_k) \le \rho^{t_{k-1}}\big(2D + \sqrt{G_g/M}\big)$ (see Lemma A.11 and Lemma A.5), we have $D(X_k) = \mathcal{O}(\|g_k\|^{1+\gamma})$. Moreover, Lemma 5.24 gives $\bar{\delta}_k = \mathcal{O}(\|g_k\|^{1+\gamma})$. By the same Jensen argument as in Lemma B.1,

$$\lambda_k \le \sqrt{M}\sqrt{\|\bar{\tilde{g}}_k\| + \Delta_k^g} + \bar{\delta}_k.$$

The bridge bound gives $\|\bar{\tilde{g}}_k\| \le \|g_k\| + L_1 D(X_k)$. The second condition in (5.7) gives $\Delta_k^g = \mathcal{O}(\|g_k\|^{1+\gamma})$. Consequently,

$$\lambda_k \le C\sqrt{\|g_k\| + \mathcal{O}(\|g_k\|^{1+\gamma})} + \mathcal{O}(\|g_k\|^{1+\gamma}) = \mathcal{O}(\sqrt{\|g_k\|}).$$

By Theorem 5.20, $\|g_k\| \to 0$ along the post-burn-in tail. Together with the preceding bound, this implies $\eta\lambda_k \le \mu/2$ for all sufficiently large $k$.

Since $H_k + \lambda_k I \succeq \mu I$,

$$\mu\|\bar{s}_k\| \le \|(H_k + \lambda_k I)\bar{s}_k\| = \|g_k - r_k\| \le \|g_k\| + \eta\lambda_k\|\bar{s}_k\|.$$

Therefore $\|\bar{s}_k\| \le \frac{2}{\mu}\|g_k\|$ for all sufficiently large $k$.

Using $g_{k+1} = g_k + H_k\bar{s}_k + e_k$ with $\|e_k\| \le \frac{L_2}{2}\|\bar{s}_k\|^2$ and $g_k + H_k\bar{s}_k = -\lambda_k\bar{s}_k + r_k$,

$$\|g_{k+1}\| \le (1+\eta)\lambda_k\|\bar{s}_k\| + \frac{L_2}{2}\|\bar{s}_k\|^2 \le \mathcal{O}(\|g_k\|^{3/2}) + \mathcal{O}(\|g_k\|^2) = \mathcal{O}(\|g_k\|^{3/2}),$$

which proves the claim. ∎

*Proof of Theorem A.16.* Under (A.8), $\tau_k = t_k = \lceil (p \log(k+2) + c_{\text{mix}})/(-\log \rho) \rceil \leq c_0 + c_1 \log(k+2)$. Hence $\sum_{k=0}^{K(\varepsilon)-1} (\tau_k + 2t_k) = \mathcal{O}(K(\varepsilon) \log(K(\varepsilon) + 2))$. Combining with Theorem 5.20 yields the claim. ∎

## Appendix C. Supplementary experiments.

This appendix collects additional experimental figures that support the main results in Section 6. All settings are identical to those described in Section 6.1 unless stated otherwise.

### C.1. Additional per-problem convergence curves.

The relF-vs.-iteration panel for all nine functions is already shown in Figure 3 of the main text. Below we collect the remaining metrics.

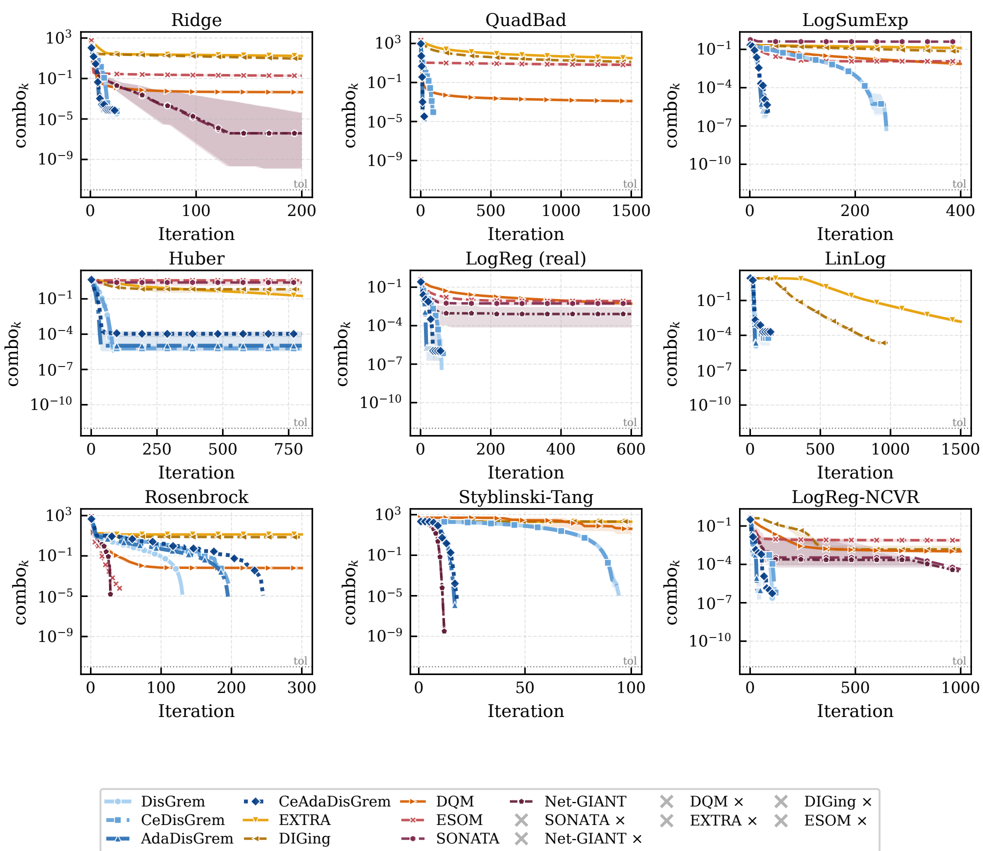


Fig. 8: Composite optimality ($\text{combo} = \|\nabla f(\bar{x}_k)\| + \text{cons}_k$) vs. iteration for all nine functions. Same setting as Figure 3; × legend entries indicate runs terminated by NaN, overflow, or divergence.

### C.2. Communication ablation details.

### C.3. Adaptive mechanism details.

### C.4. Parameter sensitivity sweeps.

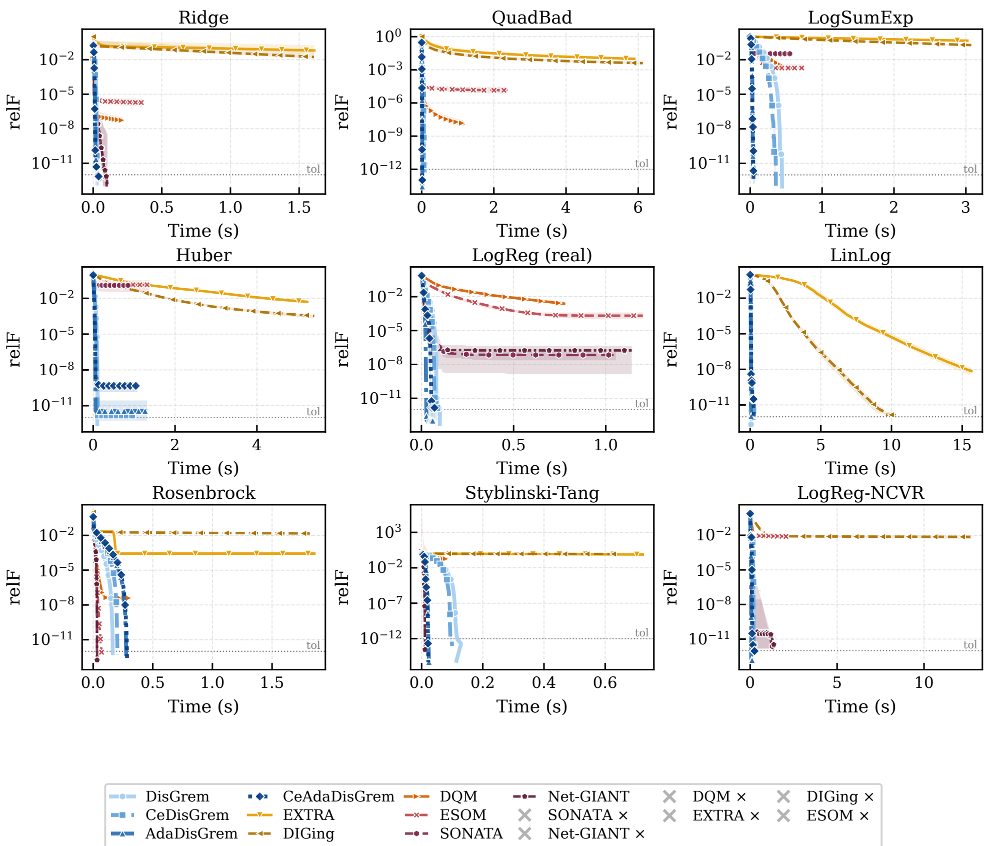


Fig. 9: relF vs. wall-clock time (seconds) for all nine functions.

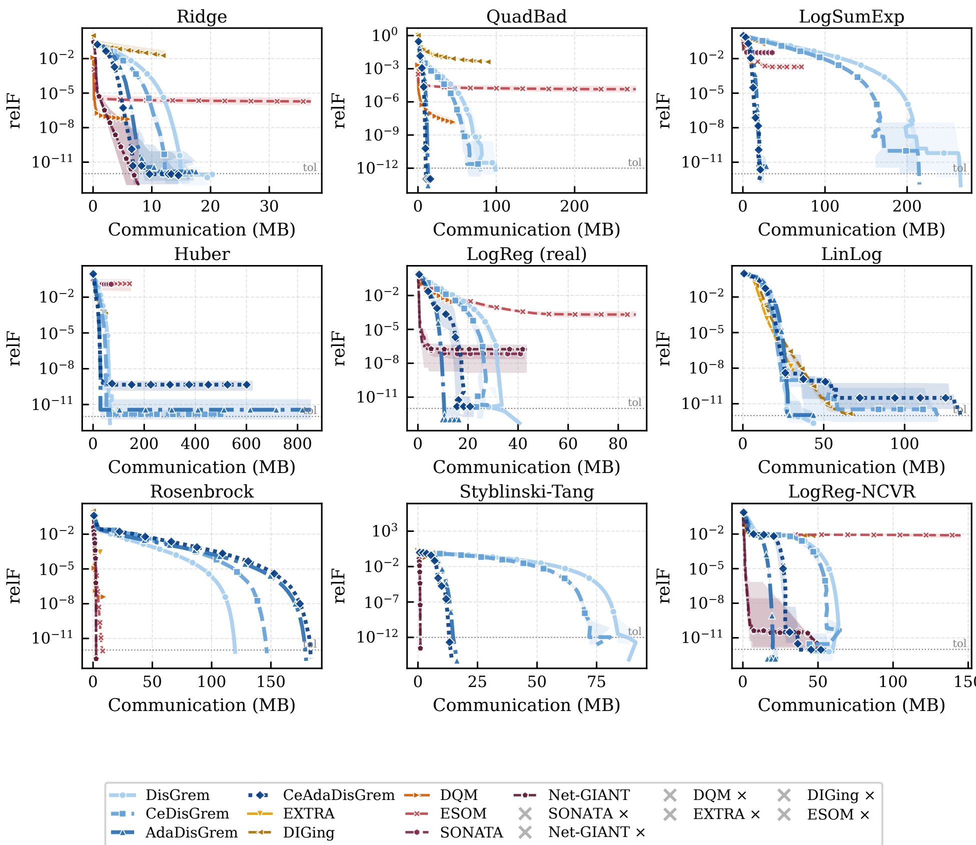


Fig. 10: relF vs. cumulative communication cost (MB) for all nine functions.

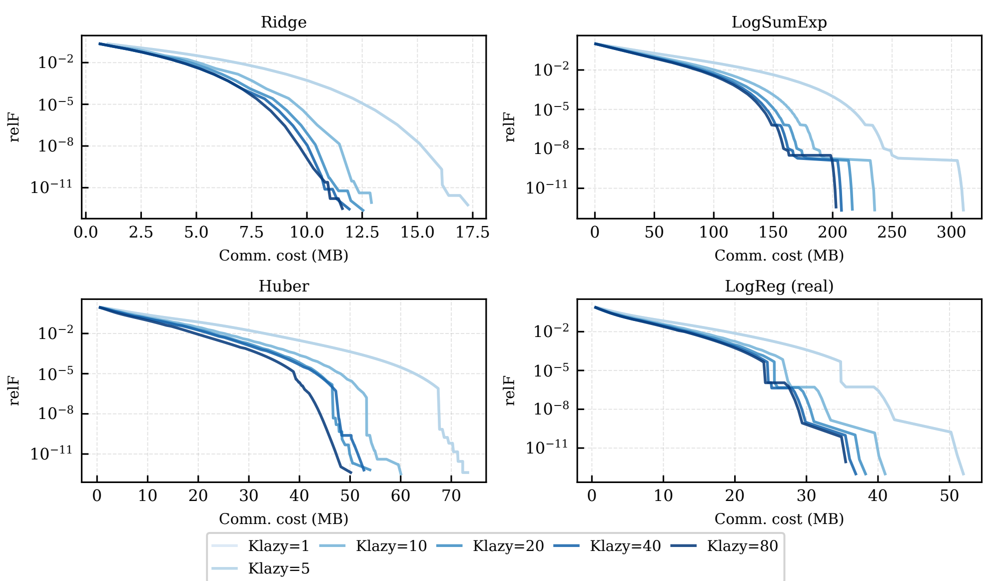


Fig. 11: Effect of $K_{\text{lazy}}$ on the communication–precision trade-off for CeDisGrem. Gradient-coloured curves from light to dark: $K_{\text{lazy}} \in \{1, 5, 10, 20, 40, 80\}$.

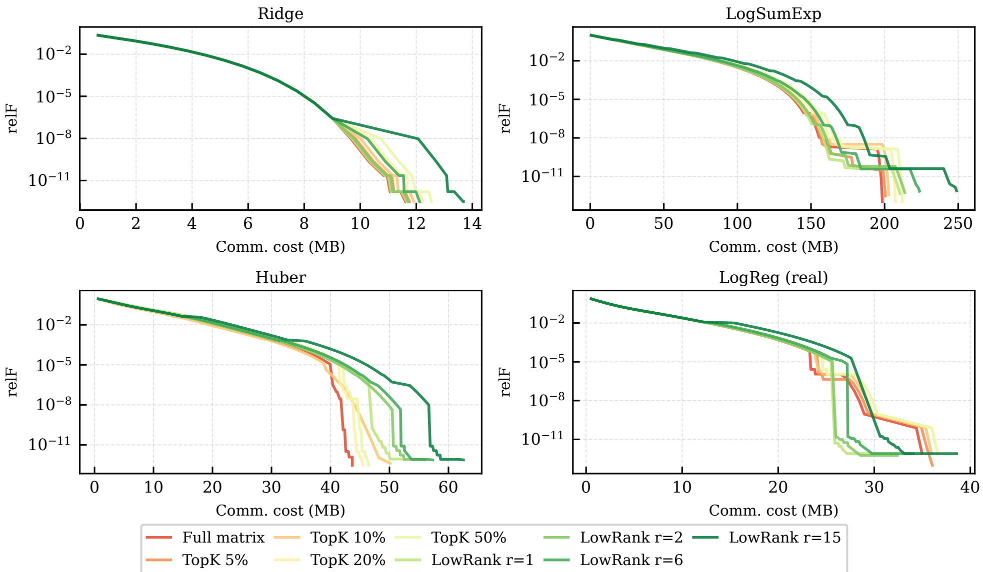


Fig. 12: Effect of compression method and rank on the communication–precision trade-off for CeDisGrem. Configurations: full (no compression), Top-$k$ ($k \in \{5\%, 10\%, 20\%, 50\%\}$), Low-Rank ($r \in \{1, 2, d/5, d/2\}$).

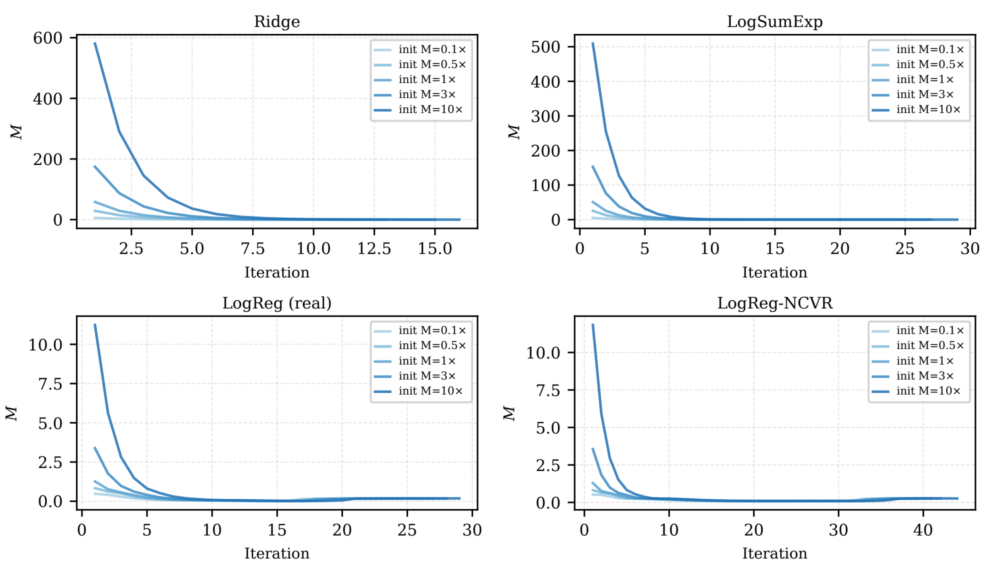


Fig. 13: Robustness of AdaDisGrem to initial $M$. Initial $M_0 \in \{0.1M^*, 0.5M^*, M^*, 3M^*, 10M^*\}$. All initializations converge to similar trajectories within 50–100 iterations.

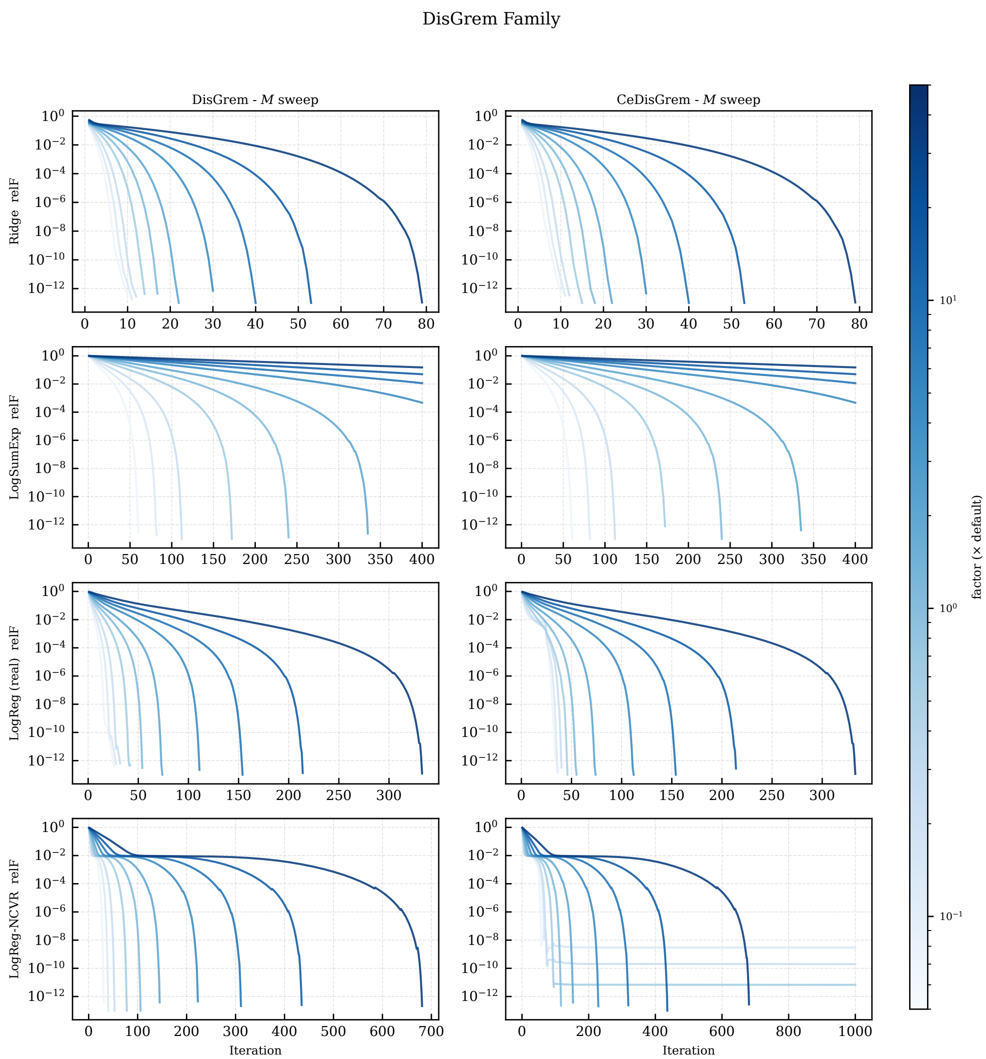


Fig. 14: Parameter sweep for the DisGrem family on four functions. 10 values of $M_{\mathrm{fac}}$ (light-to-dark for increasing $M_{\mathrm{fac}}$). Rows: functions; columns: algorithms.

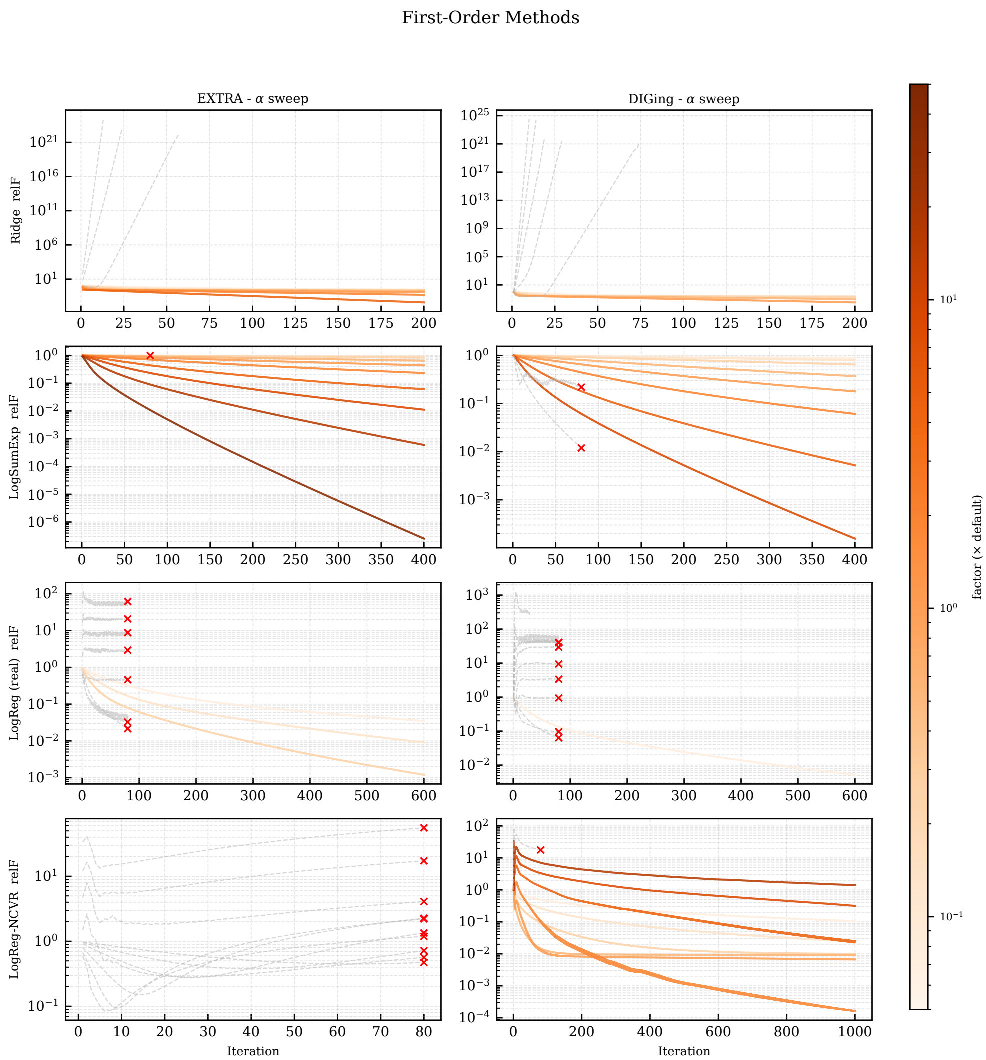


Fig. 15: Stepsize sensitivity for first-order baselines (EXTRA and DIGing) on four functions. 10 values of $\alpha$ (light-to-dark). Divergent runs are clipped at $\mathrm{relF} = 10^4$.

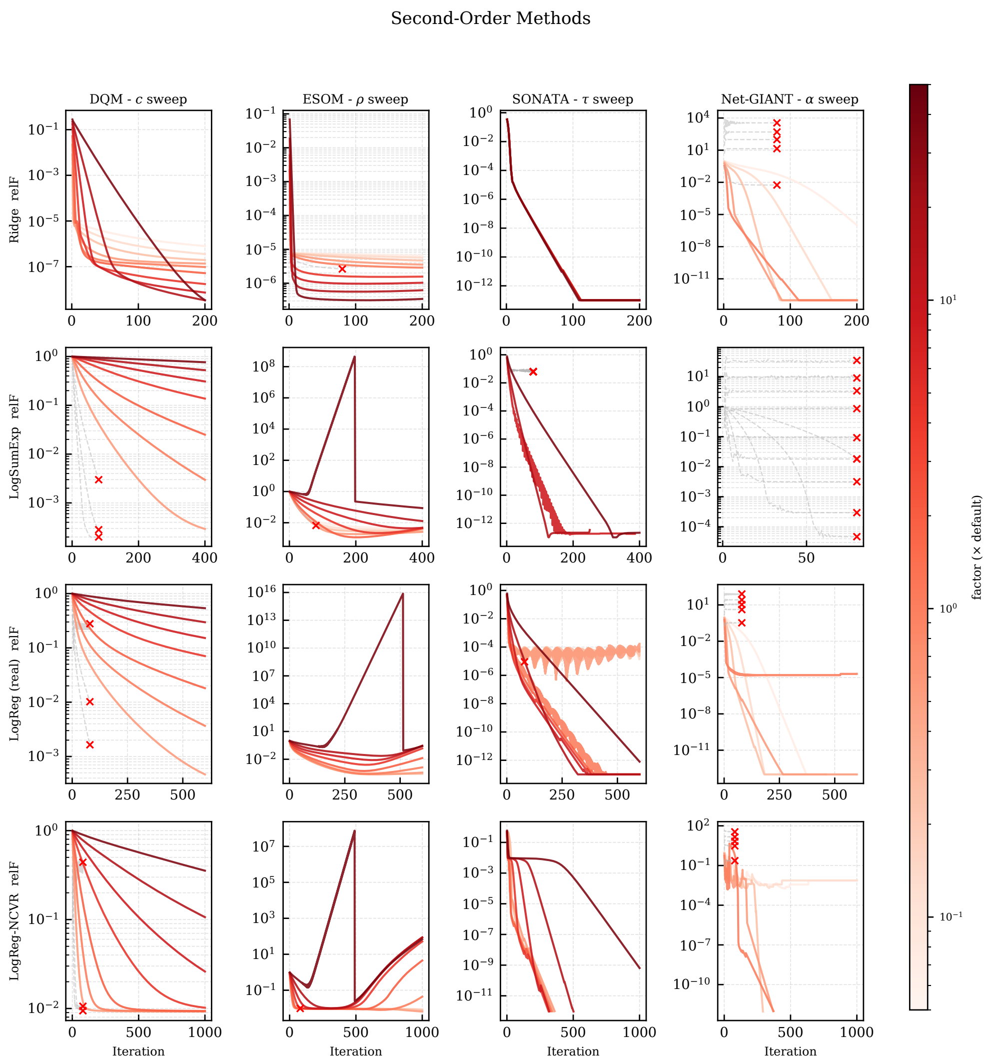


Fig. 16: Key-parameter sensitivity for second-order baselines on four functions: penalty $c$ (DQM), penalty $\rho_{\mathrm{E}}$ (ESOM), regularization $\tau$ (SONATA), stepsize $\alpha$ (Net-GIANT). Entries marked by $\times$ indicate parameter settings with NaN, overflow, or divergence.